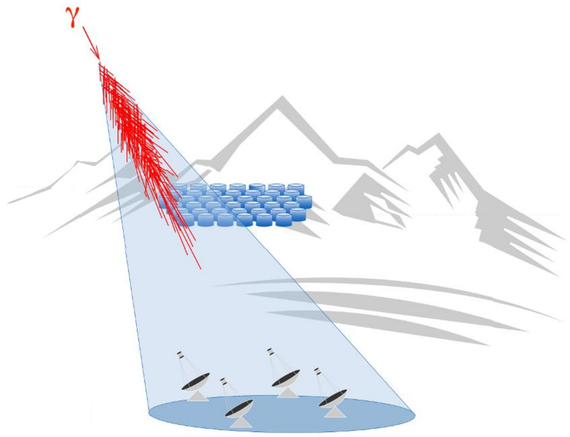


Detection and reconstruction of extensive air showers

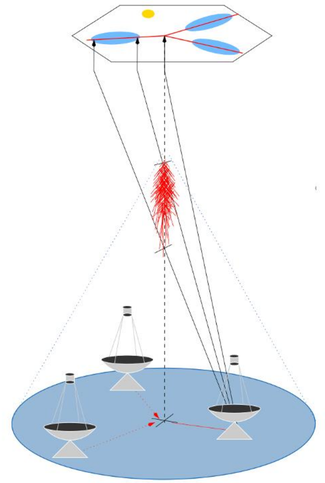
Federica Bradascio (IJCLab, Université Paris-Saclay)

CADS School - November 2026

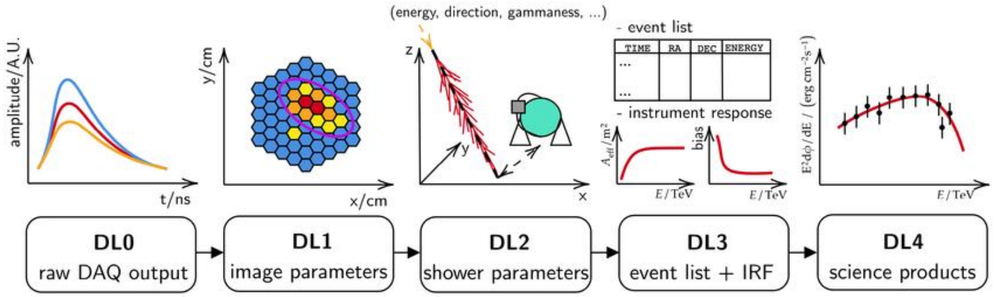
Outline



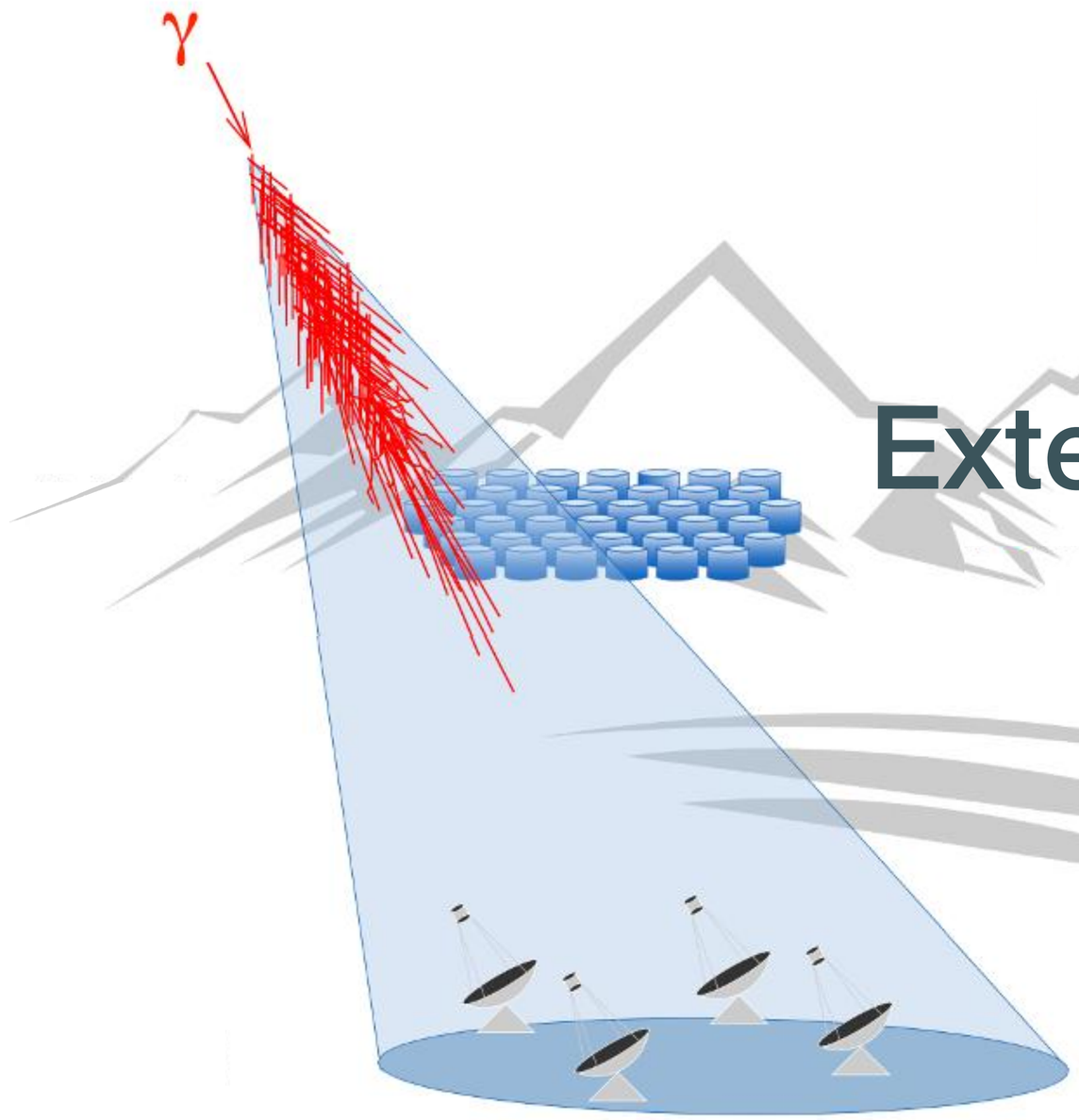
1. Extensive Air Showers:
from sources to telescopes



2. IACTs: detection principles



3. From low-level to
high-level IACTs data



Extensive Air Showers: from sources to telescopes

Gamma-rays astrophysics

What are they and why studying them?

Definitions:

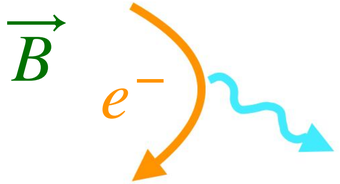
- Medium-Energy Gamma Rays (MeV)
- High-Energy (HE) Gamma Rays (100 MeV — 50 GeV)
- Very-High-Energy (VHE) Gamma Rays (50 GeV — 100 TeV)
- Ultra-High-Energy (UHE) Gamma Rays (≳100 TeV)

Astrophysical Gamma Rays:

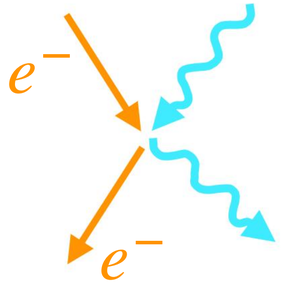
- Indicate the presence of a parental population of high-energy massive particles
- Little effect from absorption in the galaxy
- Carry information directly from the sites of acceleration

γ-rays: how are they produced?

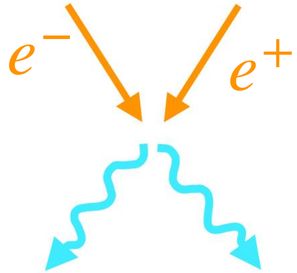
Leptonic Processes



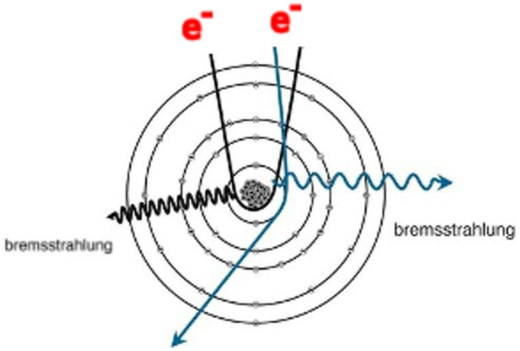
Electron Synchrotron



Inverse Compton Scattering

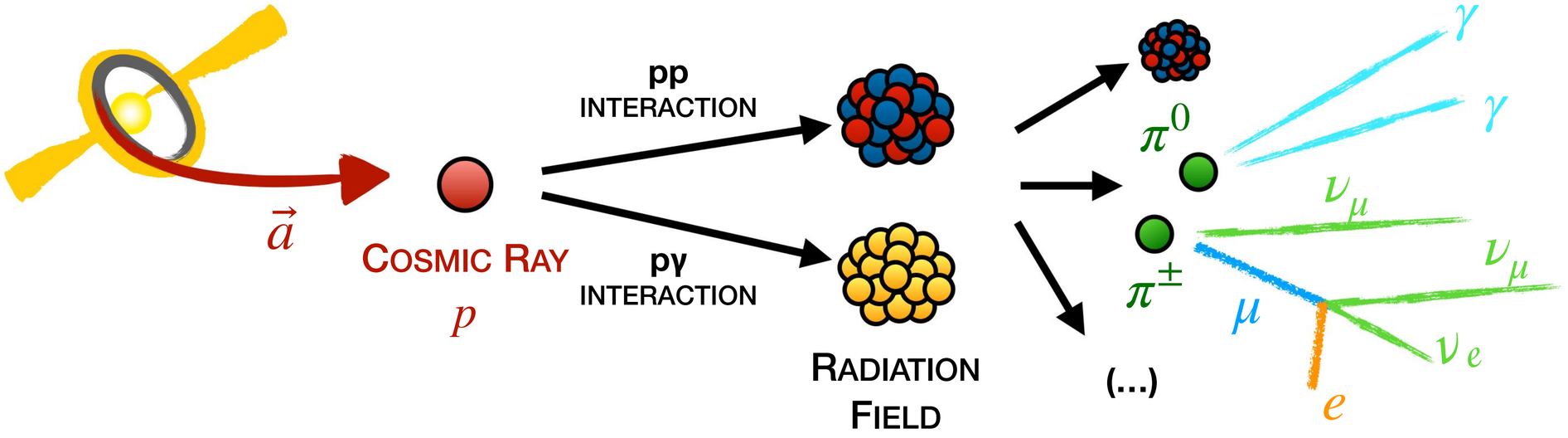


Electron-Positron Annihiliation



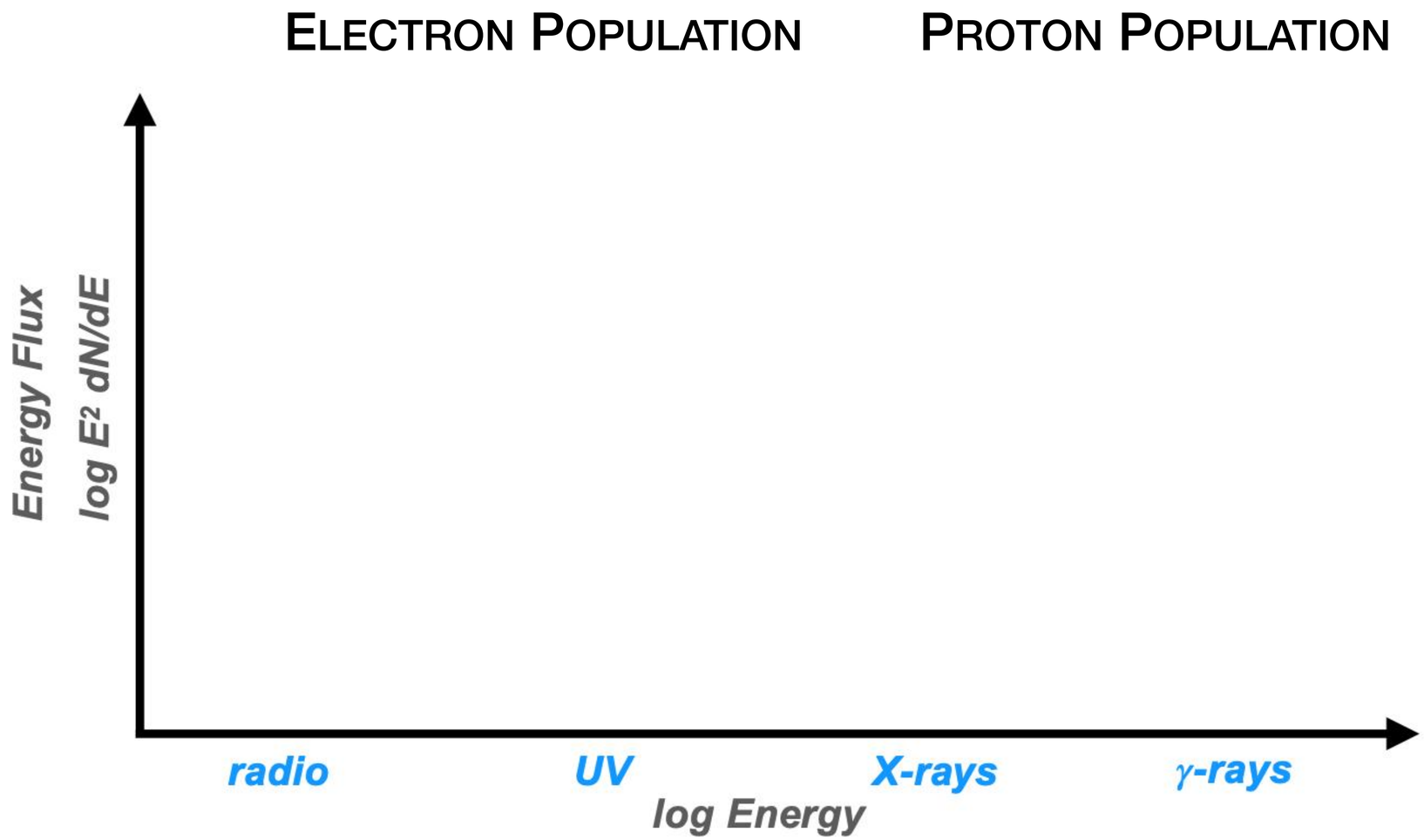
Bremsstrahlung

Hadronic Processes



γ-rays: how are they produced?

Only produced by non-thermal processes

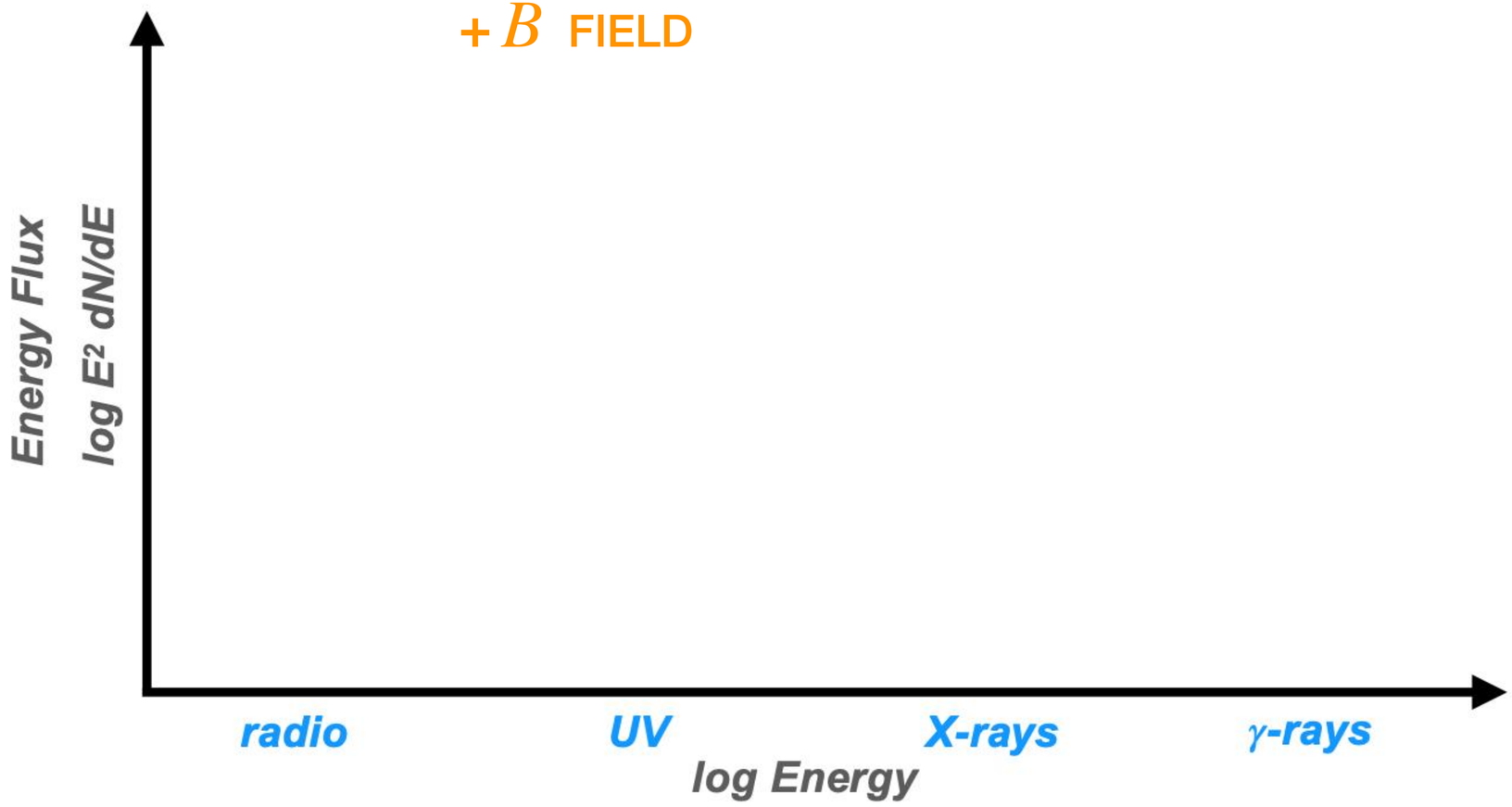


γ-rays: how are they produced?

Only produced by non-thermal processes

ELECTRON POPULATION **PROTON POPULATION**

$+\vec{B}$ FIELD

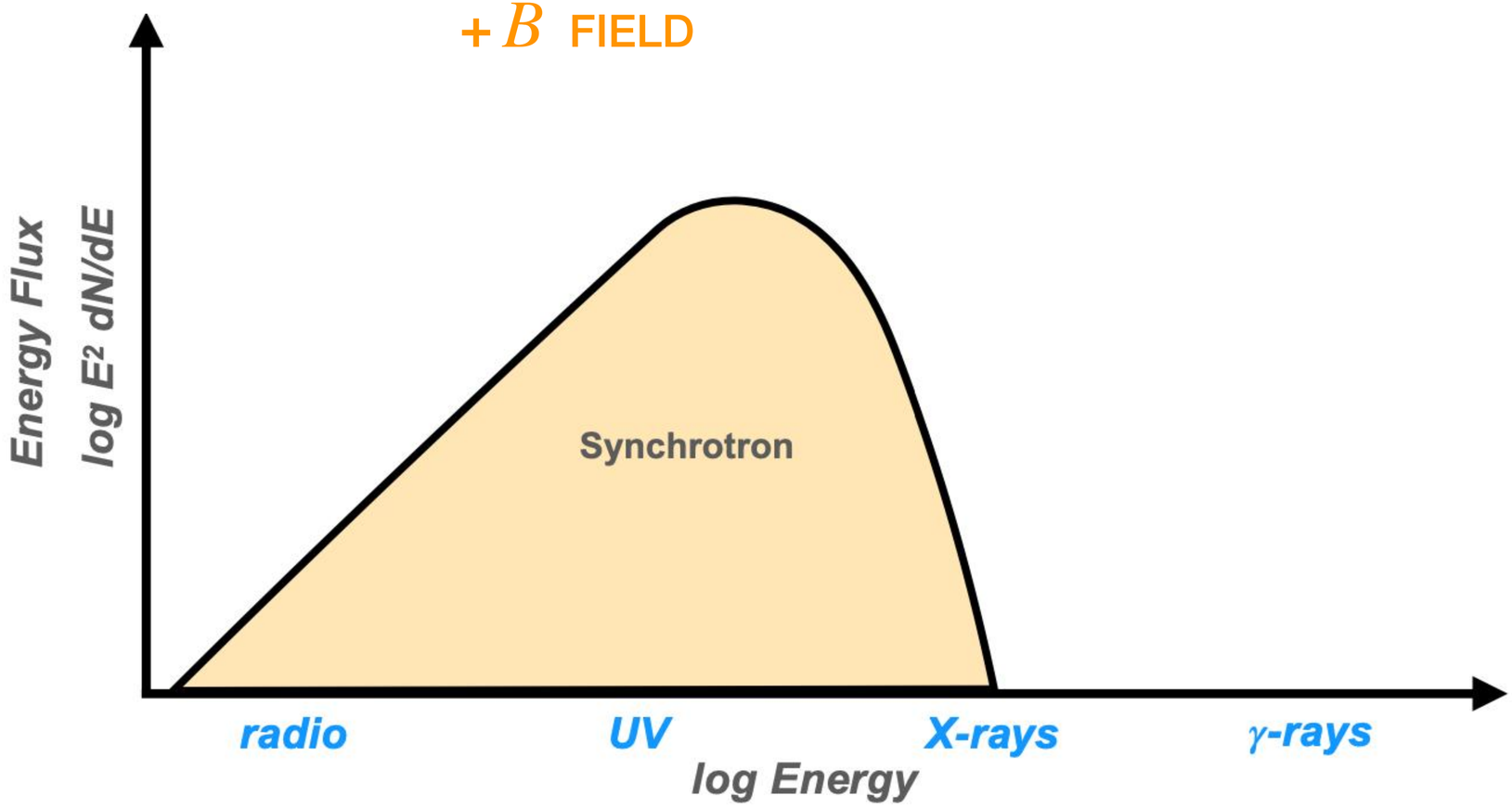


γ-rays: how are they produced?

Only produced by non-thermal processes

ELECTRON POPULATION **PROTON POPULATION**

$+\vec{B}$ FIELD



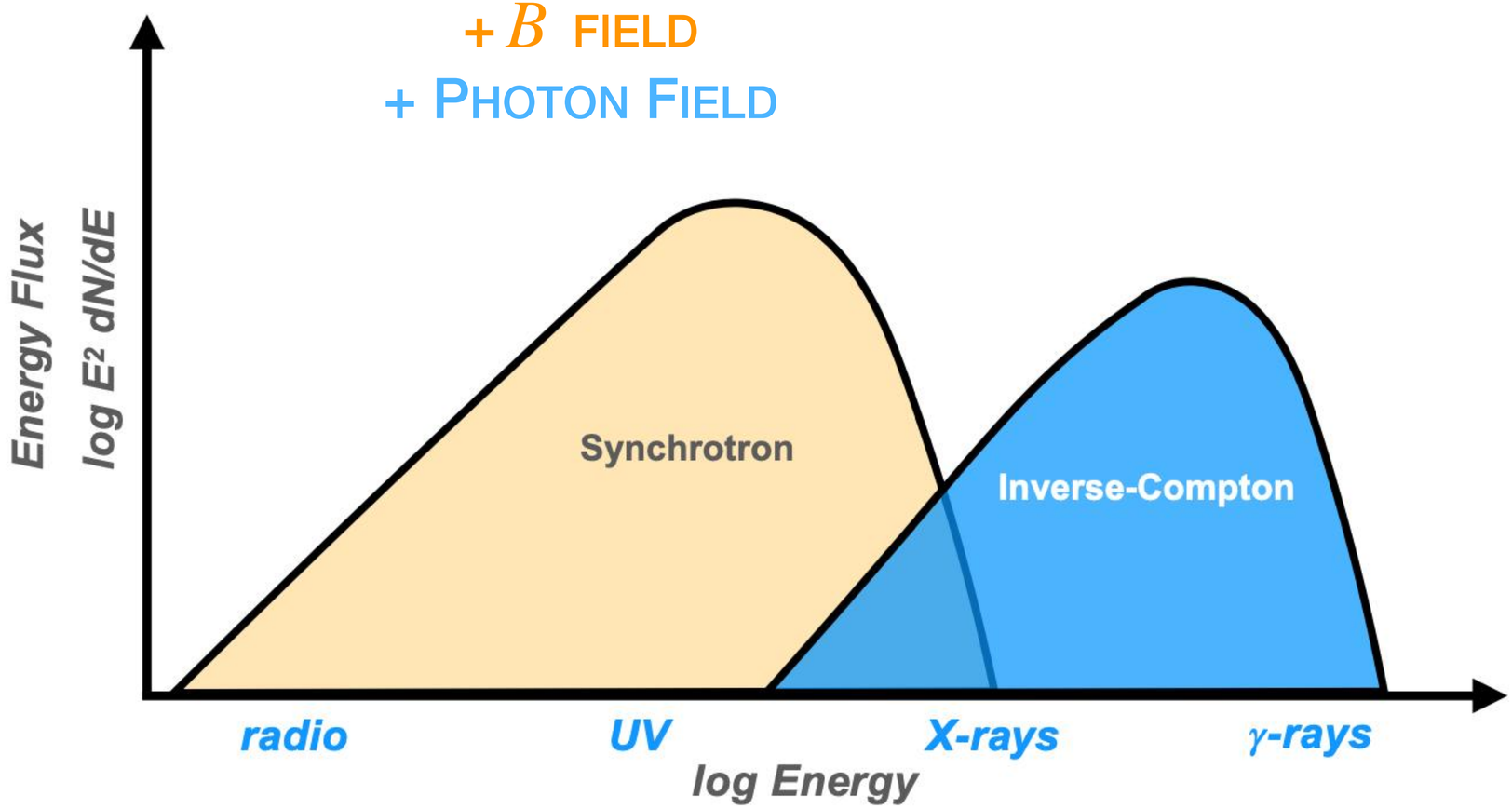
γ-rays: how are they produced?

Only produced by non-thermal processes

ELECTRON POPULATION **PROTON POPULATION**

+ $\vec{B}$ FIELD

+ PHOTON FIELD



γ-rays: how are they produced?

Only produced by non-thermal processes

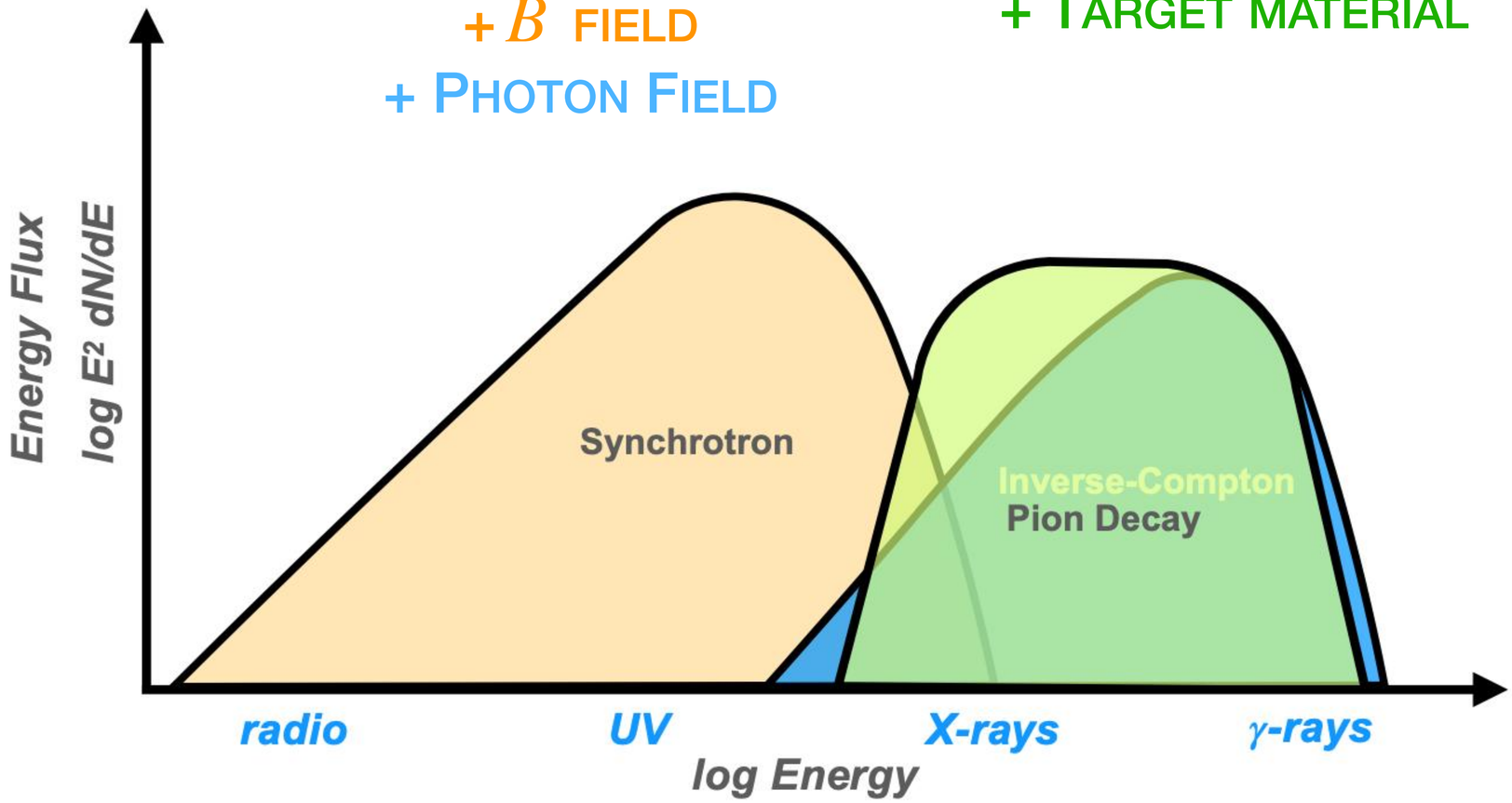
ELECTRON POPULATION

+ $\vec{B}$ FIELD

+ PHOTON FIELD

PROTON POPULATION

+ TARGET MATERIAL

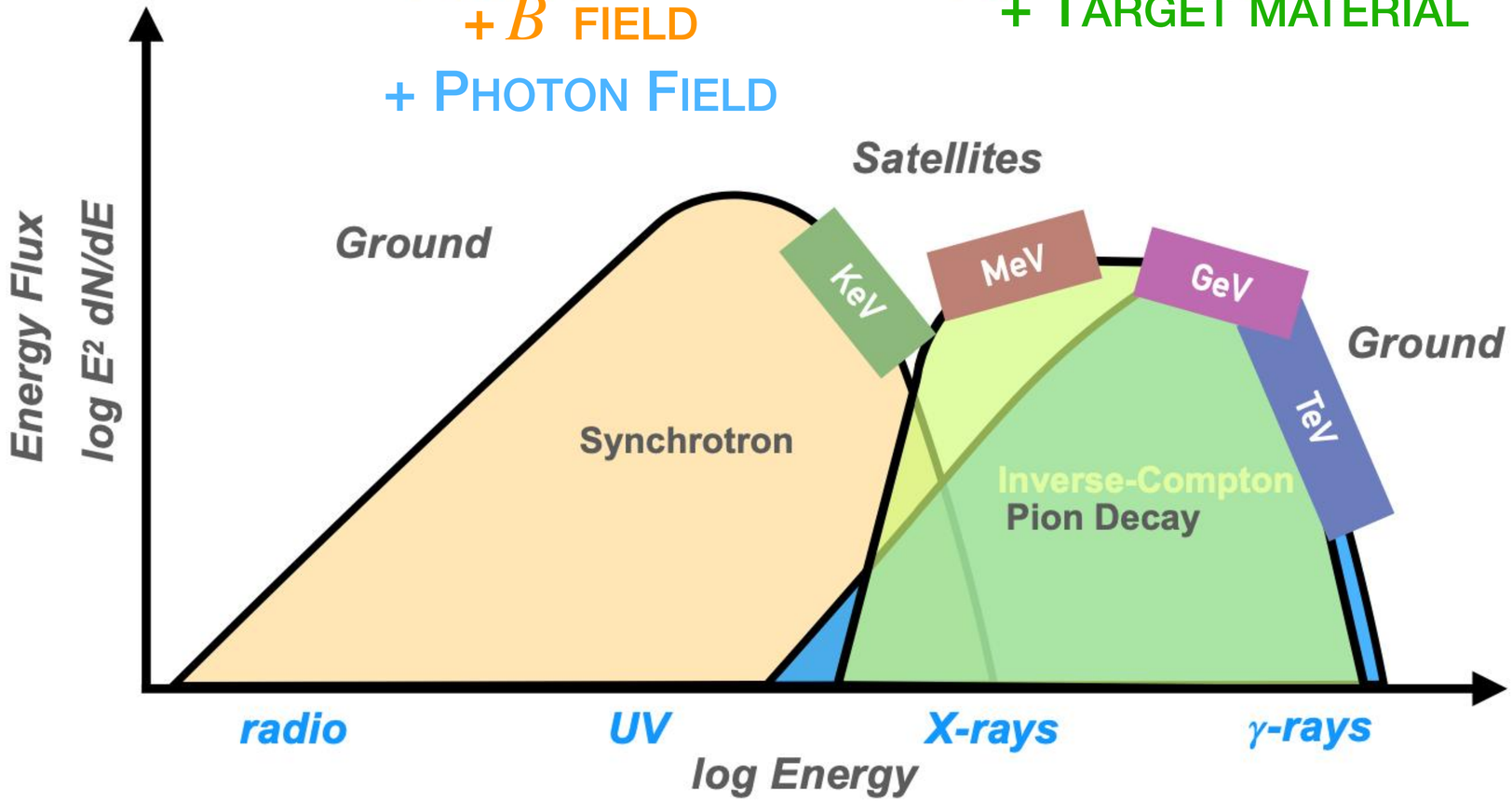


γ-rays: how are they produced?

Only produced by non-thermal processes

ELECTRON POPULATION
+ $\vec{B}$ FIELD
+ PHOTON FIELD

PROTON POPULATION
+ TARGET MATERIAL



γ-rays: how to detect them?

- Medium-Energy Gamma Rays (MeV)

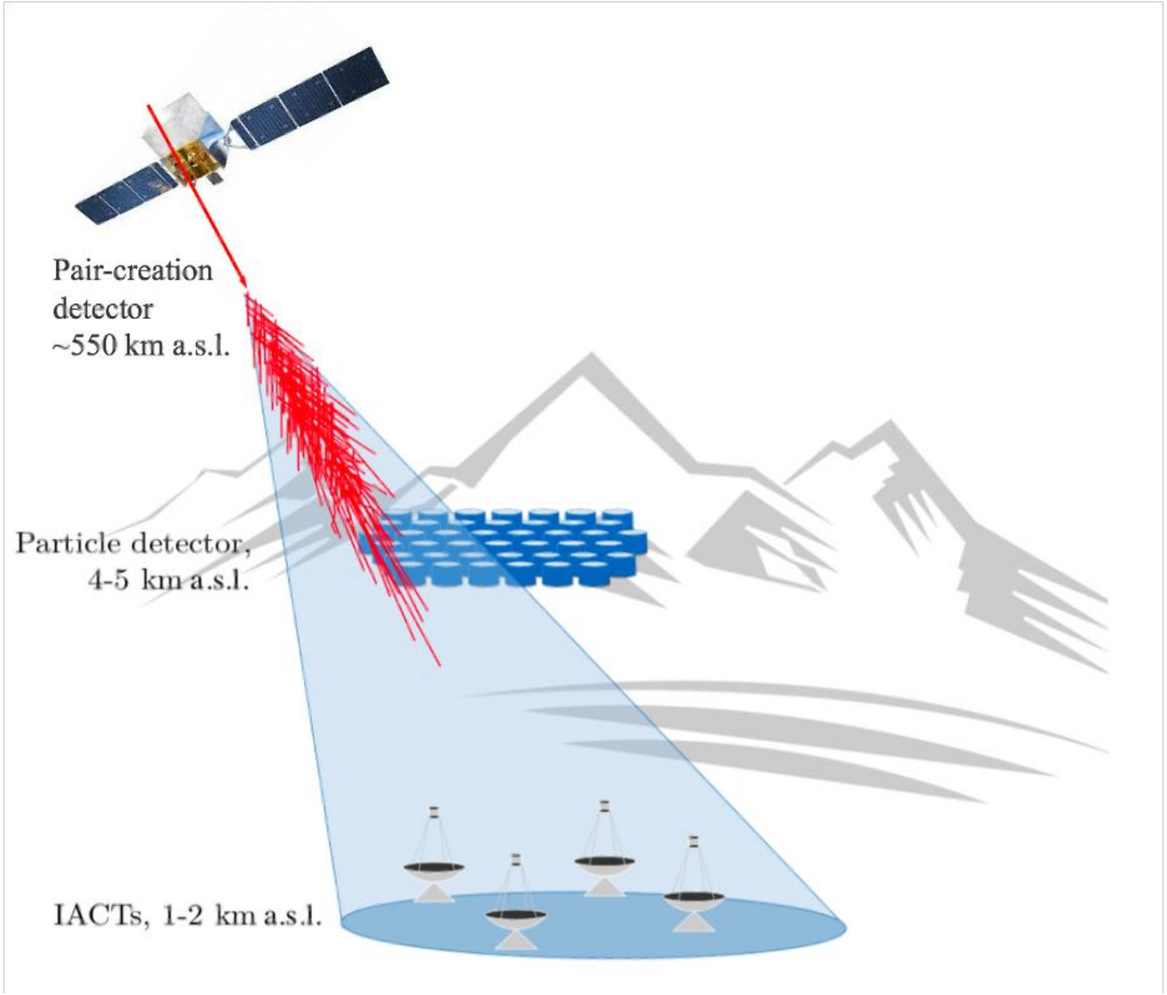
  → 🛰️ Detected from space

- High-Energy (HE) Gamma Rays (100 MeV — 50 GeV)

  → 🛰️ & 🌍 Detected from space and ground-based experiments

- Very-High-Energy (VHE) Gamma Rays (50 GeV — 100 TeV)

  → 🌍 Detected from ground-based experiments



γ-rays: how to detect them?

- Medium-Energy Gamma Rays (MeV)

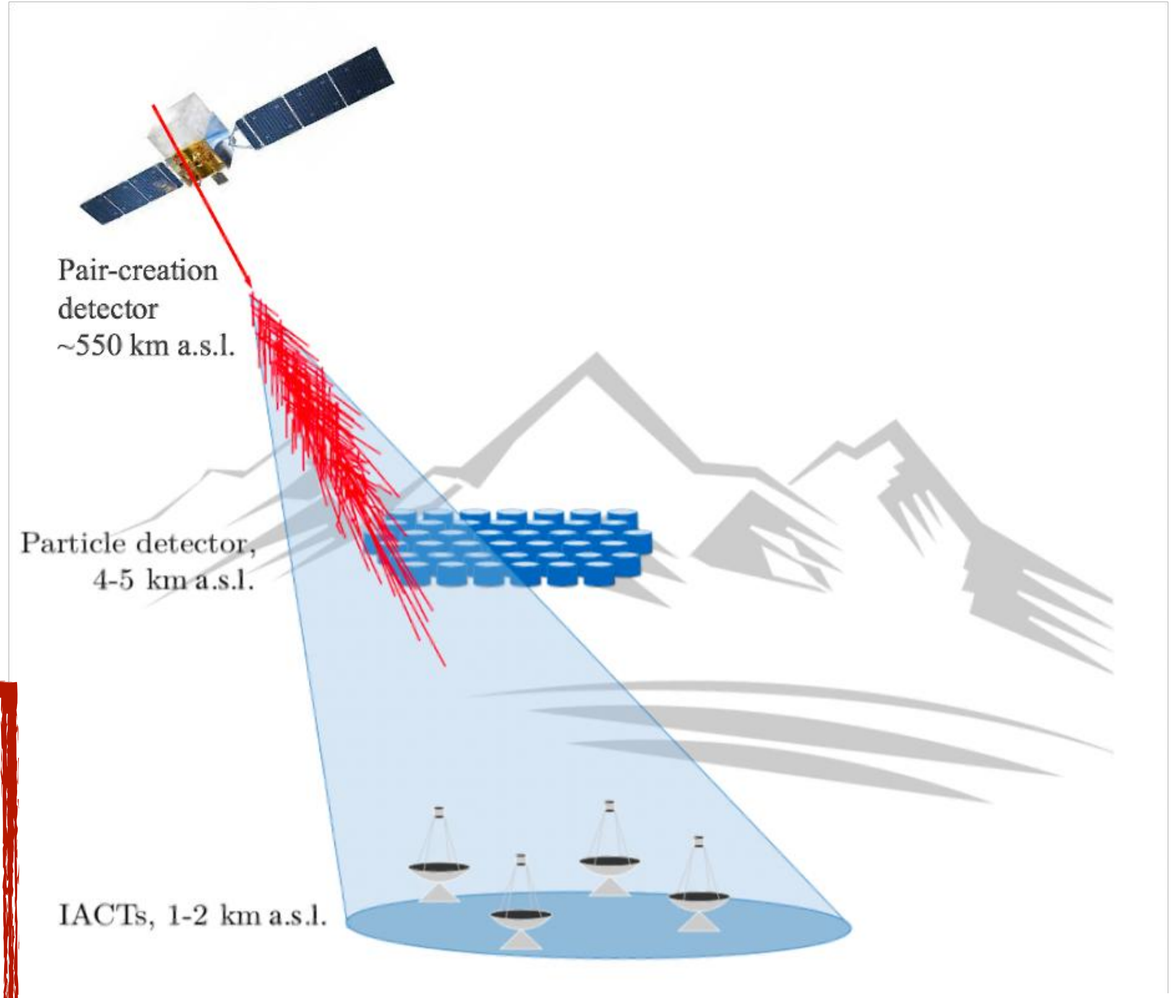
  → 🛰️ Detected from space

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  → 🛰️ & 🌍 Detected from space and ground-based experiments

- Very-High-Energy (VHE) Gamma Rays (50 GeV — 100 TeV)

  → 🌍 Detected from ground-based experiments



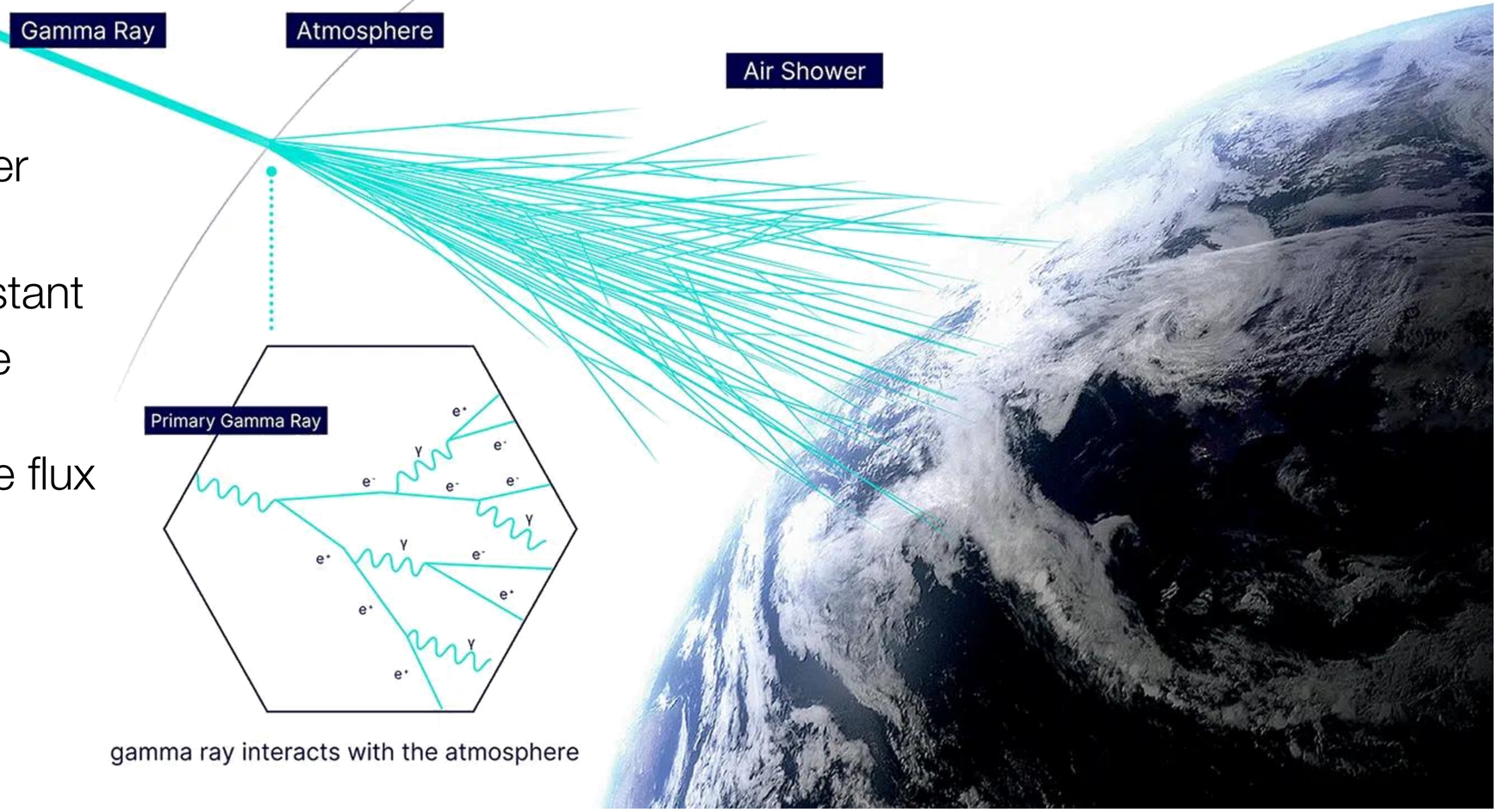


Extensive Air Showers

Interaction of primary particle in the high atmosphere

⇒ showers of secondary particles

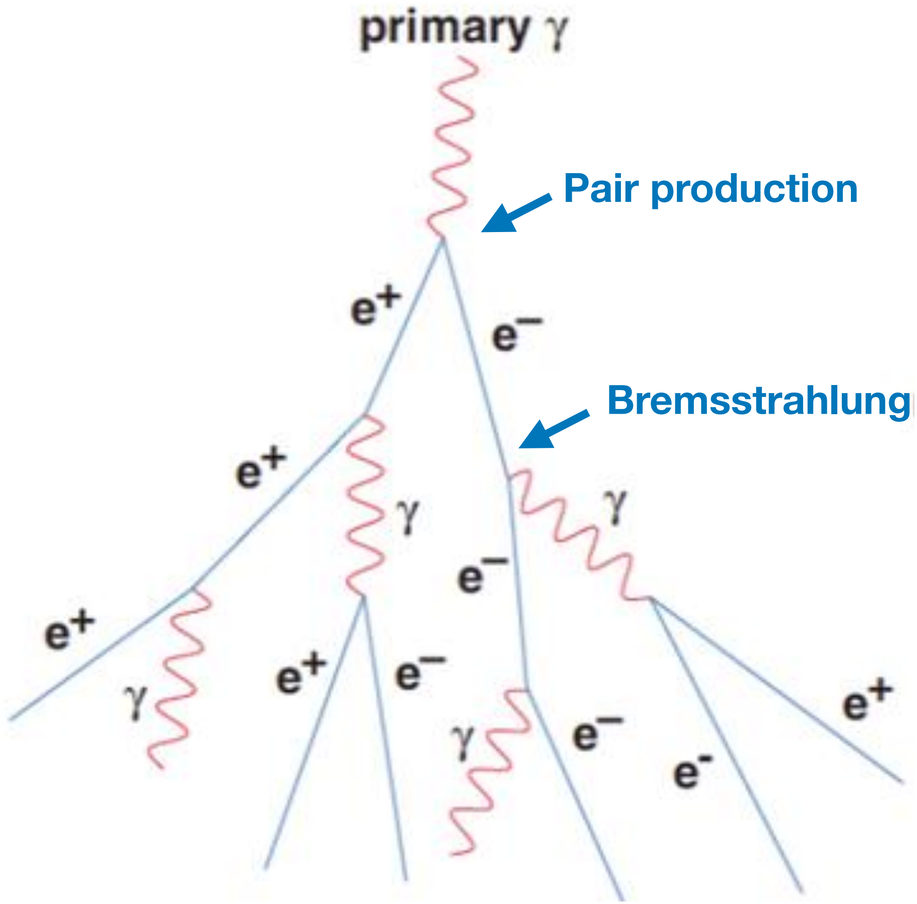
- Atmosphere acts as an inhomogeneous calorimeter and a tracking medium
- Spread on a wide area, distant detection ⇒ large effective area, can be used at high energies (≥ 100 TeV) where flux are very low
- Cascades initiated by:
  - Photons
  - Charged particles

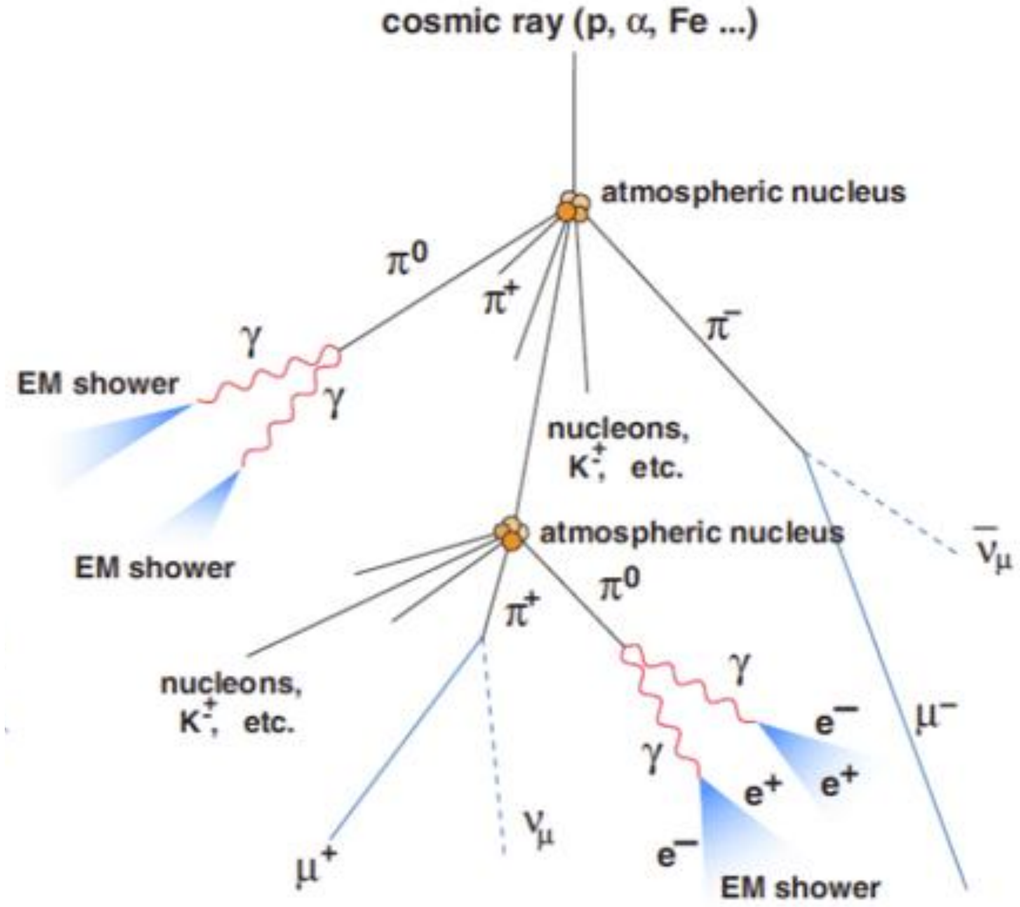


gamma ray interacts with the atmosphere

Extensive Air Showers

Electromagnetic vs Hadronic showers

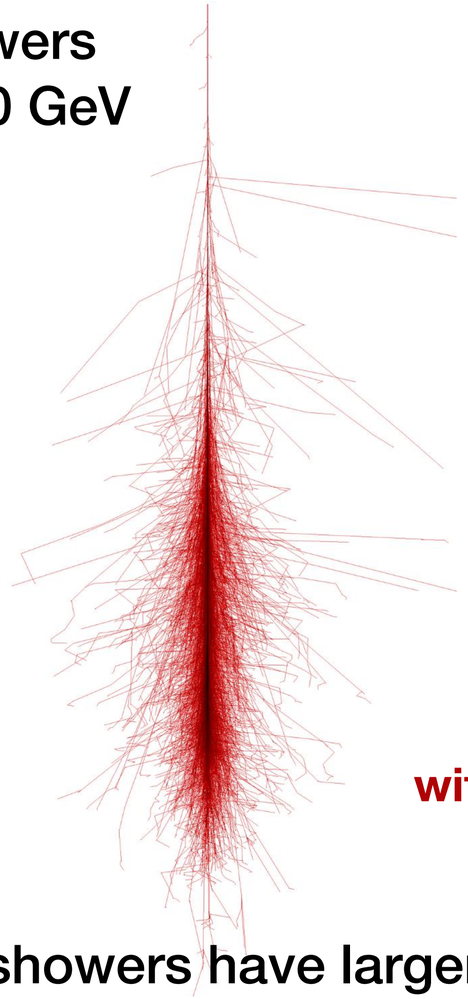




Extensive Air Showers

Electromagnetic vs Hadronic showers

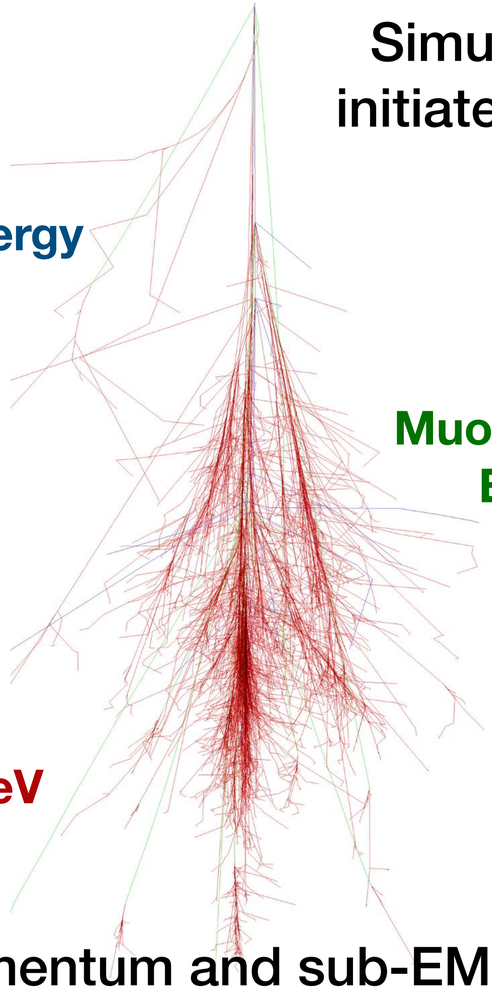
Simulated showers initiated by a 100 GeV gamma ray



Hadrons with energy E > 0.1 GeV

e^+, e^-, γ
with energy E>0.1 MeV

Simulated showers initiated by a 100 GeV proton

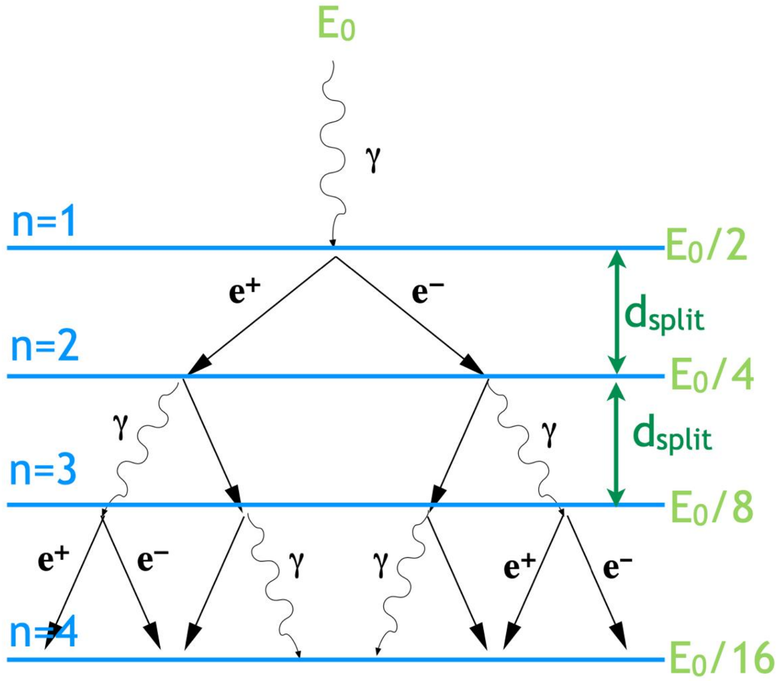


Muons with energy E > 0.1 GeV

Hadronic showers have larger transverse momentum and sub-EM showers

The Heitler model

A simplified model for EM showers



- Two processes: *pair production* + *single-particle bremsstrahlung*
- Initial energy E_0, radiation length X_0
- Distance between both interactions is a fixed length $d_{\text{split}} = X_0 \ln 2$
- Any interaction leads to two new particles of energy $E/2$
- When the energy of a particle drops below a critical energy E_{crit}, the cascade stops abruptly

Shower depth: $x = n d_{\text{split}} = n X_0 \ln 2$

Total # particles ($e^{\pm}, \gamma$): $N = 2^n = e^{x/X_0}$

Maximum # particles: $N_{\text{max}} = E_0 / E_{\text{crit}}$

$$\Rightarrow x_{\text{max}} = X_0 \ln\left(\frac{E_0}{E_{\text{crit}}}\right)$$

The Heitler model

How good is it?

- Doesn’t account for particle loss
- Assumes abrupt stop of shower after E_{crit}
- Particle interactions are stochastic processes
- Assumes single-photon emitted during bremsstrahlung
  - In reality is several, so overestimates lepton fraction
  - Approximately: $N_\gamma \approx 10 N_e$ (can be used as a simple correction factor)
- Atmosphere is complicated (e.g., density profile)

Effect of Earth's atmosphere

Longitudinal development of electromagnetic air showers

The number ***N*** of secondary electrons and positrons at a given atmospheric depth ***x***:

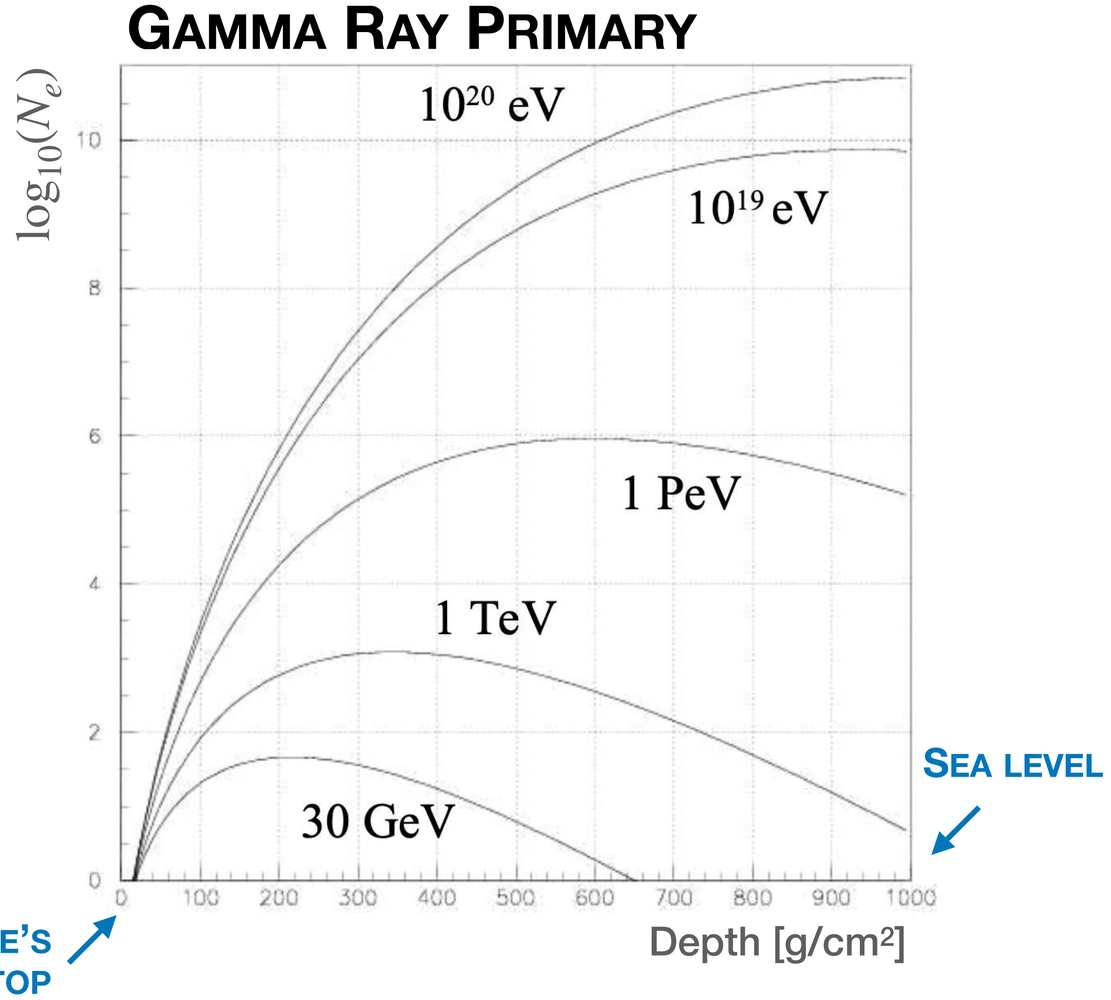
$$N(x) = \frac{0.31}{\sqrt{\ln(E_\gamma/E_c)}} \exp\left[\frac{x}{X_0}(1 - 1.5 \ln s)\right]$$

where ***s*** is the shower age parameter that describes the stage of development of the shower

$$s = \frac{3}{1 + 2\ln(E_\gamma/E_c)/(x/X_0)}$$

At the shower maximum, the total number of particles is approximately:

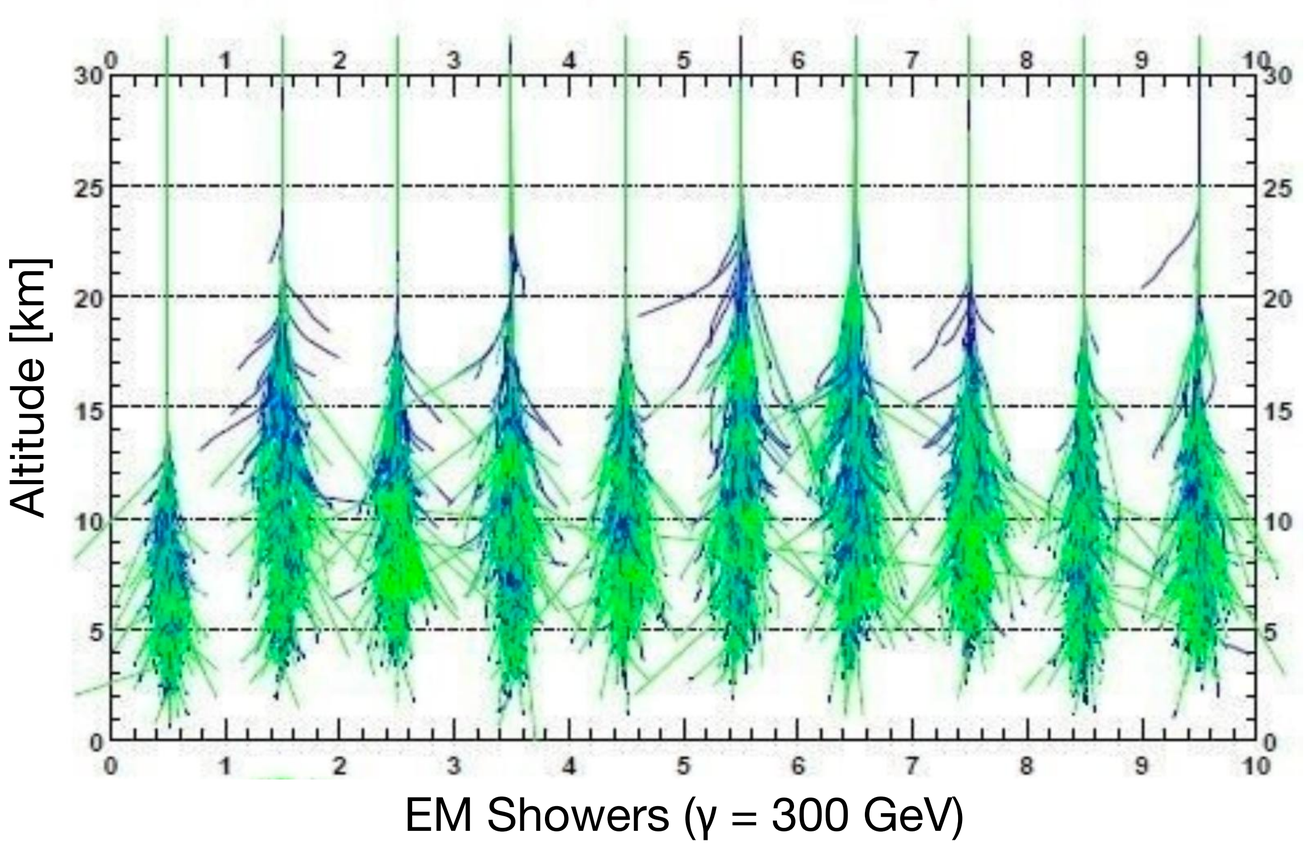
$$N(x_{max}) \sim 10^3 E_\gamma/\mathrm{TeV}$$



Shower variability

10 simulated EM showers initiated by a 300 GeV photon

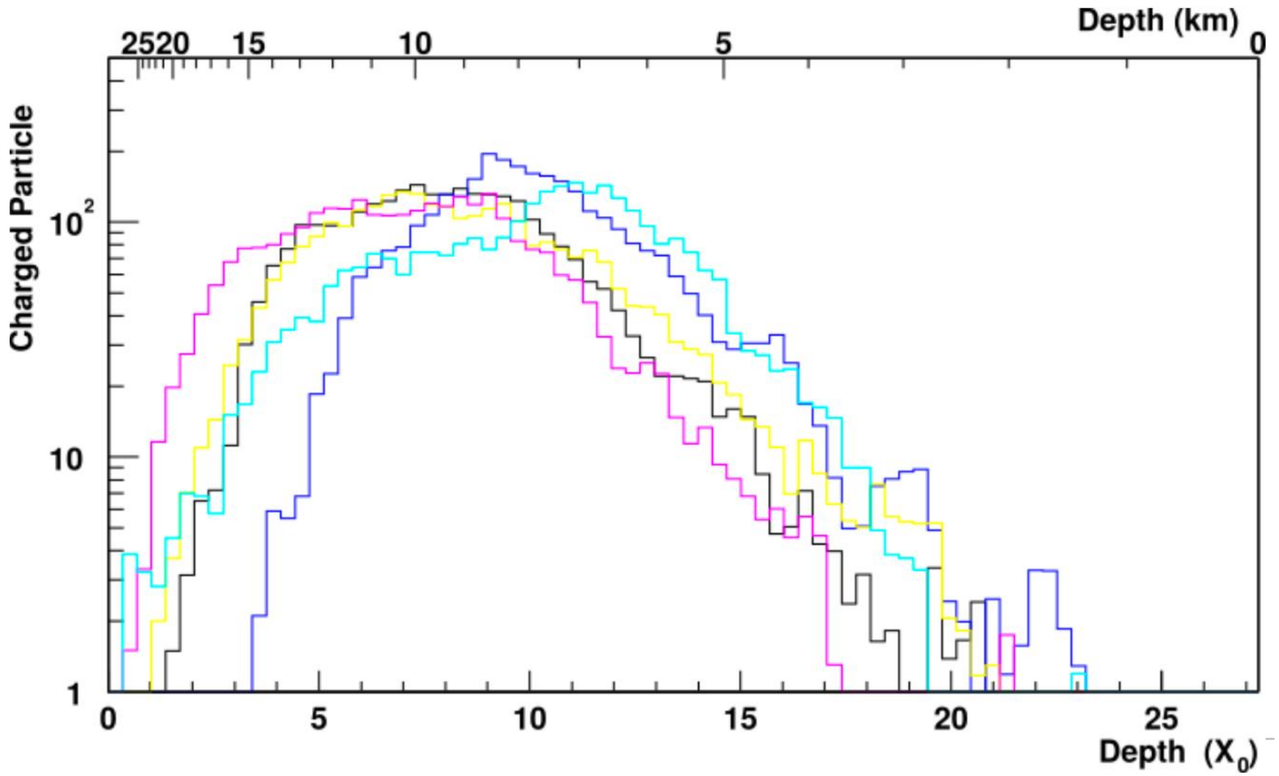
- Large fluctuations in showers, especially at low energy
- Fluctuations mainly due to the first interaction depth
- At high altitude, one radiation length ~several km → first point of interaction can vary by > 5 km
- At lower altitudes, radiation length <1km → shower development becomes similar



Shower variability

10 simulated EM showers initiated by a 300 GeV photon

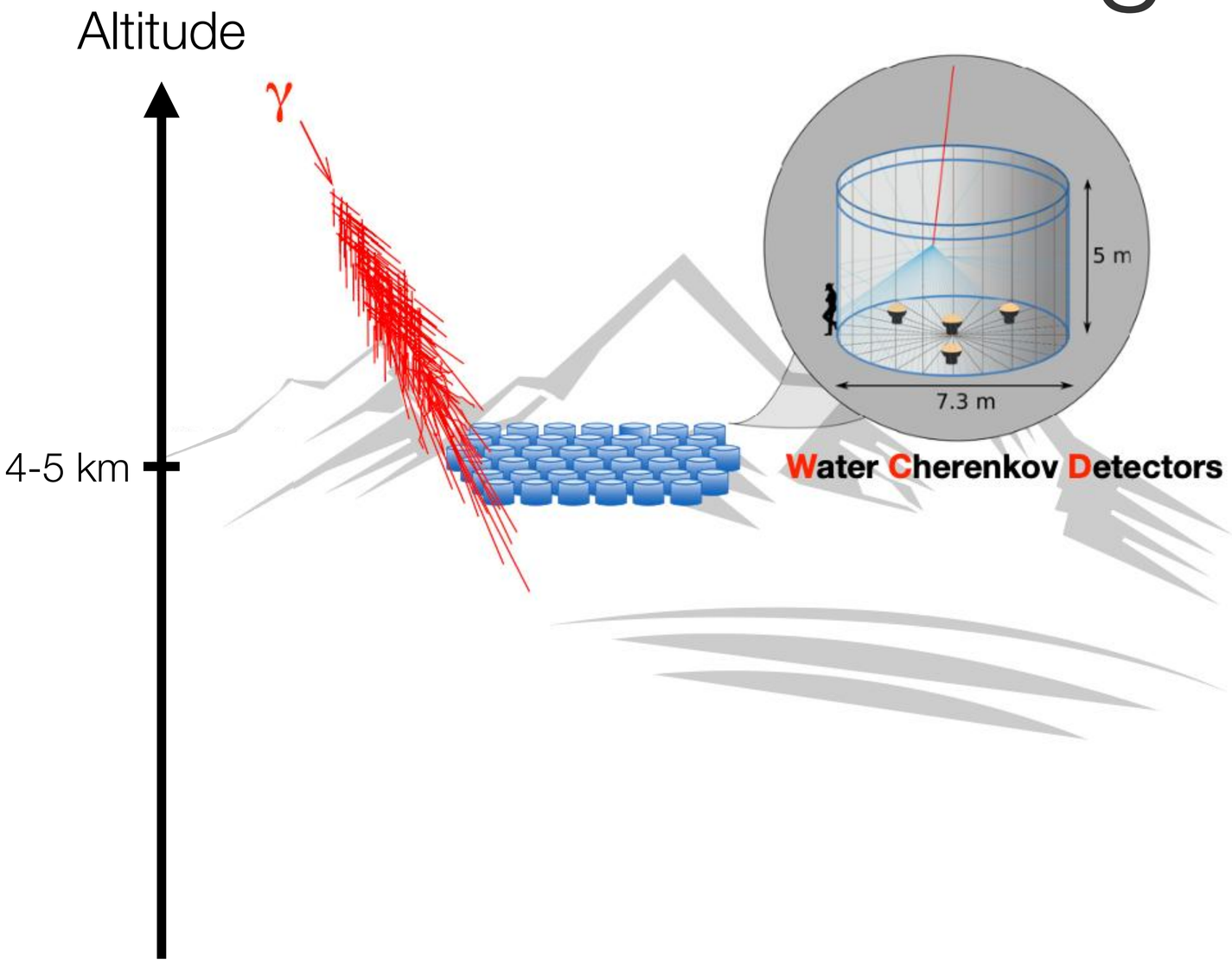
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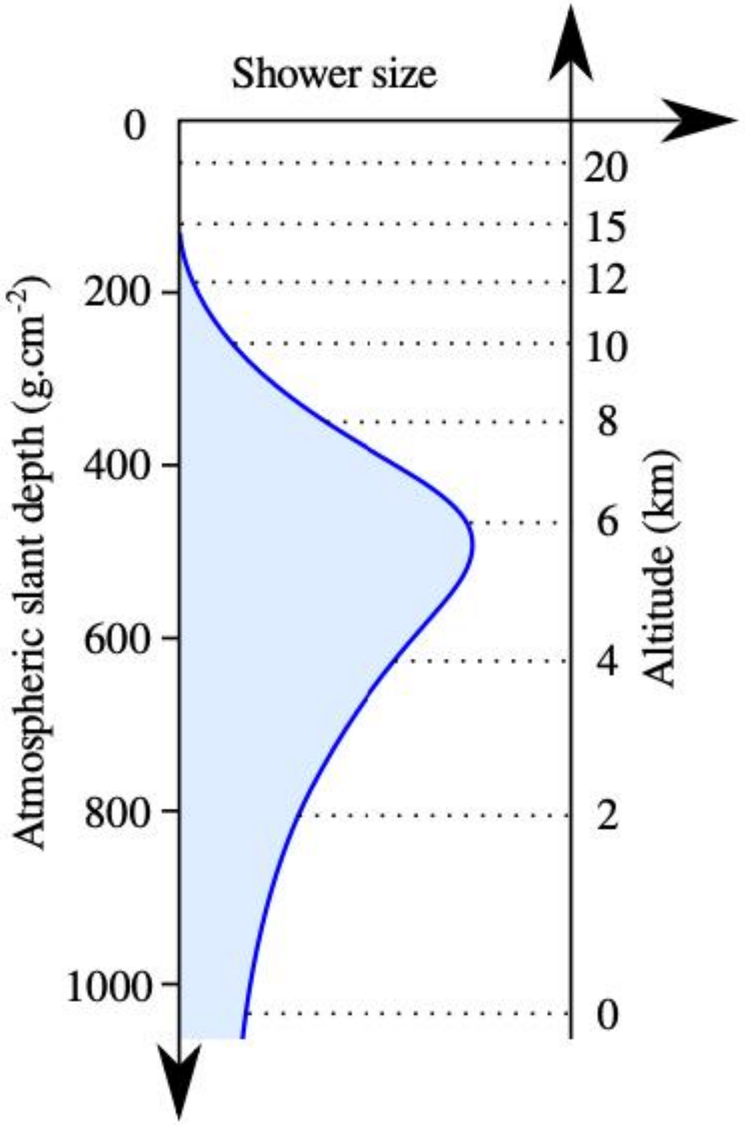


Similar shower profiles, only shifted by few radiation lengths

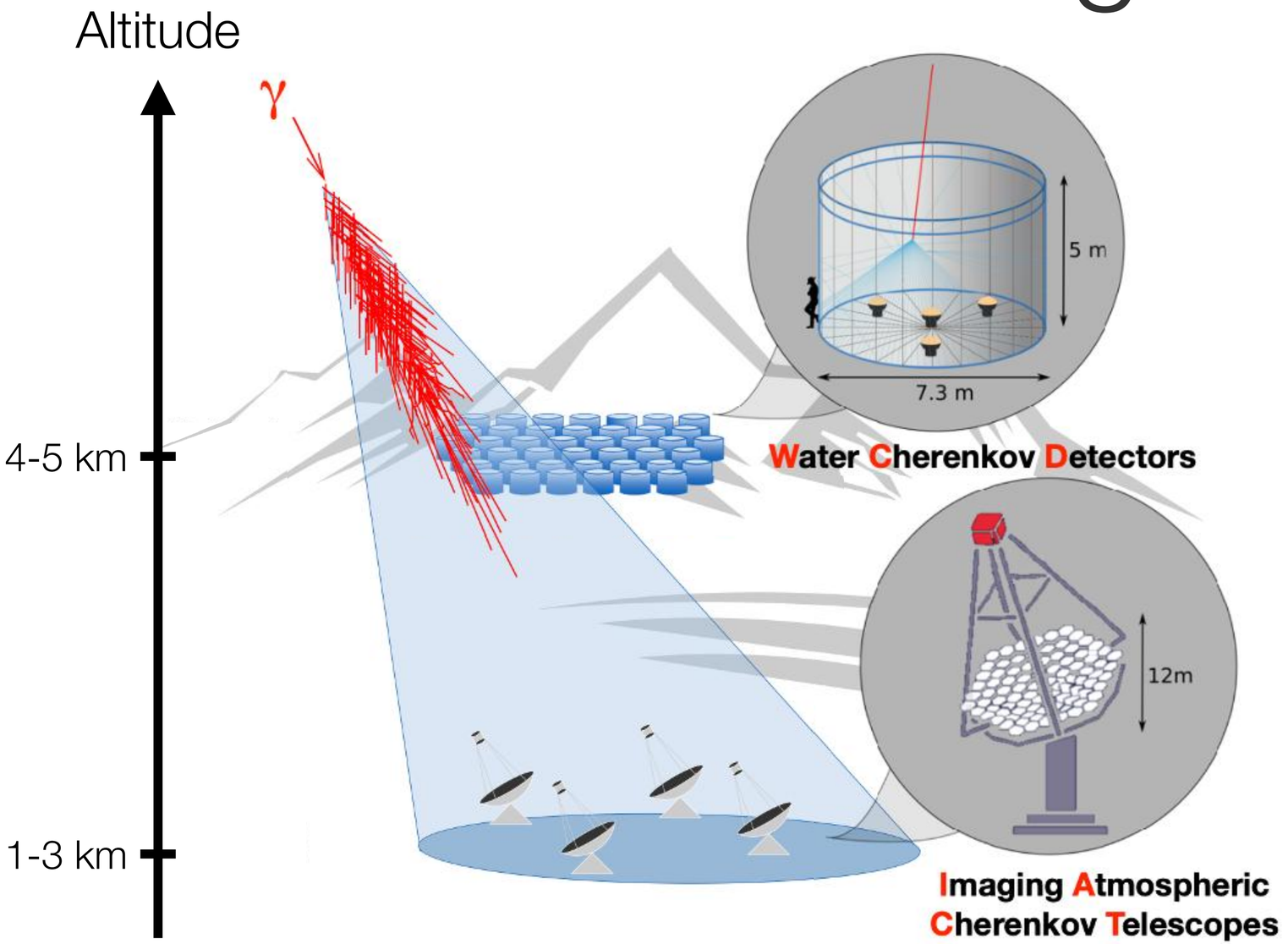
→ relevant parameter is altitude

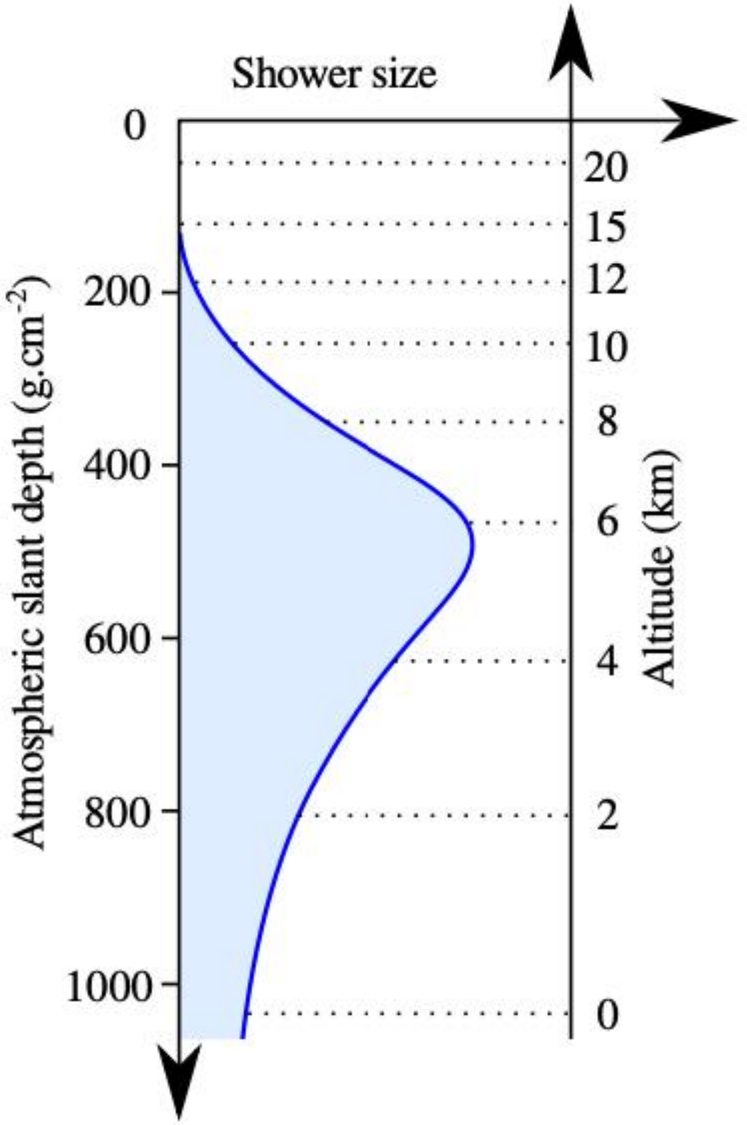
Detecting EASs



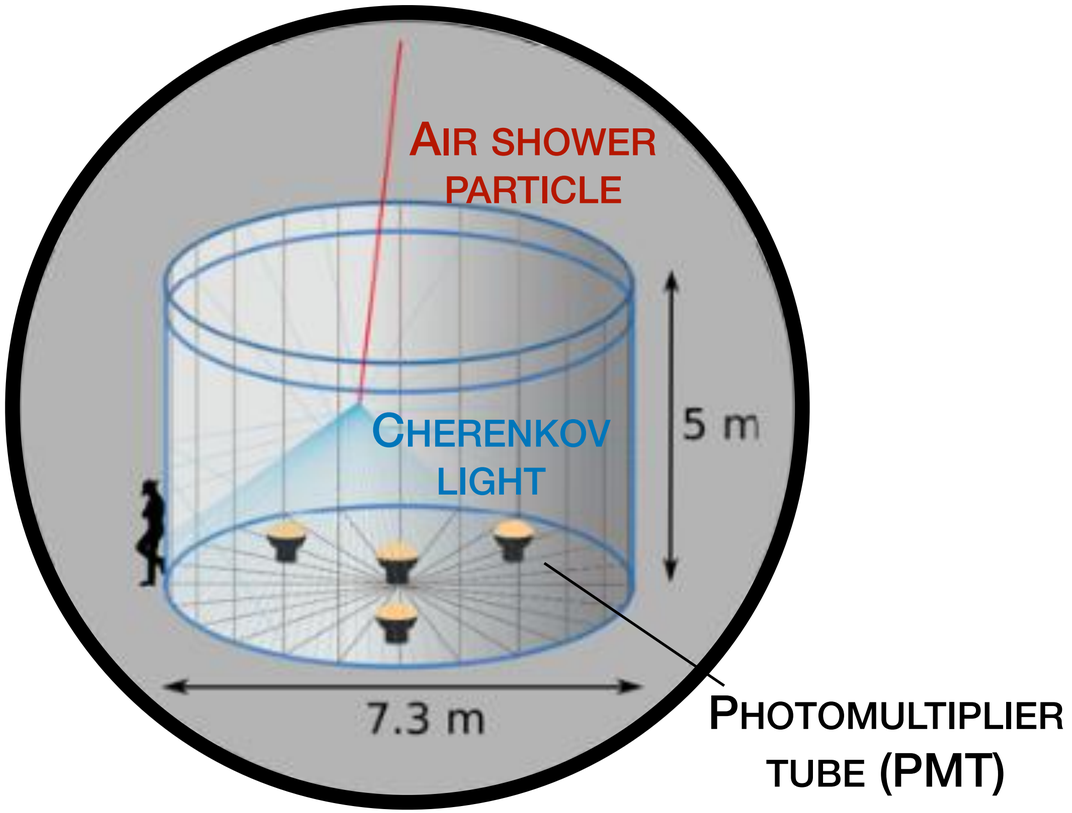


Detecting EASs





Water Cherenkov Detectors



Detecting method: showers particles reaching the ground ($\mu^{\pm}, e^{\pm}, \gamma, \nu$), in water, ground-based

Field of View (FOV): 50°

Duty cycle: 100% (operates in daylight too)

Angular resolution: $[0.1,1]^{\circ}$

Energy resolution: ~30%

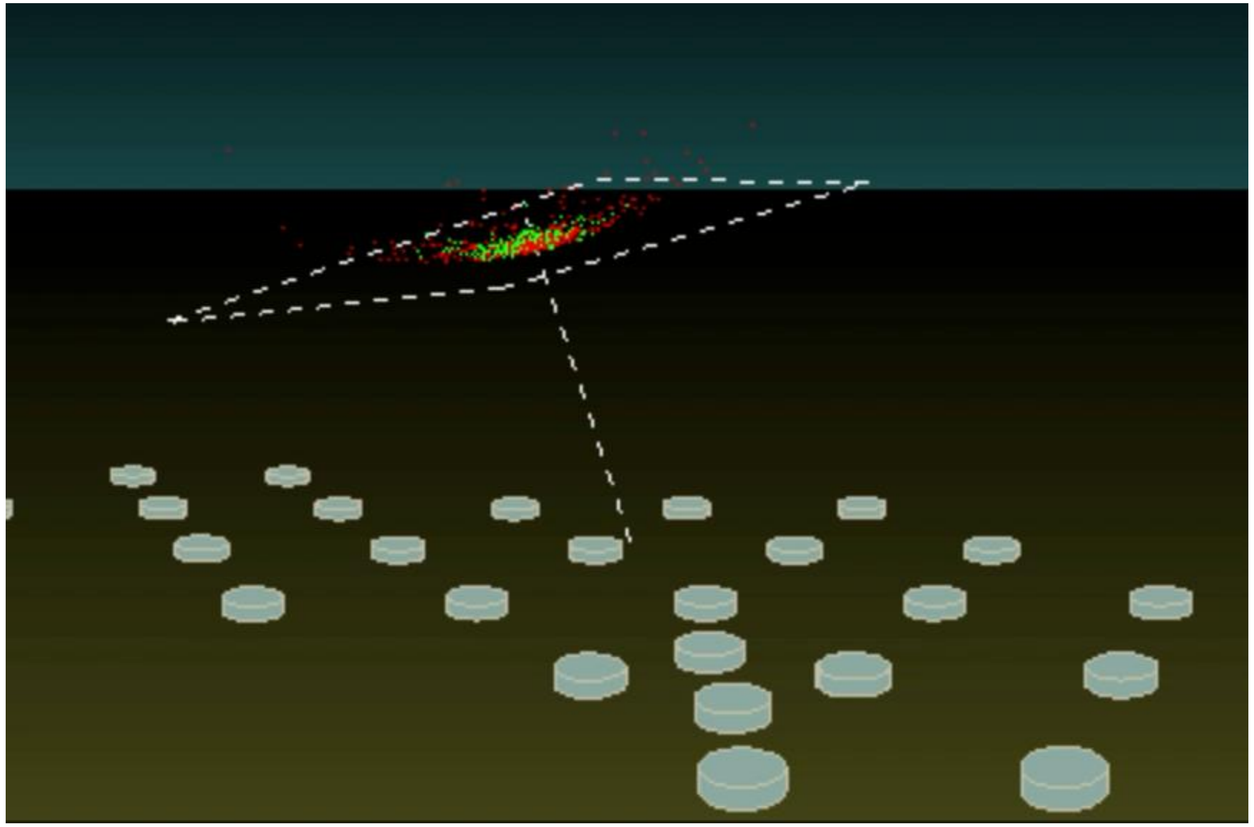
Effective aerea: ~10^5 m^2

Energy range: ultra-high-energy regime (UHE, E > 100 TeV)

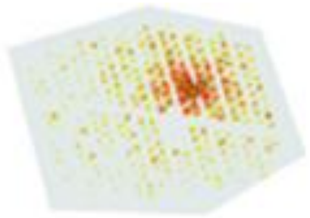
Water Cherenkov Detectors

Reconstruction

- **Timing** → Direction (determined by arrival time differences recorded at different detectors as the shower front sweeps through the detector plane)
- **Size** → Energy
- **Shape** → Nature

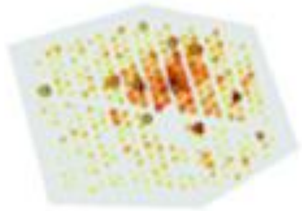


GAMMA-RAY SHOWER



“Hot” spots concentrate around the core

COSMIC-RAY SHOWER



“Hot” spots are more dispersed

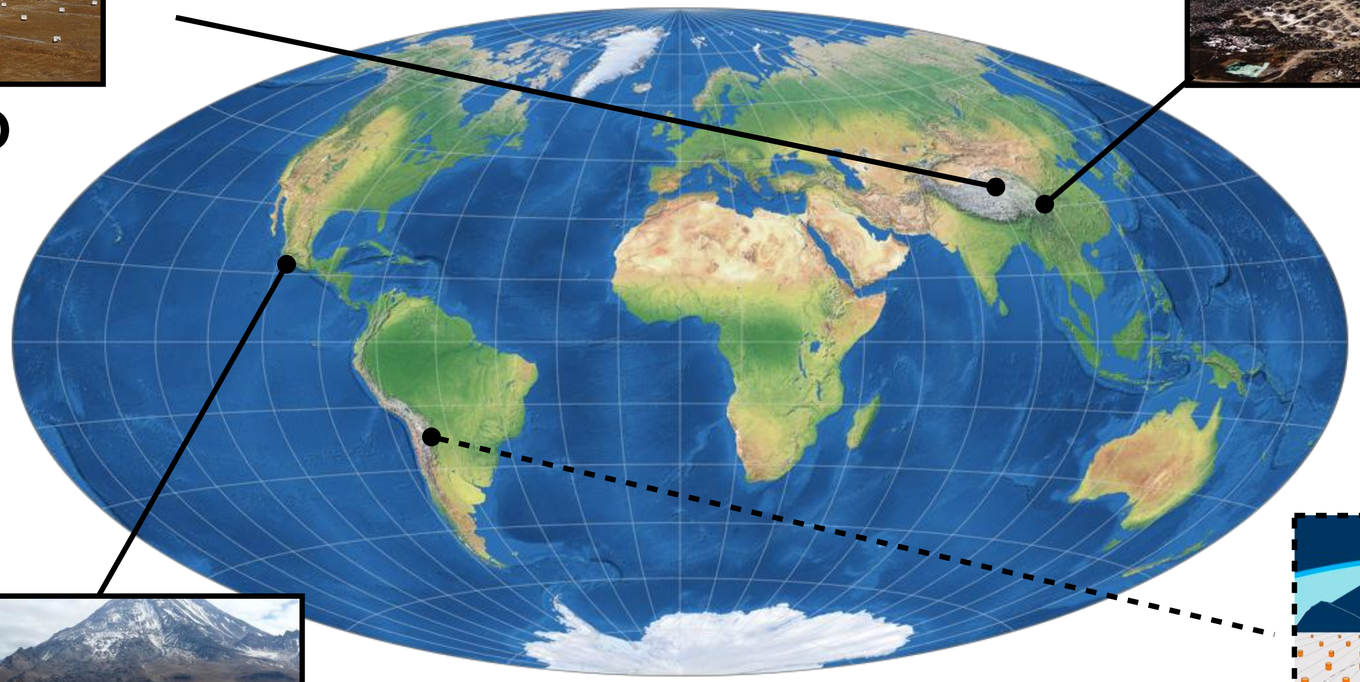
Current and future generation of WCDs



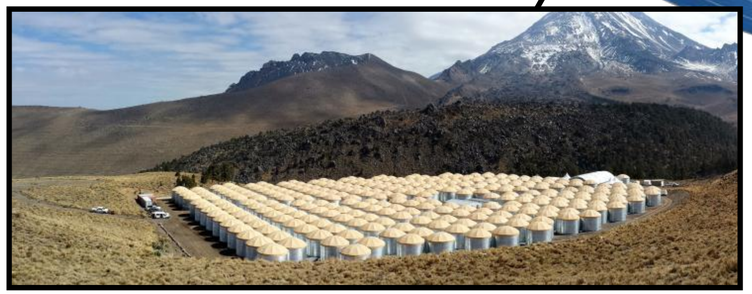
Tibet ASγ+ARGO (1990s)
Tibet (4300 m)



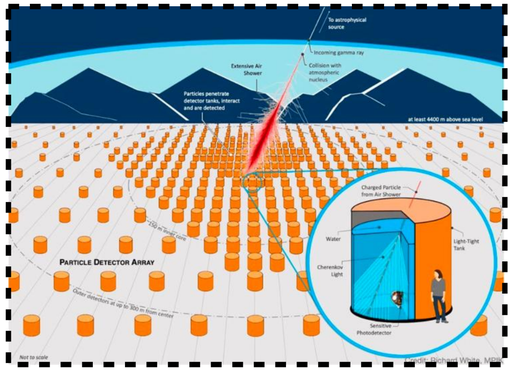
LHAASO (2021)
Daocheng, China
(4410 m)



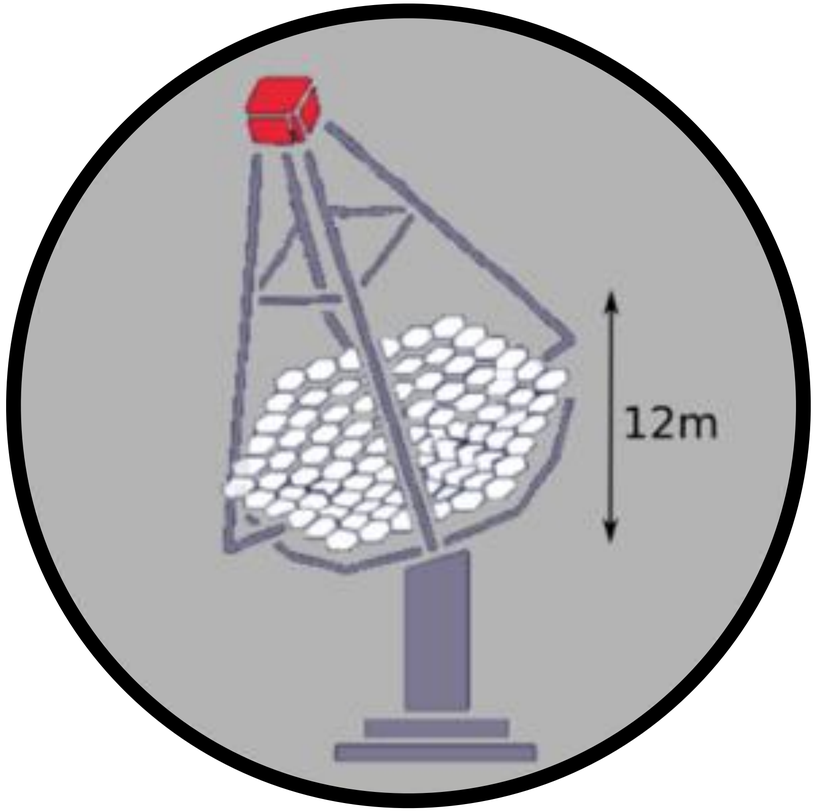
HAWC (2015)
Puebla, Mexico
(4100 m)



SWGO
Atacama, Chile
(4770 m)



Imaging Atmospheric Cherenkov Telescopes (IACTs)



Detecting method: Cherenkov radiation induced by EAS in atmosphere, ground-based, pointing

Field of View (FOV): 2-6°

Duty cycle: 10-15% (clear, moonless nights)

Angular resolution: $[0.04, 1]^\circ$

Energy resolution: ~15%

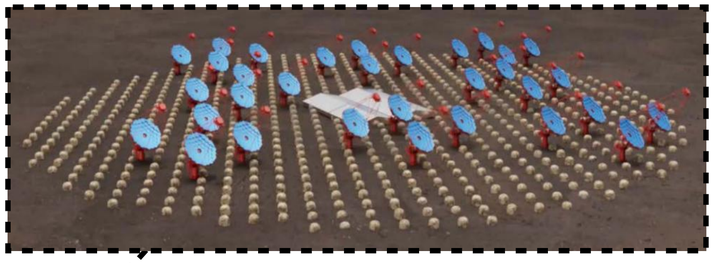
Effective aerea: ~10^5 m^2

Energy range: wide (50-100 TeV)

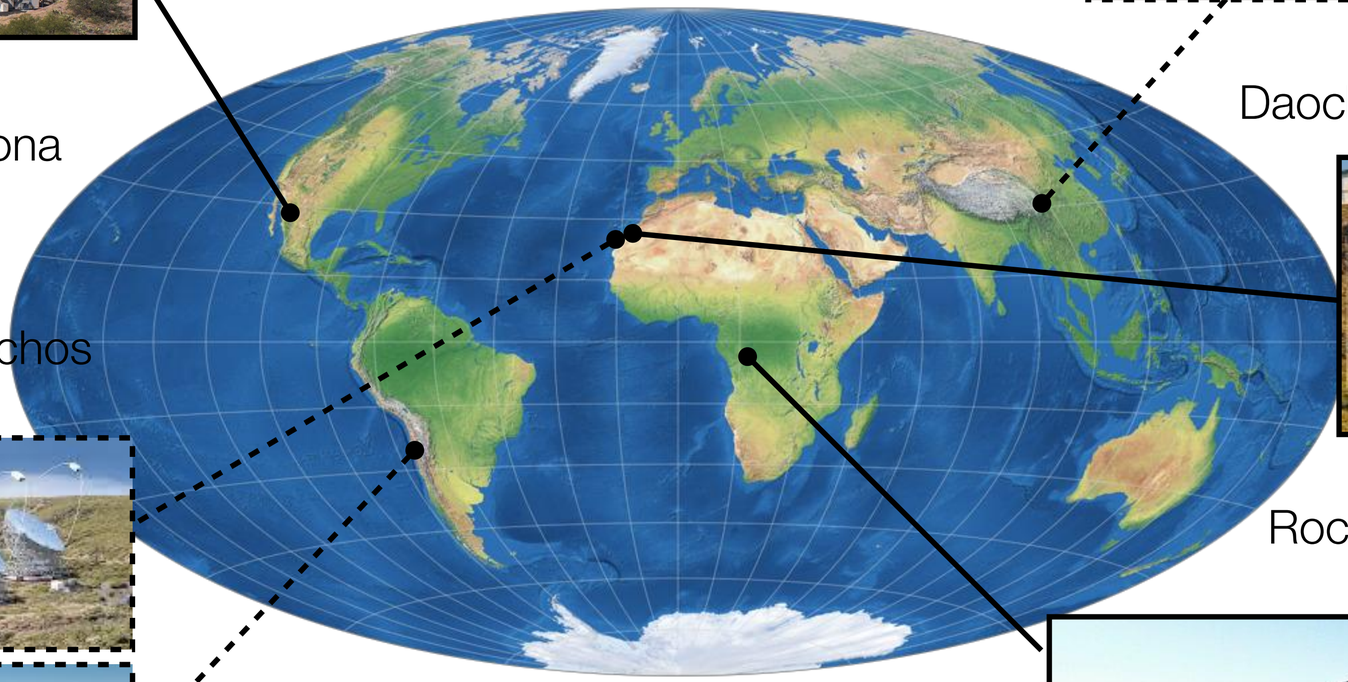
Current and future generation of IACTs



VERITAS
Mount Hopkins, Arizona



LACT
Daocheng, China (4410 m)

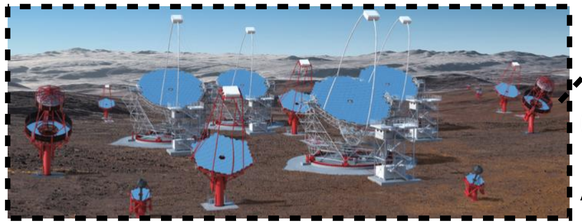




MAGIC
Roche de los Muchachos
Canary Island

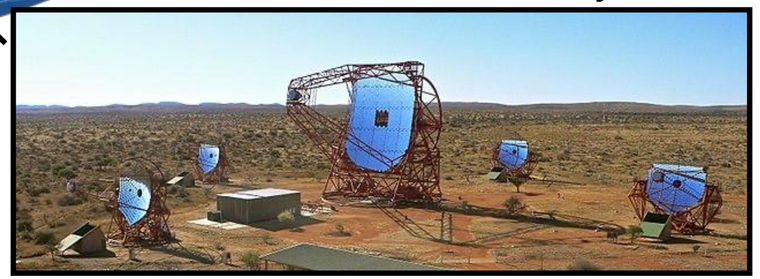
CTAO - North
Roche de los Muchachos
Canary Island

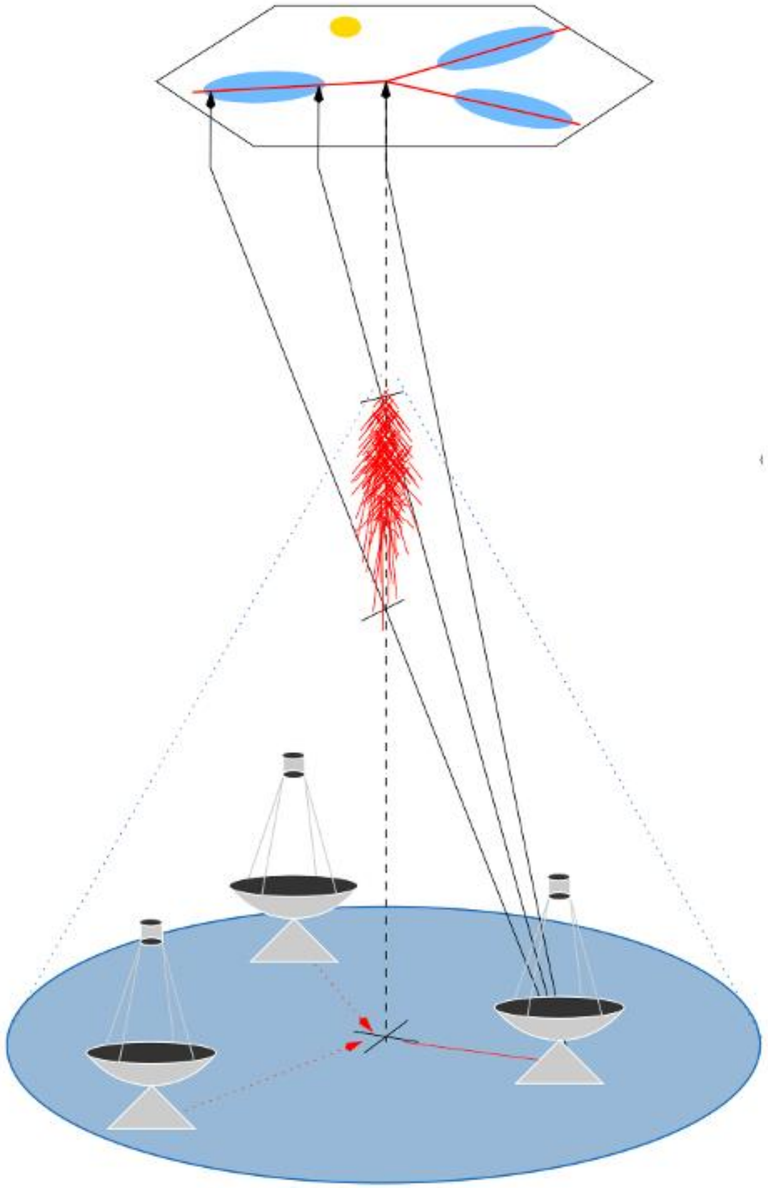




CTAO - South
Atacama, Chile

H.E.S.S. (2002)
Khomas Highland,
Namibia





IACTs: detection principles

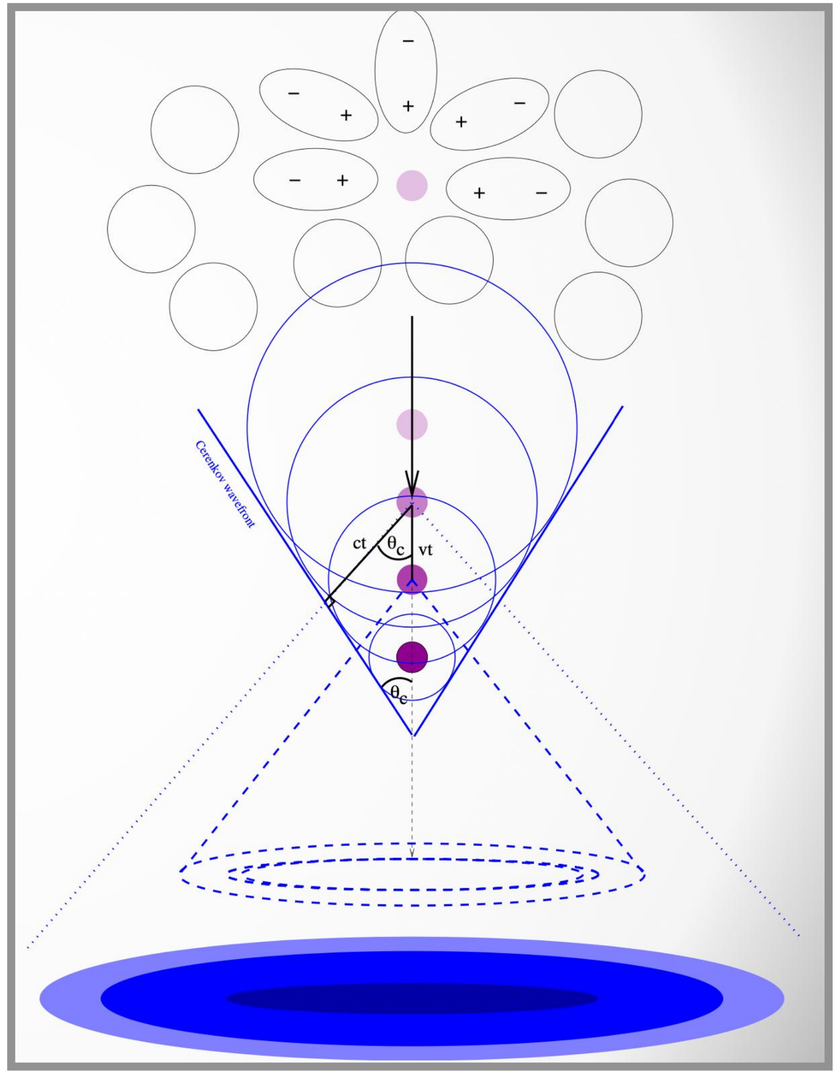
Cherenkov radiation

- Polarization of dielectric medium by charged particles
- Constructive interference when particle is faster than the emitted radiation (c/n)
- Emission in a cone with respect to the particle direction

Cherenkov condition: $n\beta > 1$

Light is emitted along a cone with half opening angle θ:

$$\cos\theta = \frac{1}{n\beta}$$



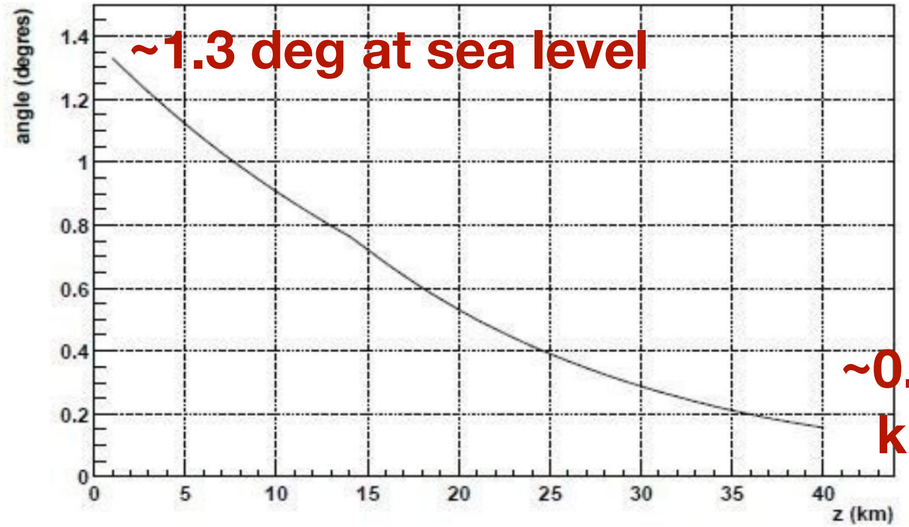
Cherenkov radiation

In the atmosphere

- Refractive index: $n - 1 \propto \rho(z)$
- Atmospheric density profile:

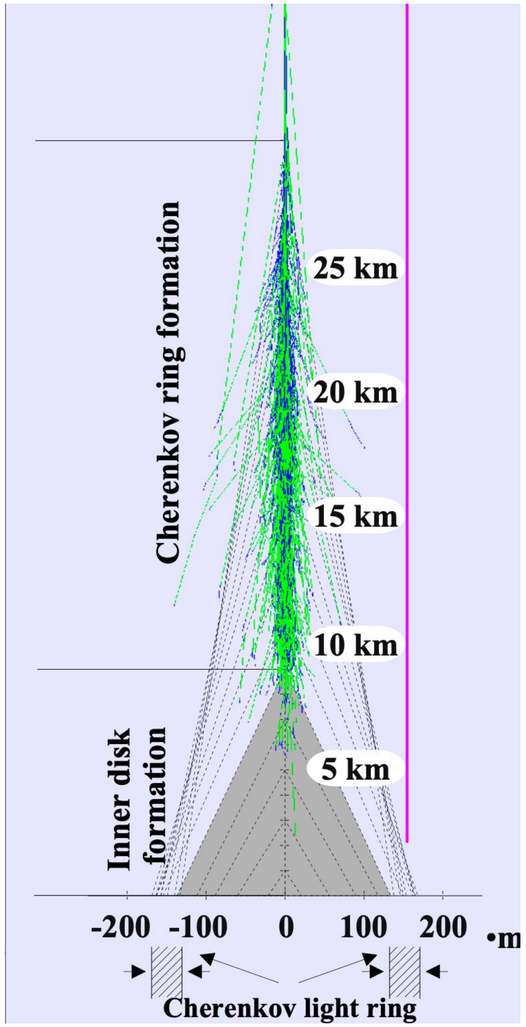
$$\rho(z) \propto \rho_0 \exp\left(-\frac{z}{z_0}\right)$$

- Evolution of the Cherenkov angle with altitude ***z***:



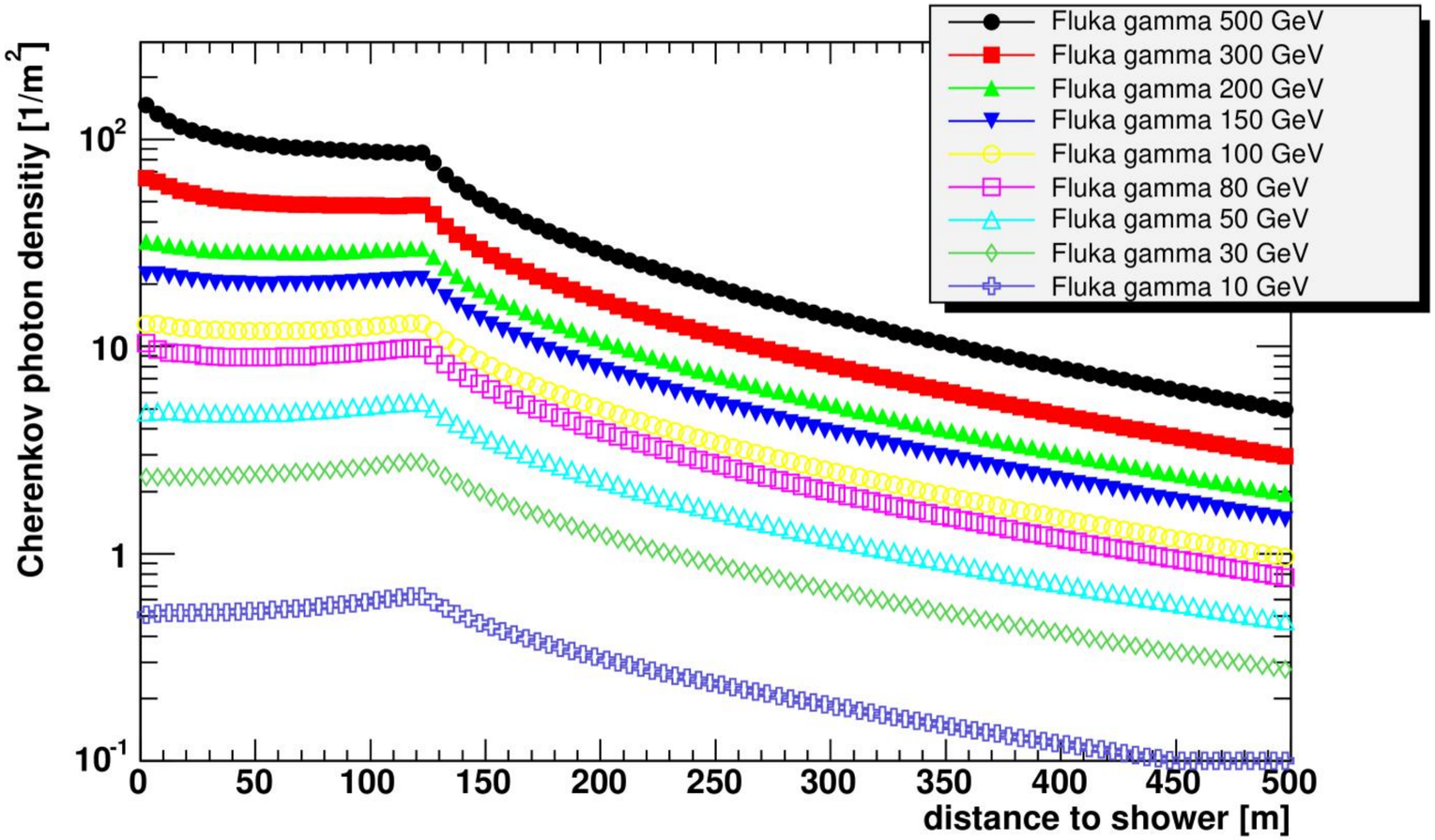
~0.2 deg at 40 km altitude

Credit: M. De Naurois



Cherenkov radiation

Cherenkov photon density on ground

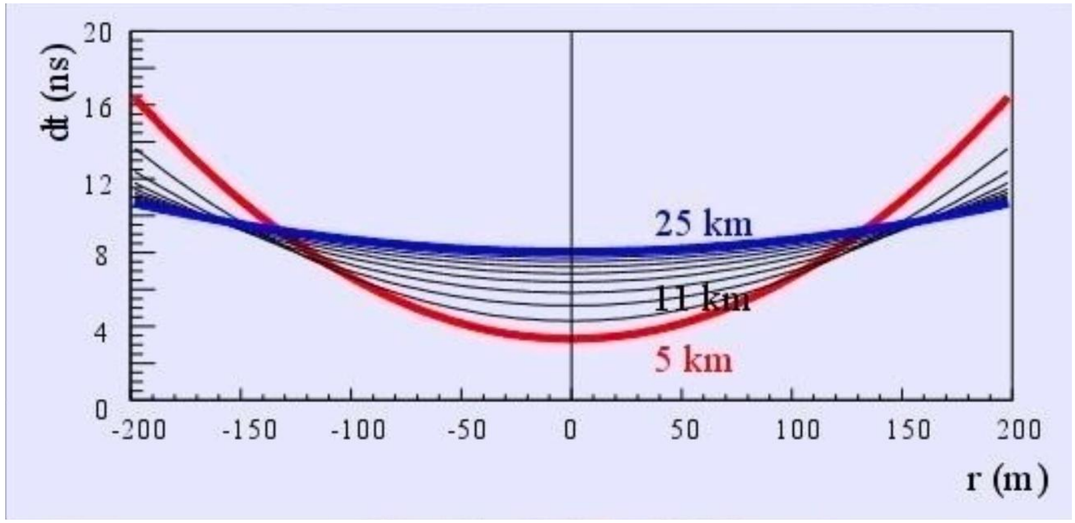


Cherenkov radiation

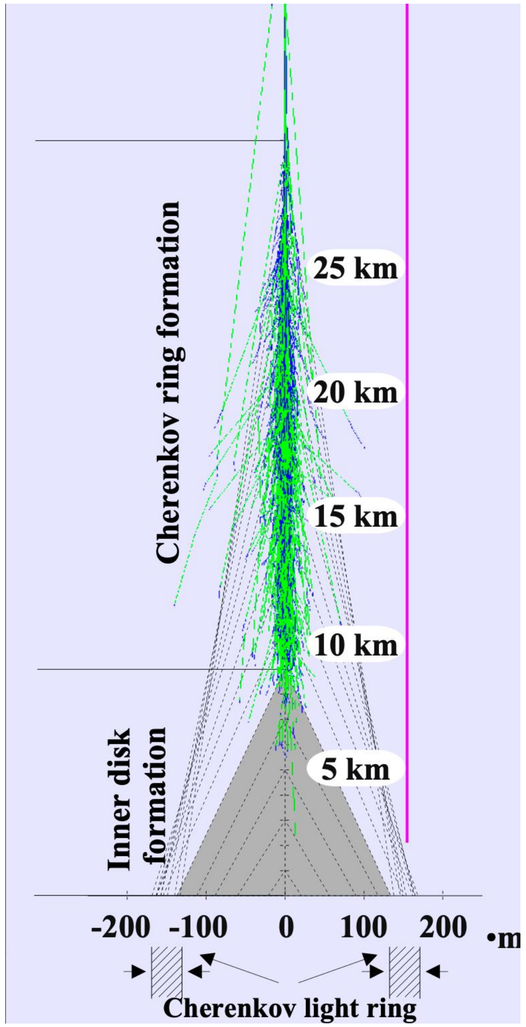
Temporal aspect

Competition between geometrical path and particle speed:

- Close to shower axis, light emitted at the bottom is first
- Far away, light emitted at high altitude arrives first
- Transition ~ 125 m (depends on altitude) , short duration flash (~ 2 ns)



Credit: M. De Naurois



Cherenkov radiation at IACTs

Orders of magnitude for $E_\gamma \sim 1$ TeV

Shower maximum:

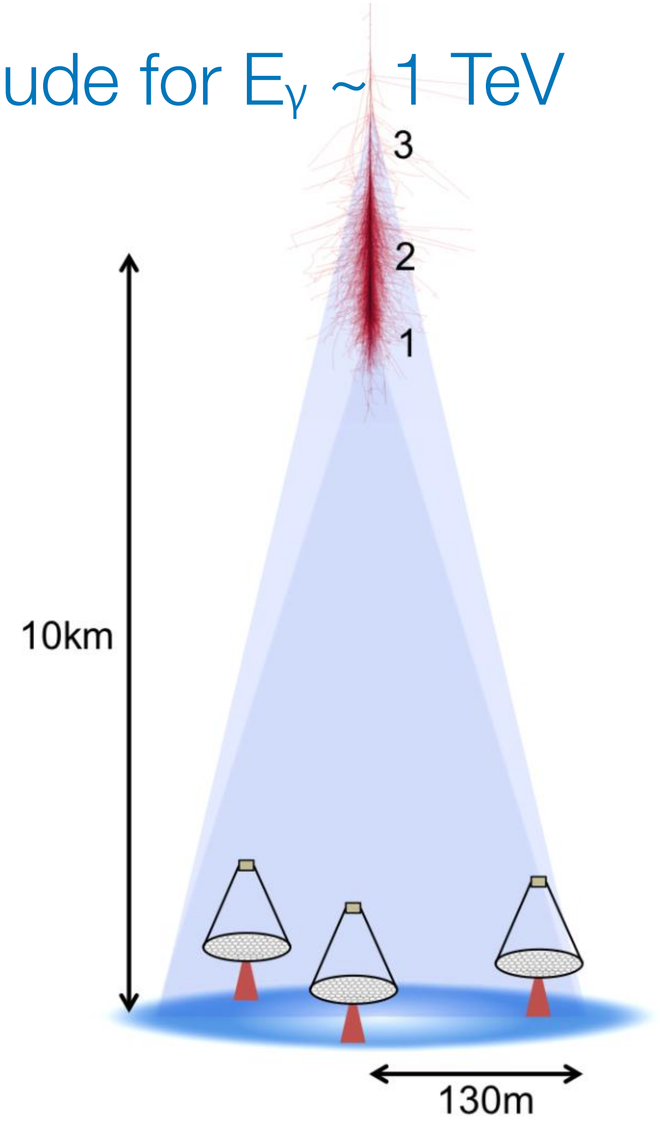
- Altitude ~10 km
- ~10^3 charged particles

Cherenkov emission:

- Altitude $\theta_{Cherenkov}$ ~1°
- Duration < 10 ns
- UV light

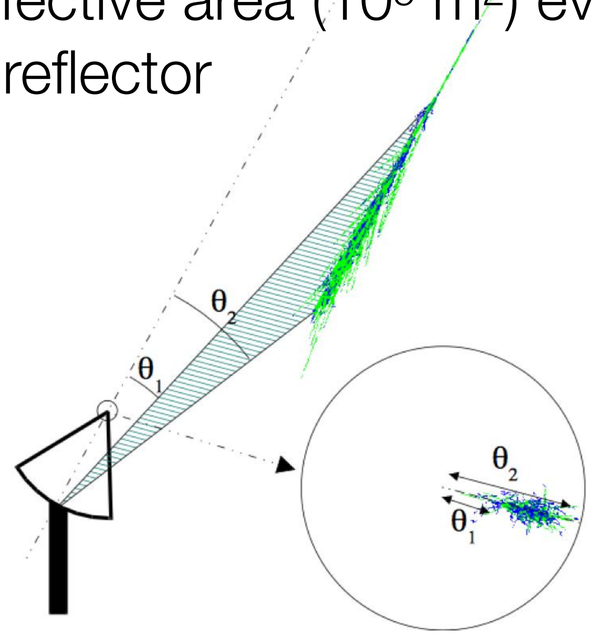
Cherenkov light-pool on ground:

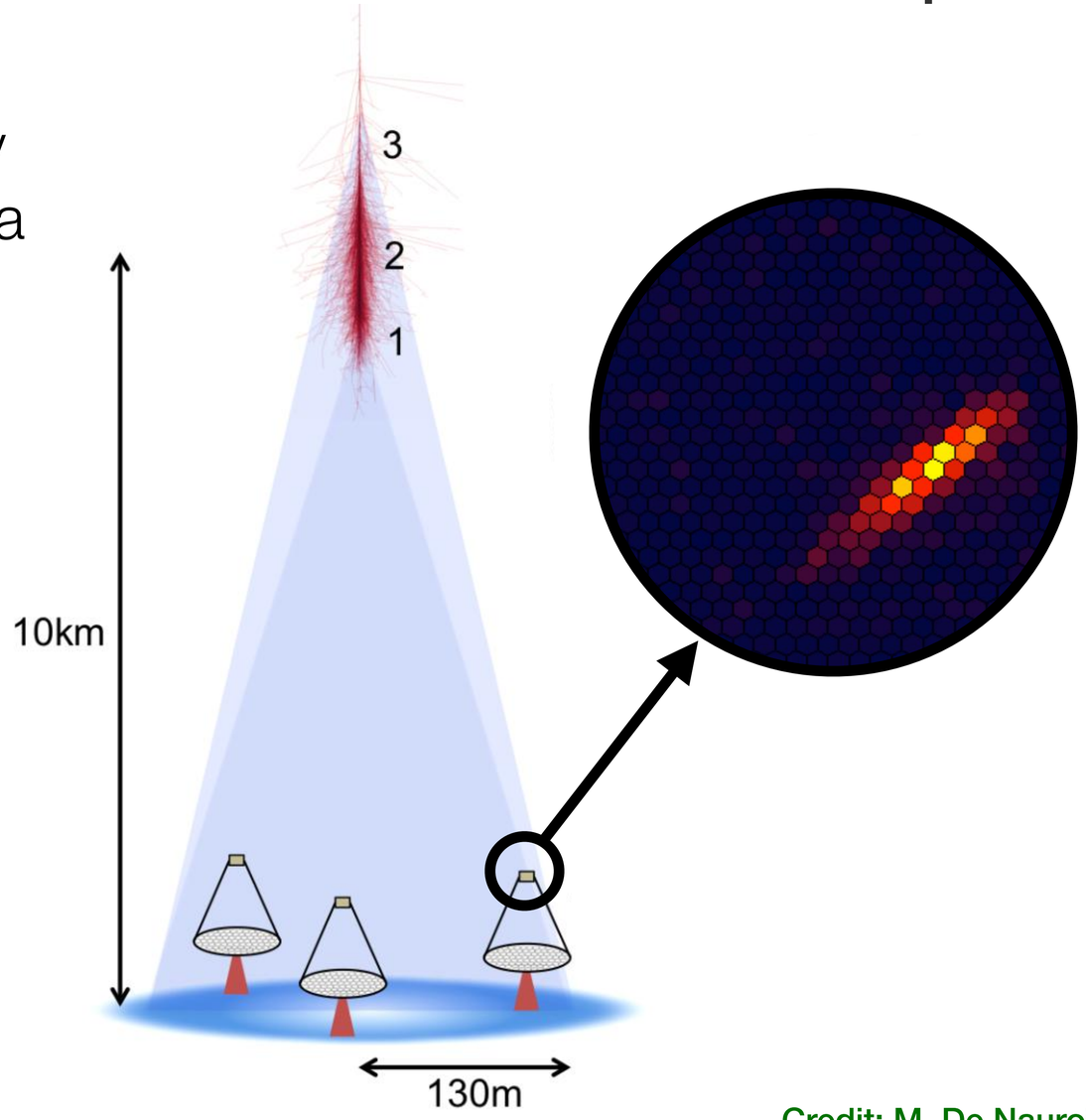
- R ~100 m
- ~100 $\gamma_{Cherenkov}/m^2$



Imaging Atmospheric Cherenkov Telescopes

- The telescope mirror focuses Cherenkov light from the air shower onto the camera
  → elliptical image
- Cherenkov light-pool ~120 m
- Large effective area (10^5 m^2) even with modest reflector

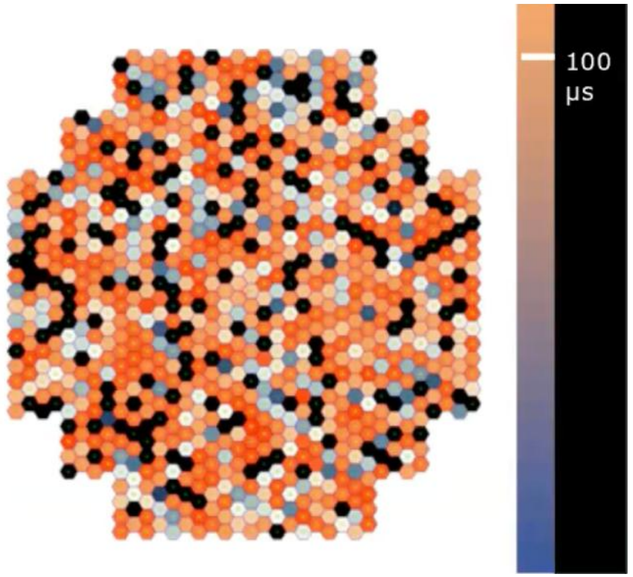


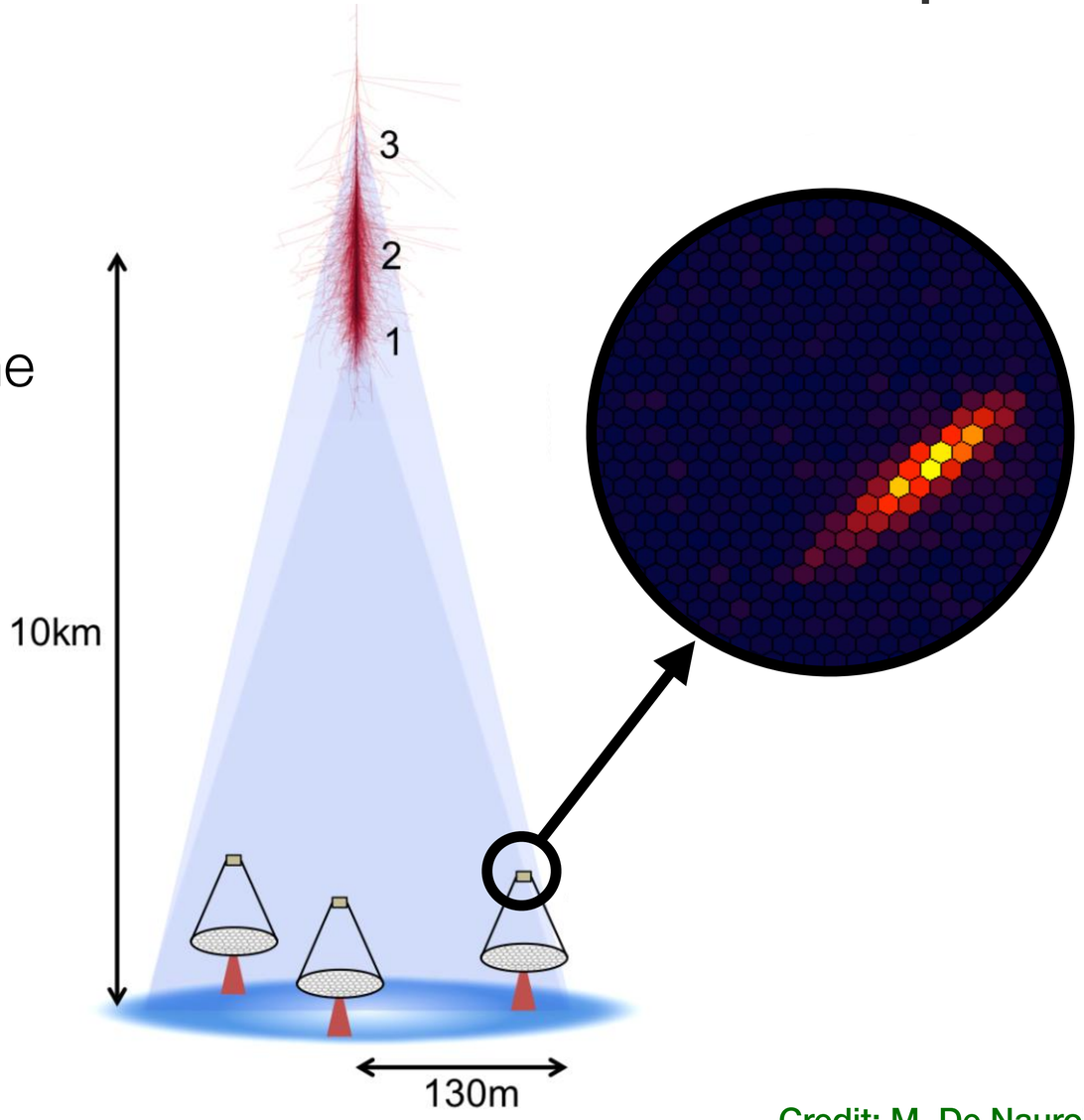


Credit: M. De Naurois

Imaging Atmospheric Cherenkov Telescopes

- Image the shower on a fast camera (Δt ~2ns)
- Fast timing is essential: a short integration window (< 10 ns) captures the Cherenkov flash while minimizing the night-sky background

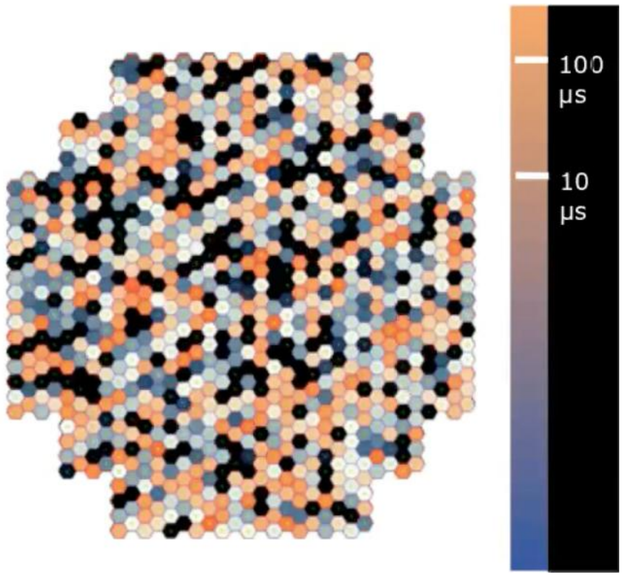


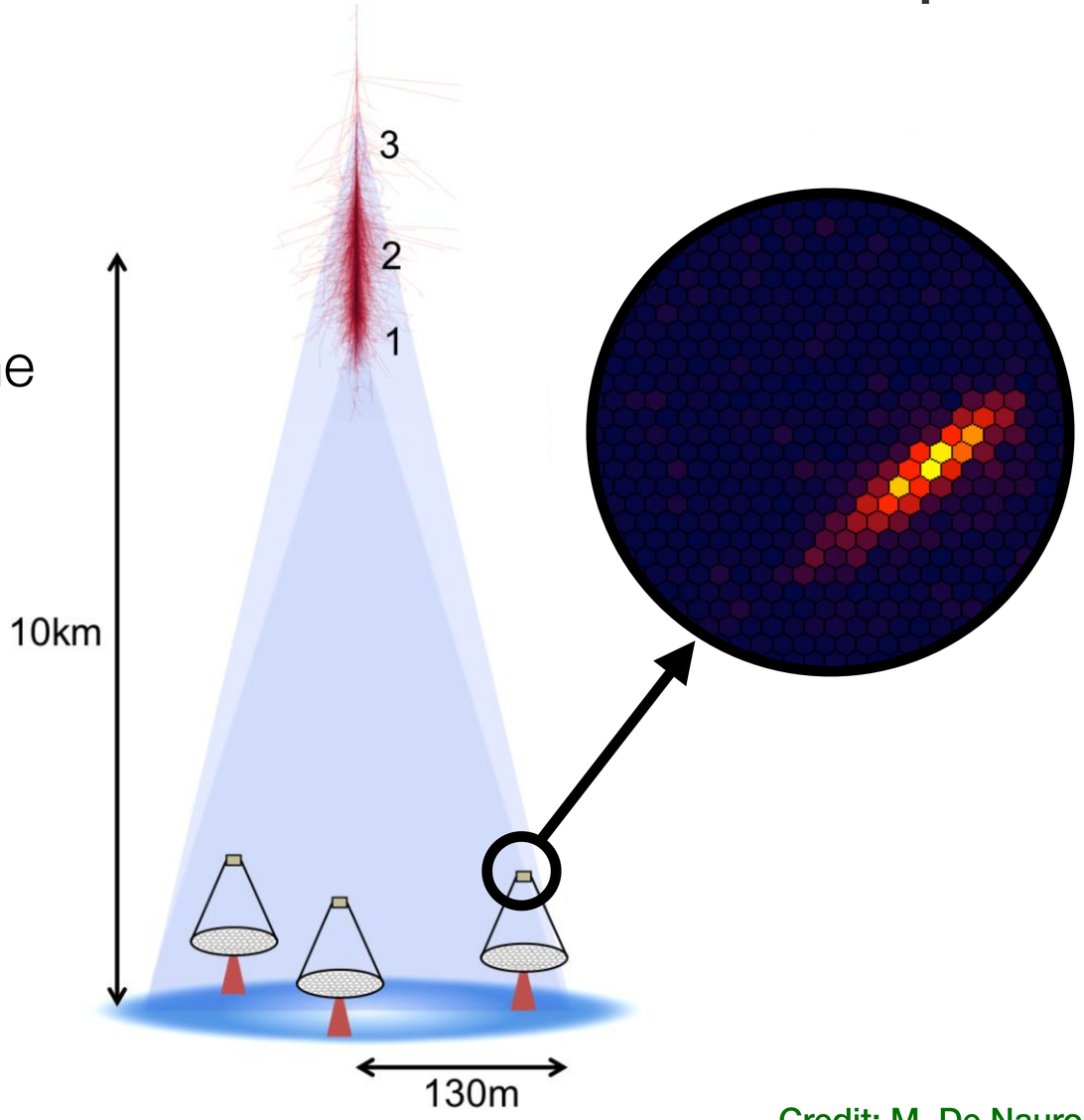


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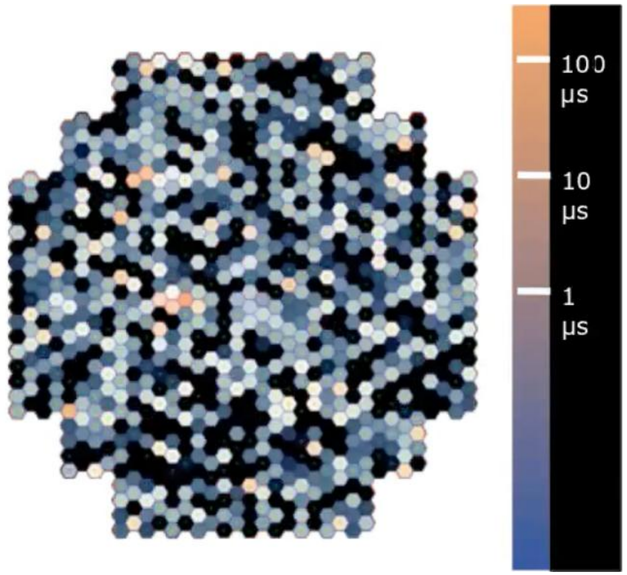


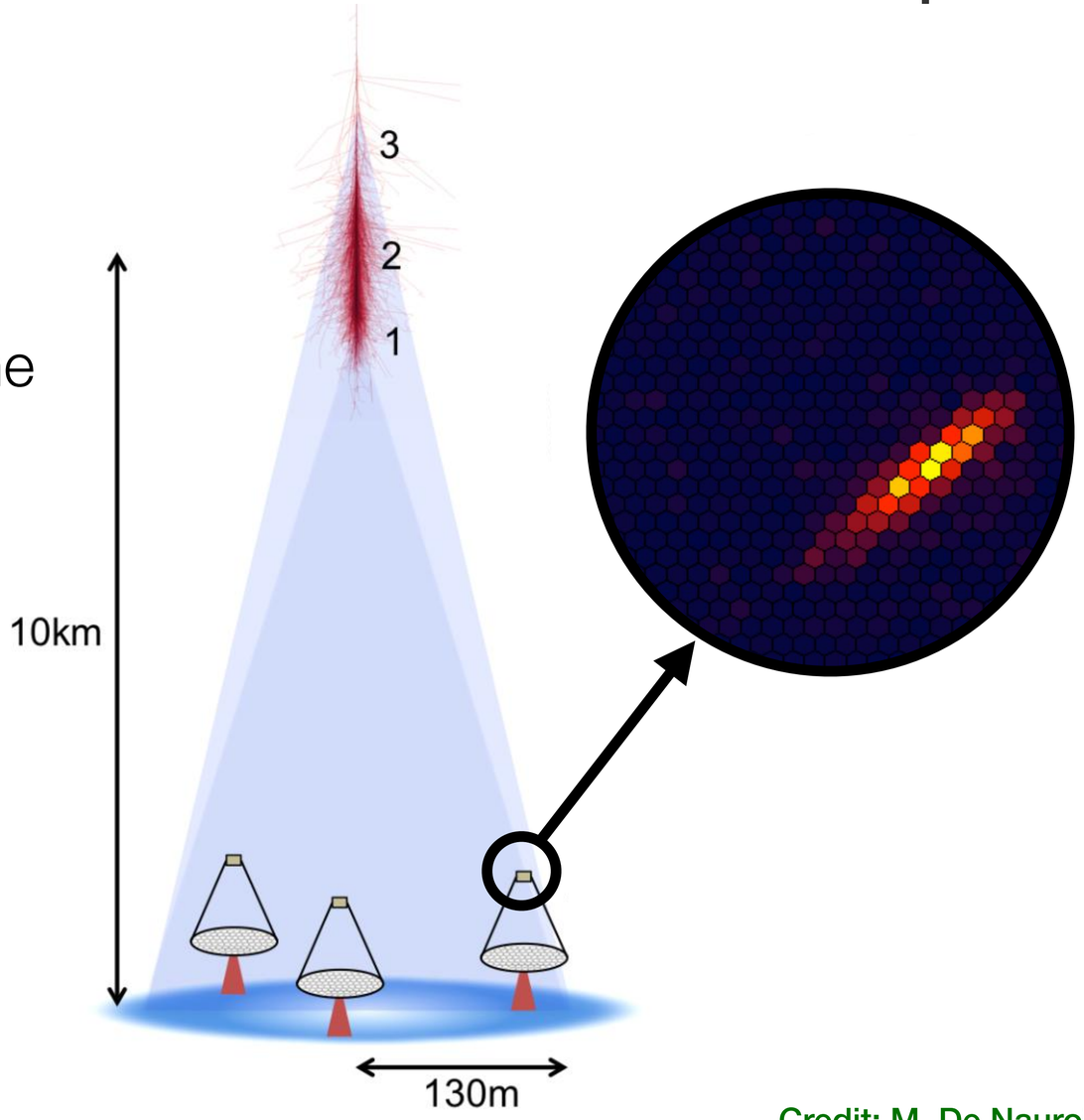


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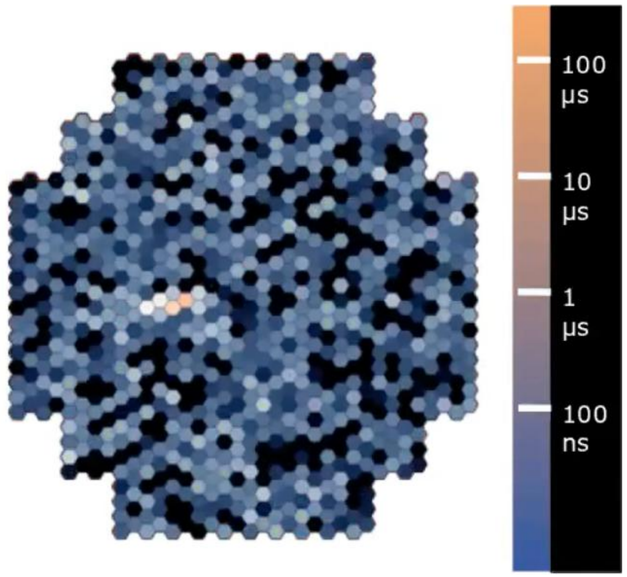


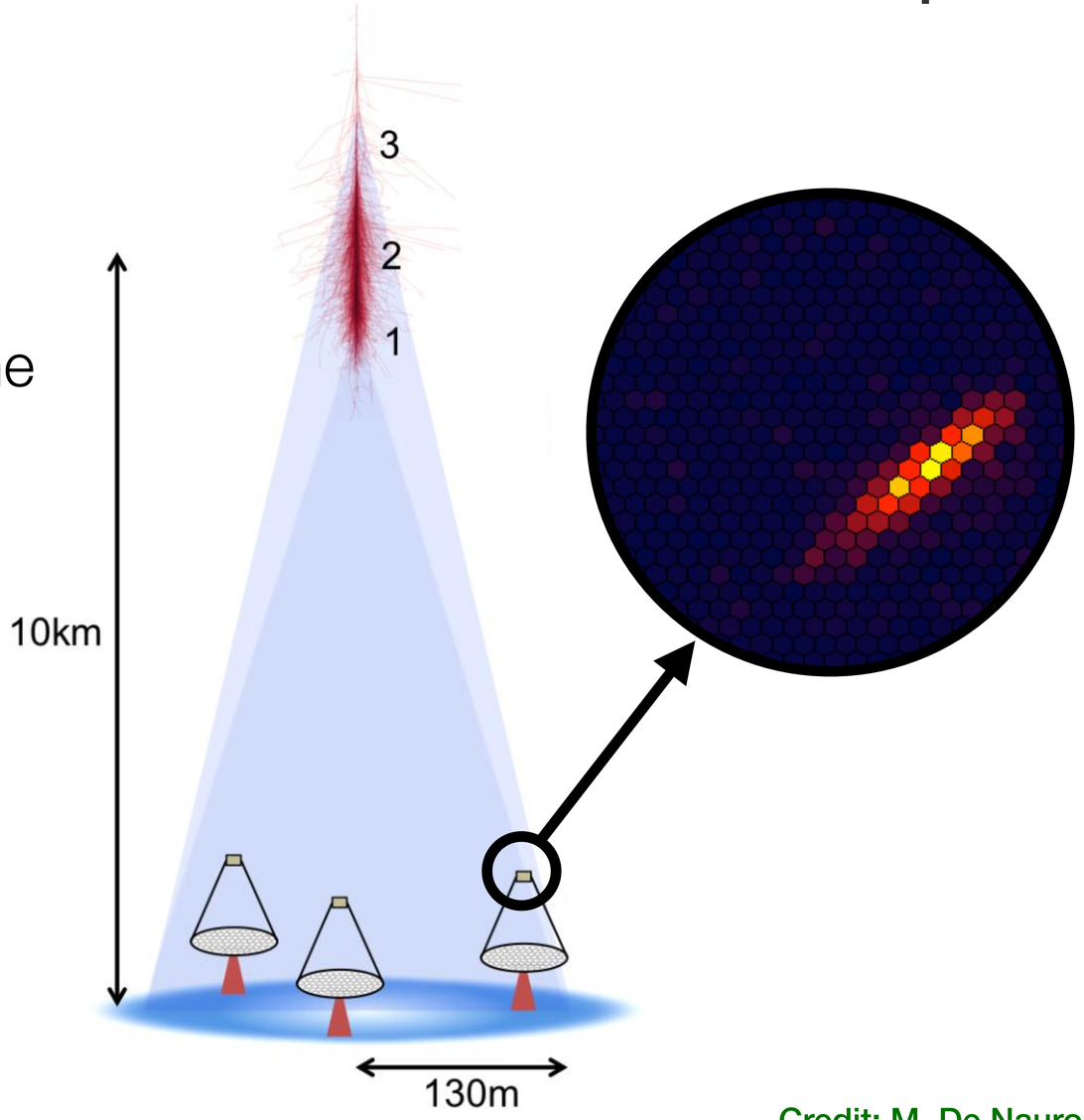


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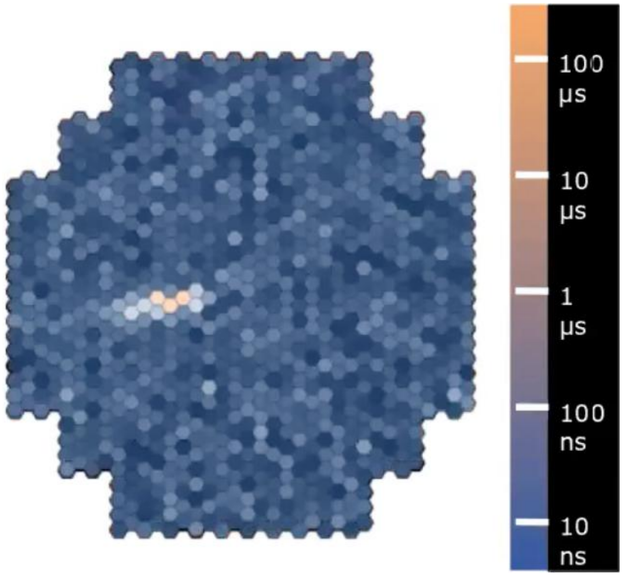


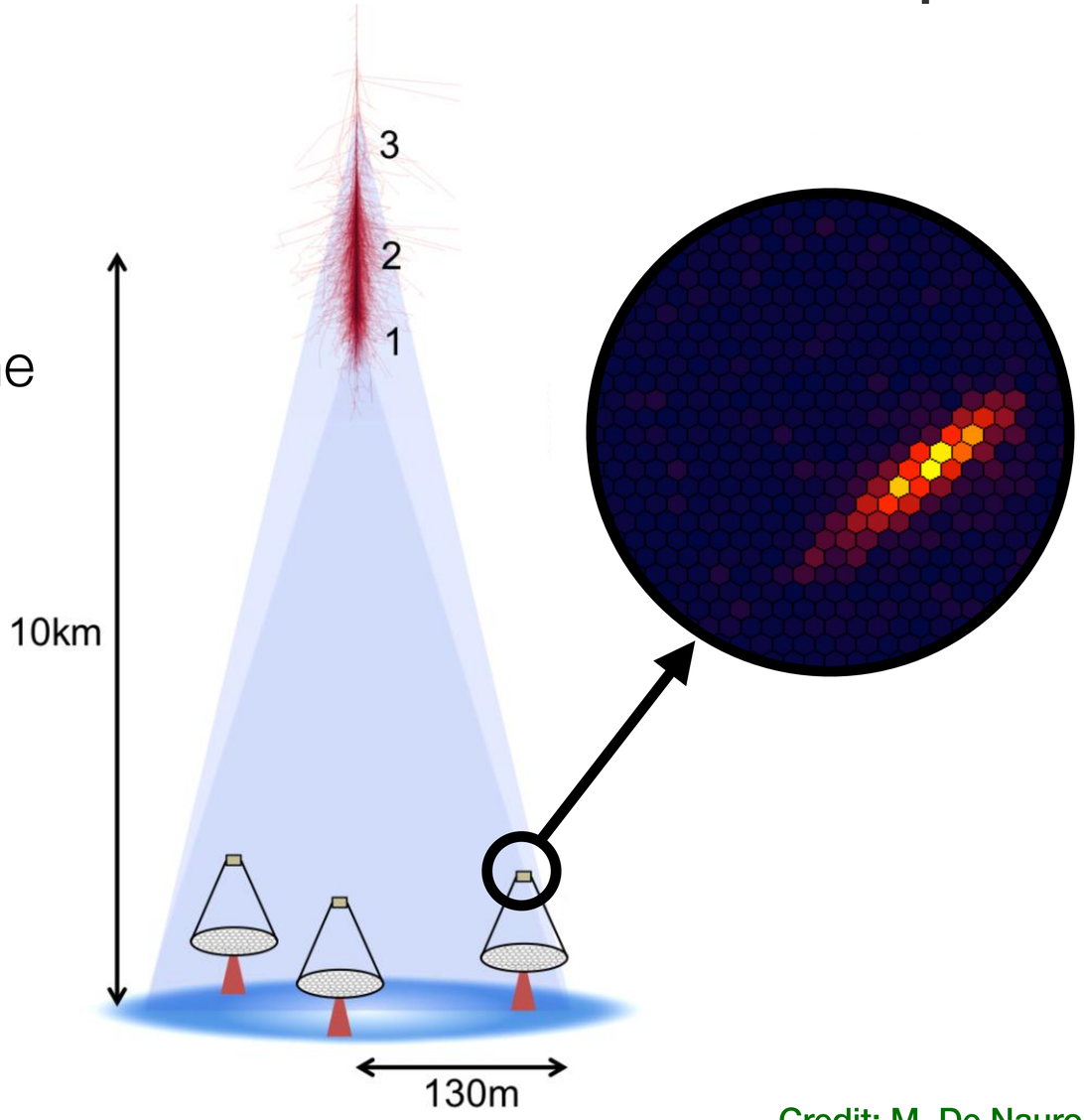


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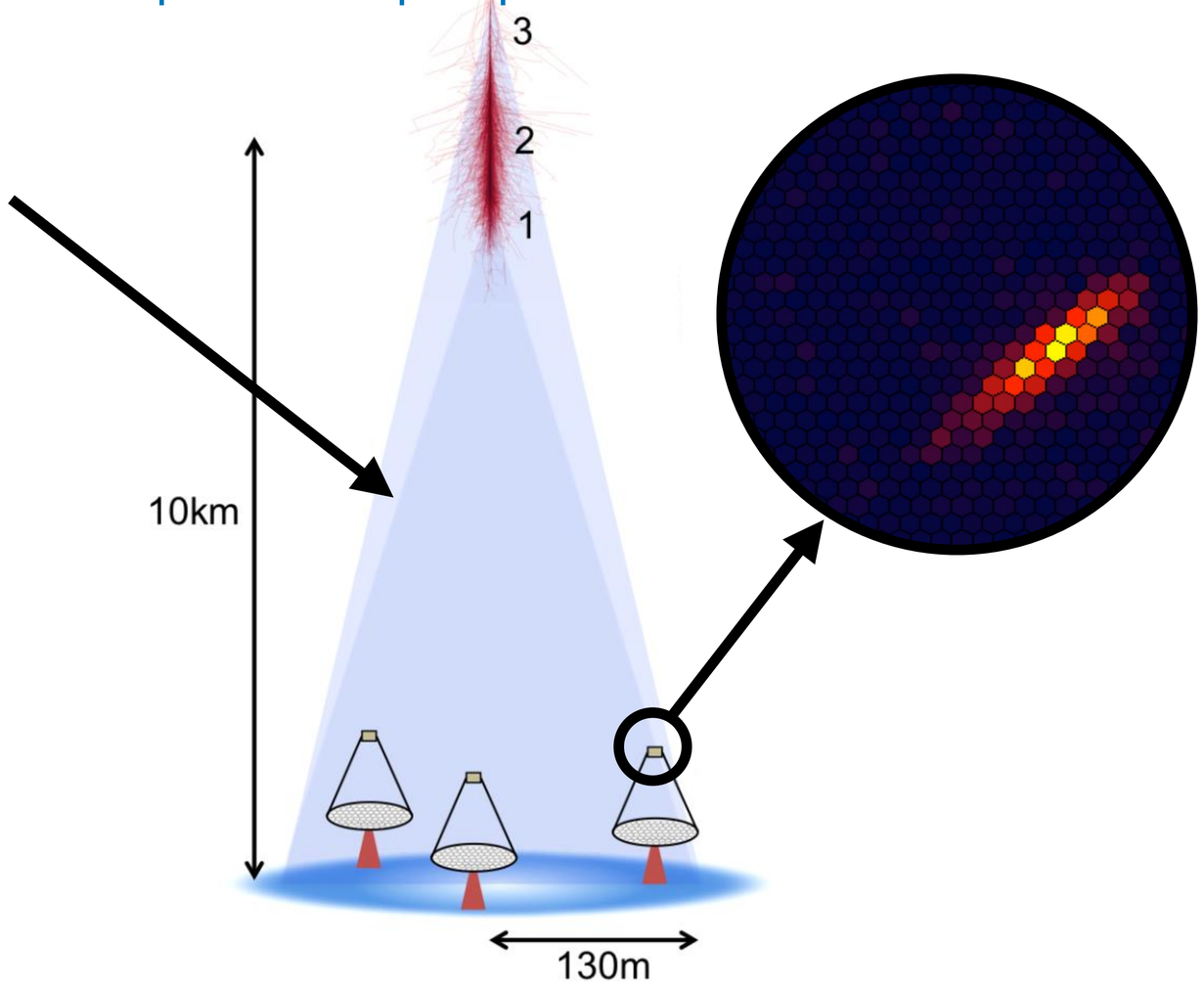


Credit: M. De Naurois

Imaging Atmospheric Cherenkov Telescopes

Reconstruction of particle properties

Energy: amount of Cherenkov light (calorimeter)

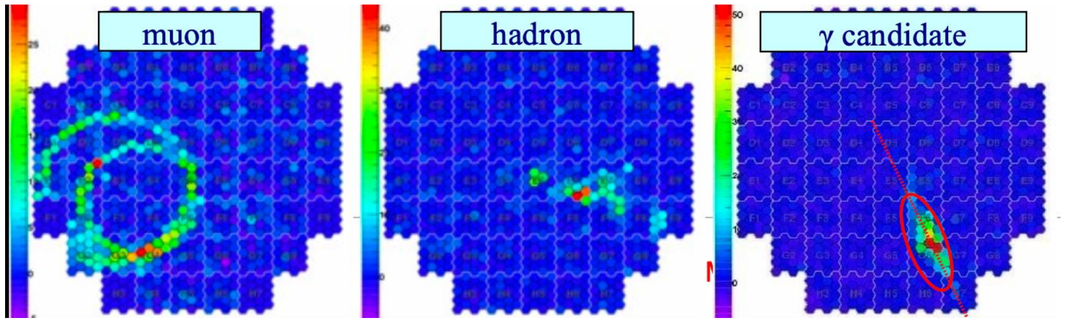


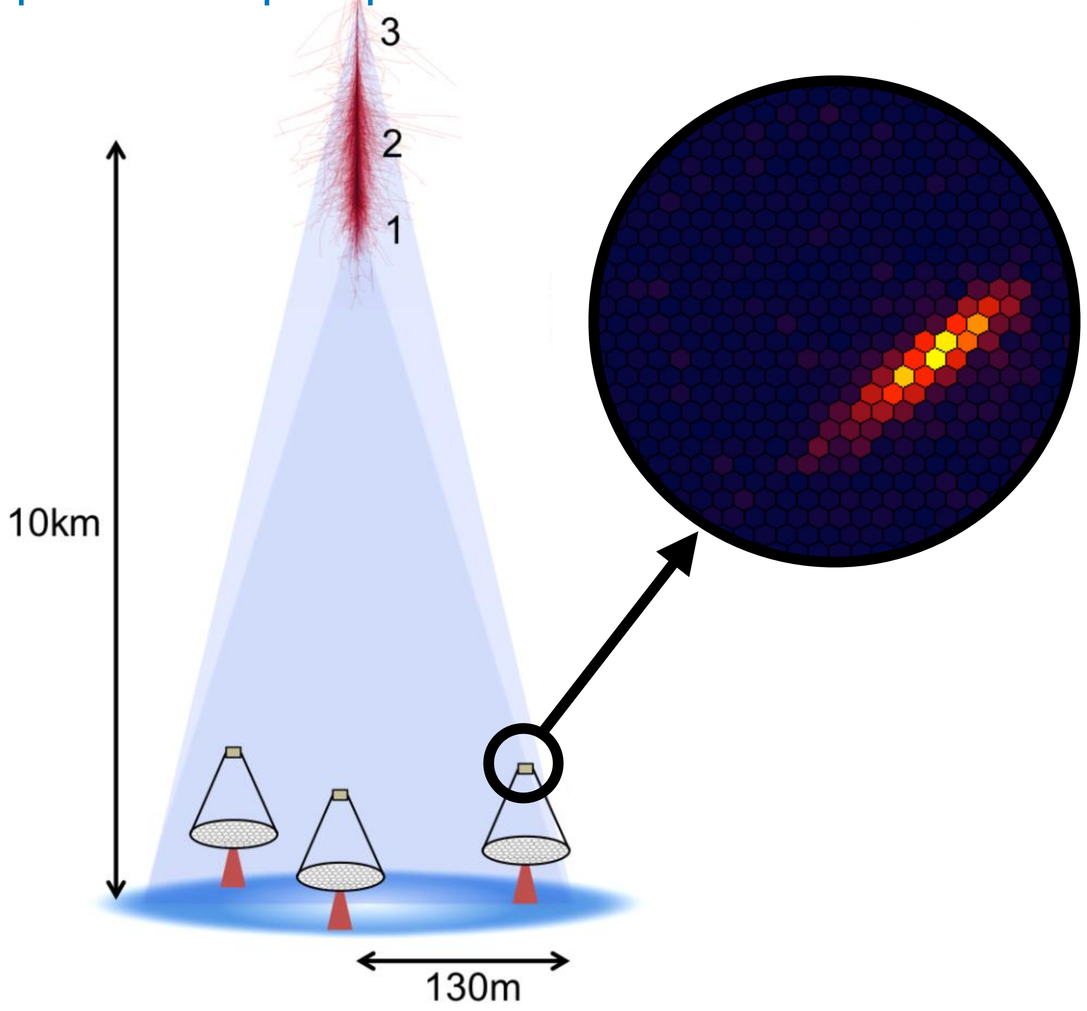
Imaging Atmospheric Cherenkov Telescopes

Reconstruction of particle properties

Energy: amount of Cherenkov light (calorimeter)

Nature: shape of the shower





Imaging Atmospheric Cherenkov Telescopes

Reconstruction of particle properties

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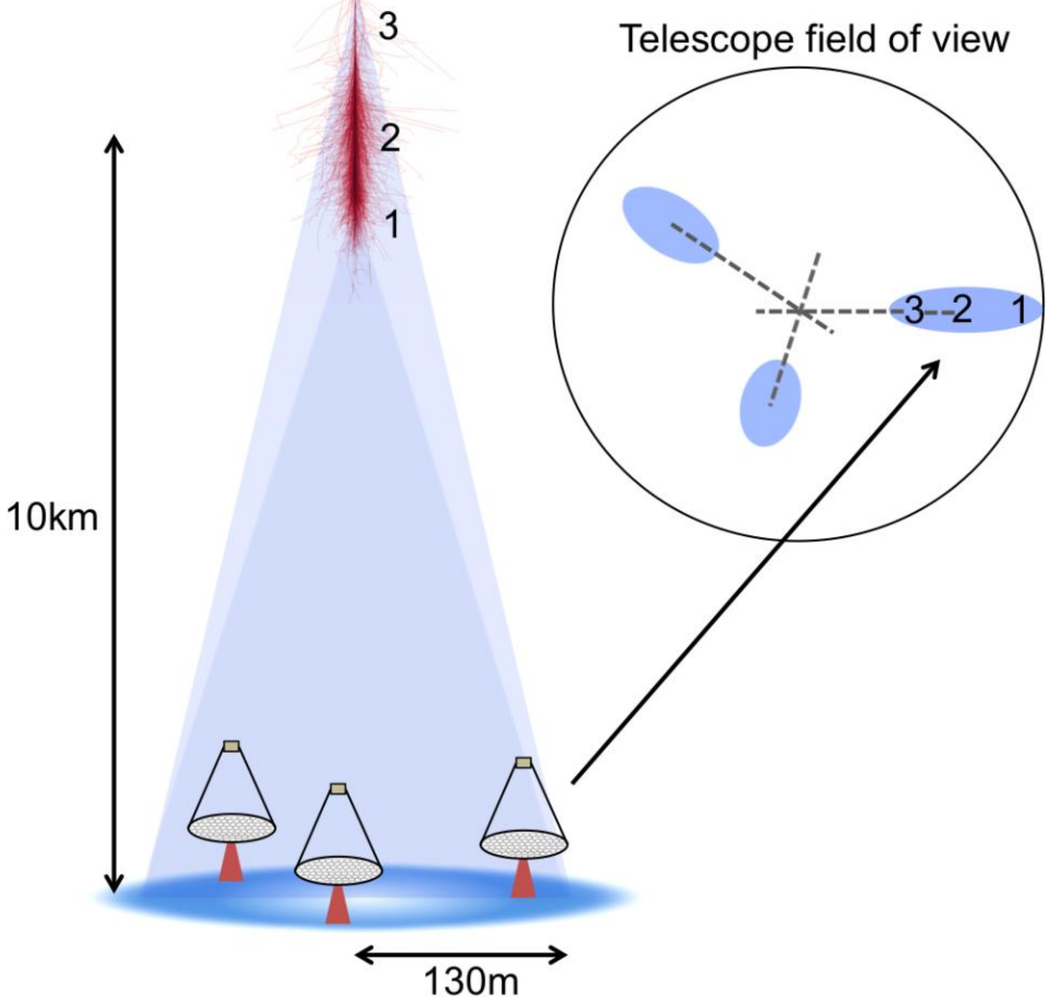
Nature: shape of the shower

Direction: stereoscopic observation

↓

Directional reconstruction by geometrical intersection of image main axes

- Greatly reduces the background at the trigger level (requiring 2-tel coincidence)
- Significantly improves shower reconstruction, PSF, etc



Imaging Atmospheric Cherenkov Telescopes

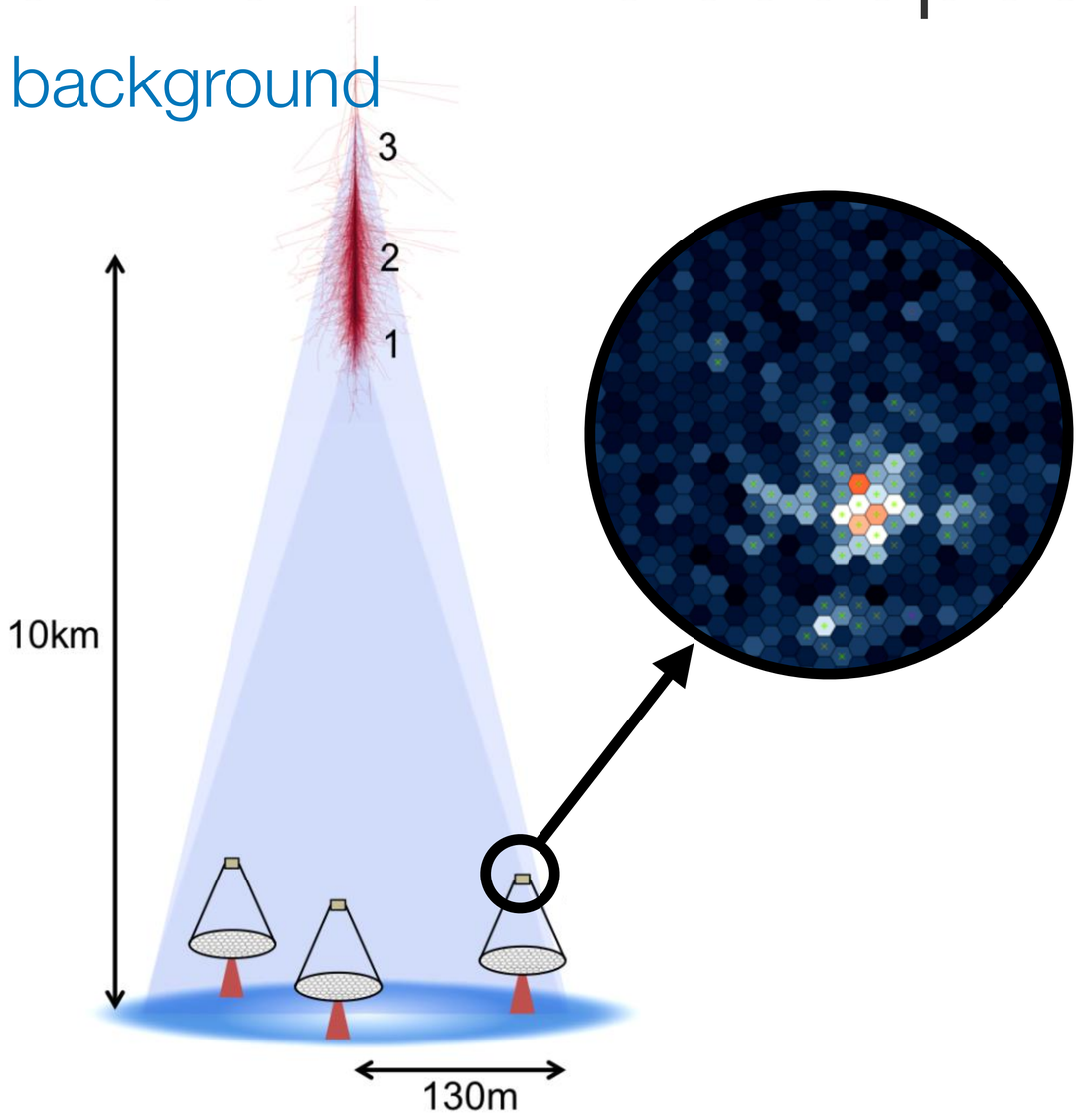
Cosmic-ray background

$10^4 - 10^5$ **more cosmic rays than gamma rays**

Stereo trigger: $\approx 1/4$

Image analysis/shape: $\approx 1/250$

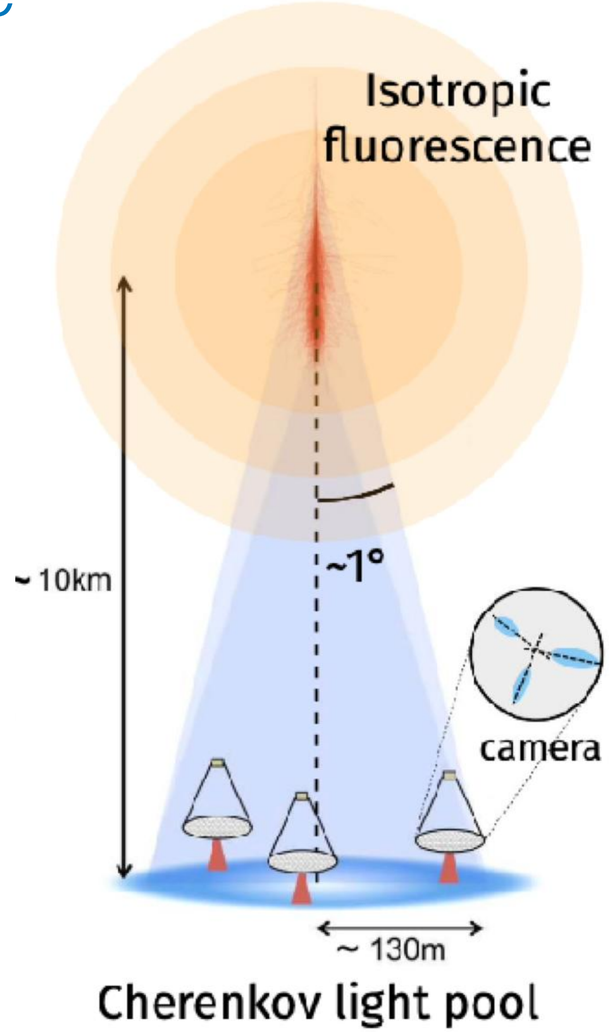
Arrival direction: $\approx 1/100$



Backgrounds

Fluorescence

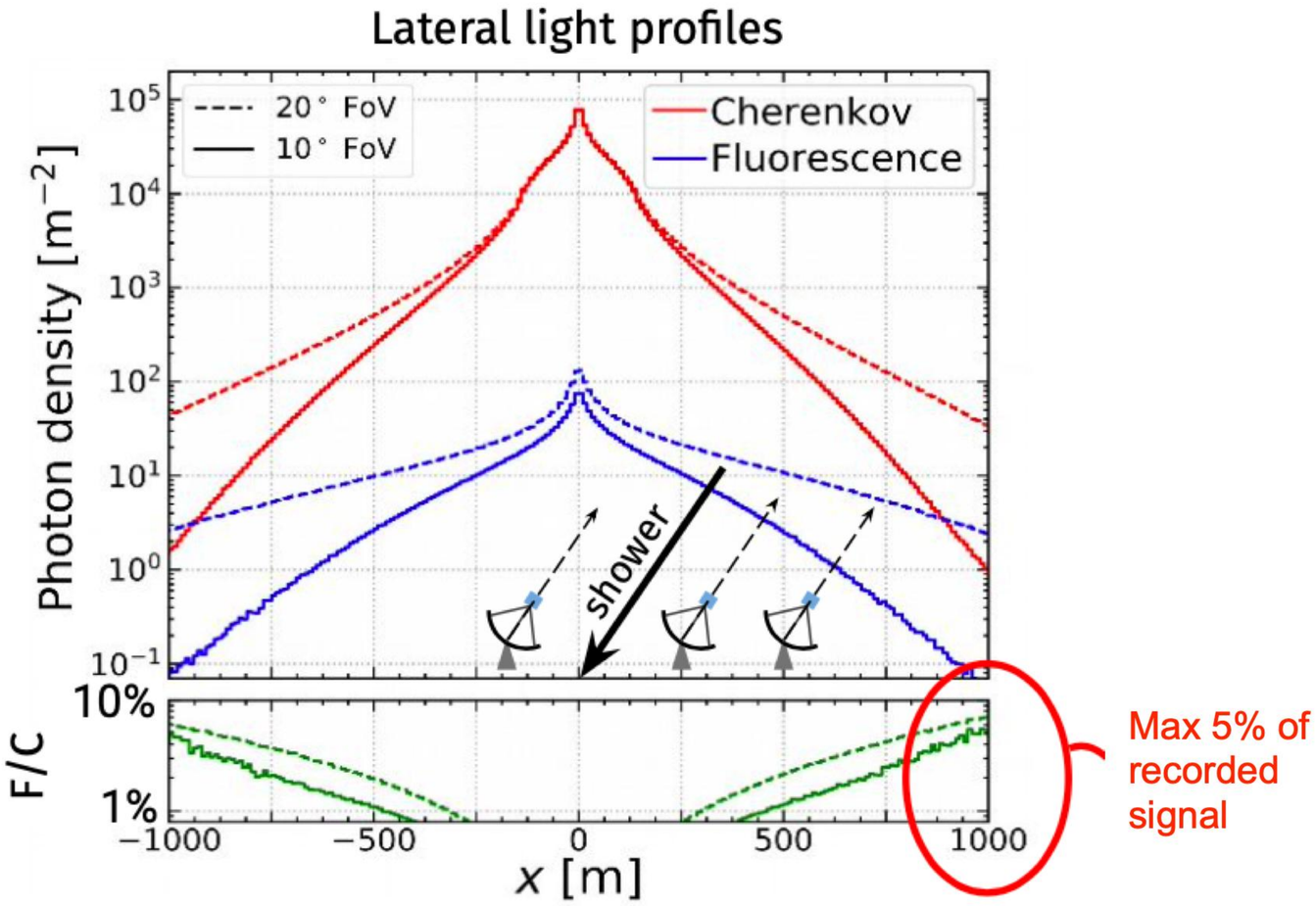
- Fluorescence emission from de-excitation of N_2 states (290-430 nm)
- Less efficient light emission:
  - For 1 GeV electron near ground, 30 photons from Cherenkov light vs 4 photons from fluorescence light (per m track length)
- Longer time profile: microseconds vs 10s of nanoseconds



Backgrounds

Fluorescence

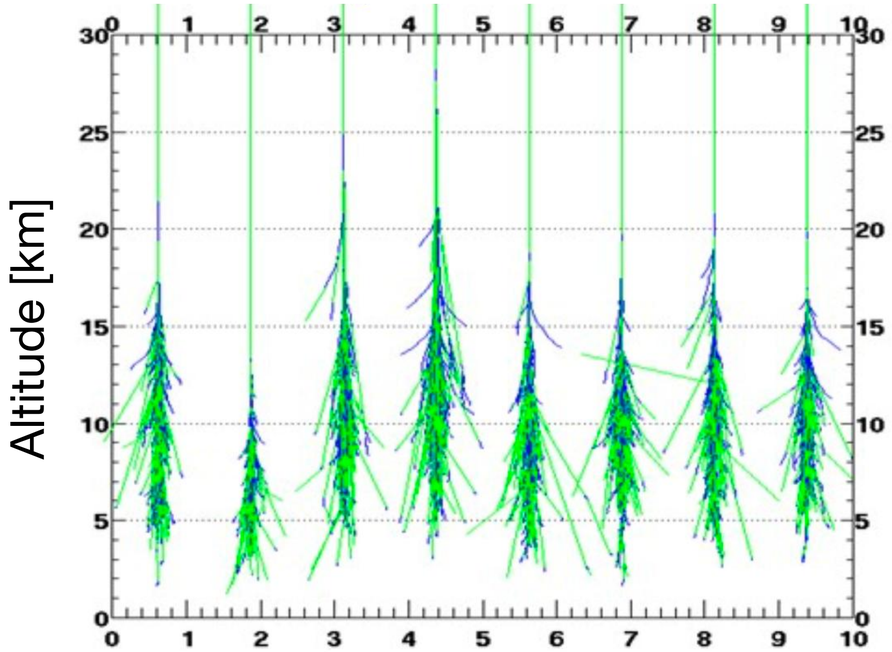
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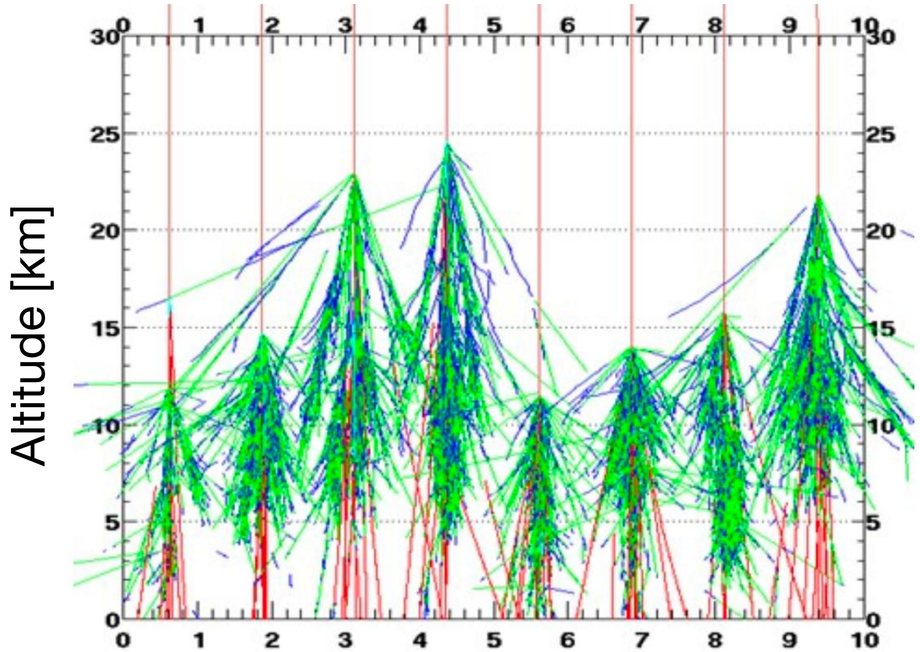
Backgrounds

Hadronic showers

GAMMAS, 100 GeV



PROTONS, 500 GeV

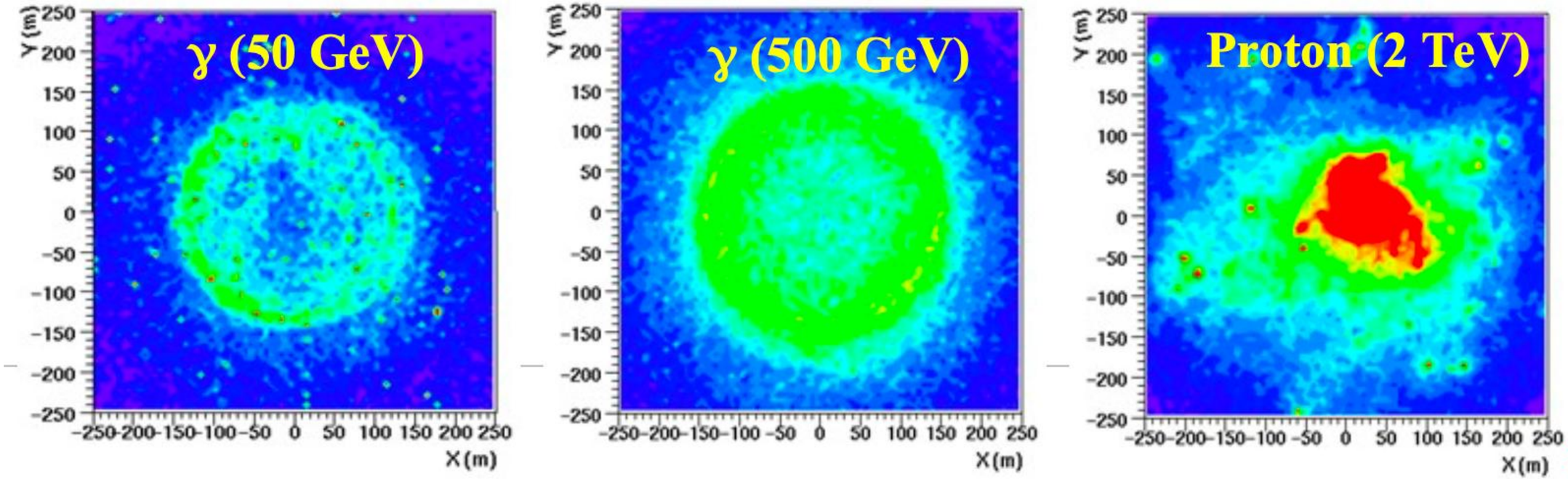


For hadronic showers, the higher transverse momentum
from pion production and sub-showers produce a wider shower

→ the width of the shower can be used to discriminate gammas from hadrons

Backgrounds

Hadronic showers



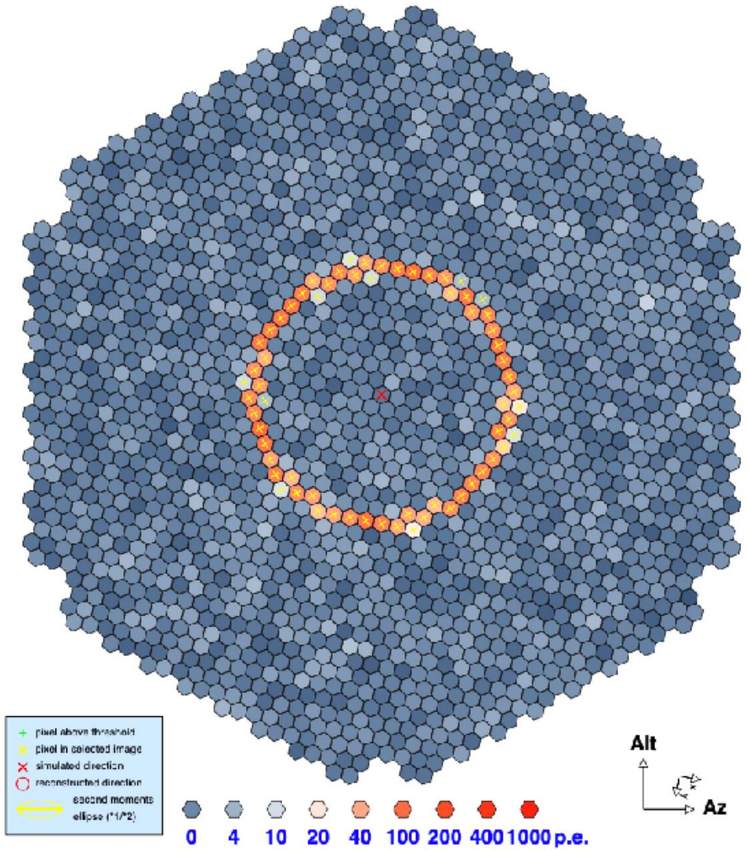
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Backgrounds

Hadronic showers: muons

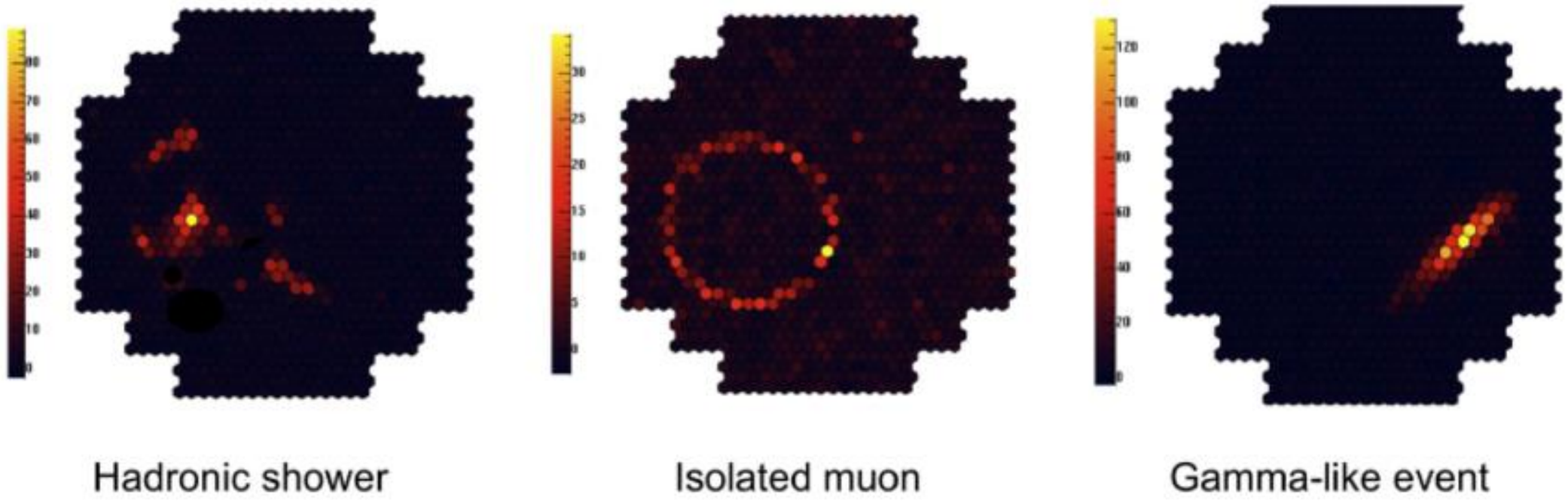
- **Produced in hadronic sub-showers**
- Most muons detected by IACTs are minimum ionizing particles with a long lifetime → large penetration depth
- Travel in straight lines towards the ground with Cherenkov radiation emitted under constant opening angle in a cone around the muon direction of travel
- As IACTs image in angular space, the Cherenkov light from muons traveling parallel to the telescope's optical axis forms a **ring-shaped image in the camera**



⇒ Partial muon arcs look much like EM showers, particularly a problem at ≈100 GeV

Backgrounds

Hadronic showers and muons



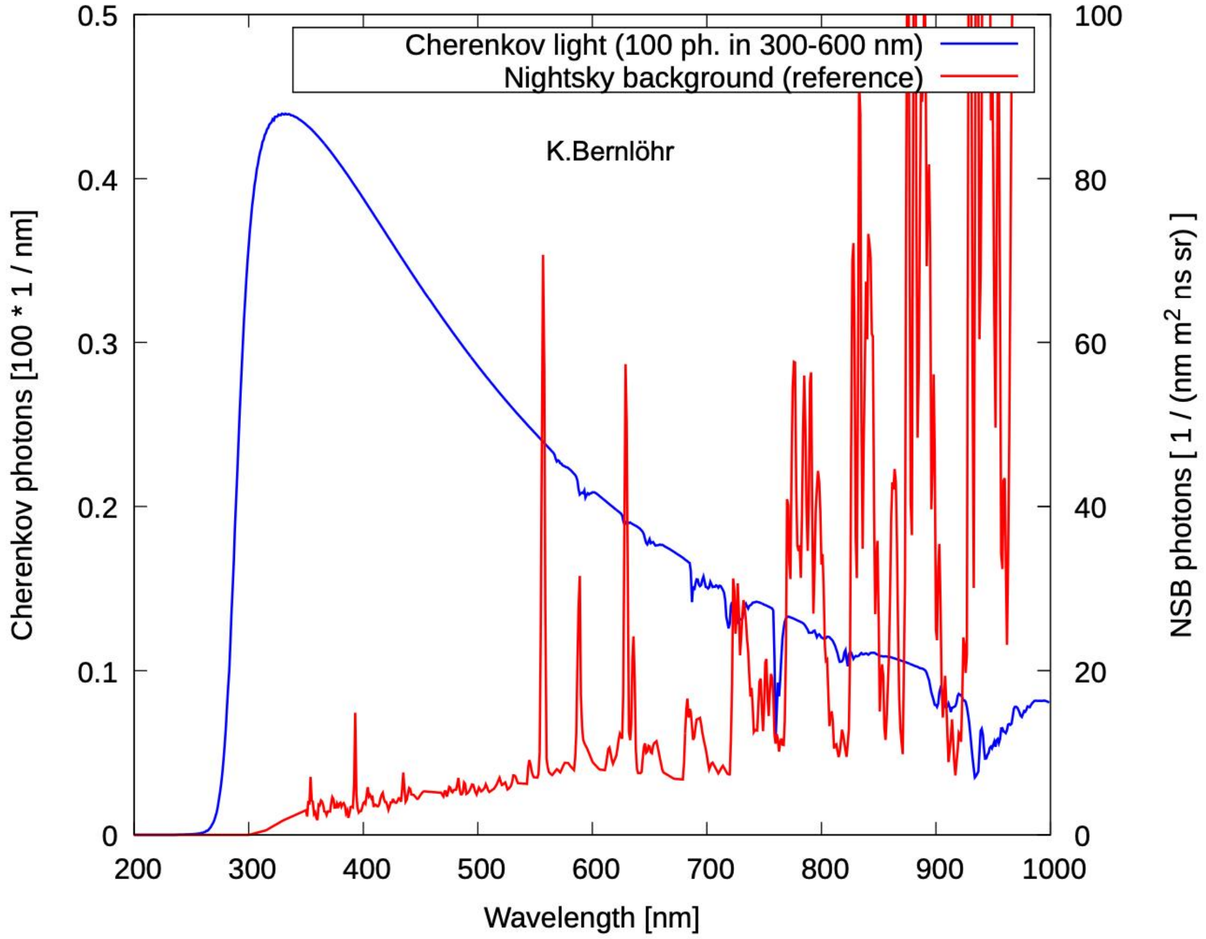
Gamma / hadron discrimination through image topology

Backgrounds

NSB

Even ignoring stars, the sky is not dark:

- Night Sky Background (NSB) light is always present
- Collective unresolved visible/UV light from optical sources (stars, Milky Way, etc)
- Glow from moon, nearby light pollution



How to build an IACT?

General astronomical requirements

Angular Resolution

What is the smallest object we can resolve?

→ Ability to resolve two nearby objects

Telescope optics, camera pixel sizes

Light Collecting Power

What is the faintest object we can detect?

→ Ability to detect weak Cherenkov flashes

Mirror area and reflectivity, camera sensitivity

Field of View (FOV)

What is the largest object we can image?

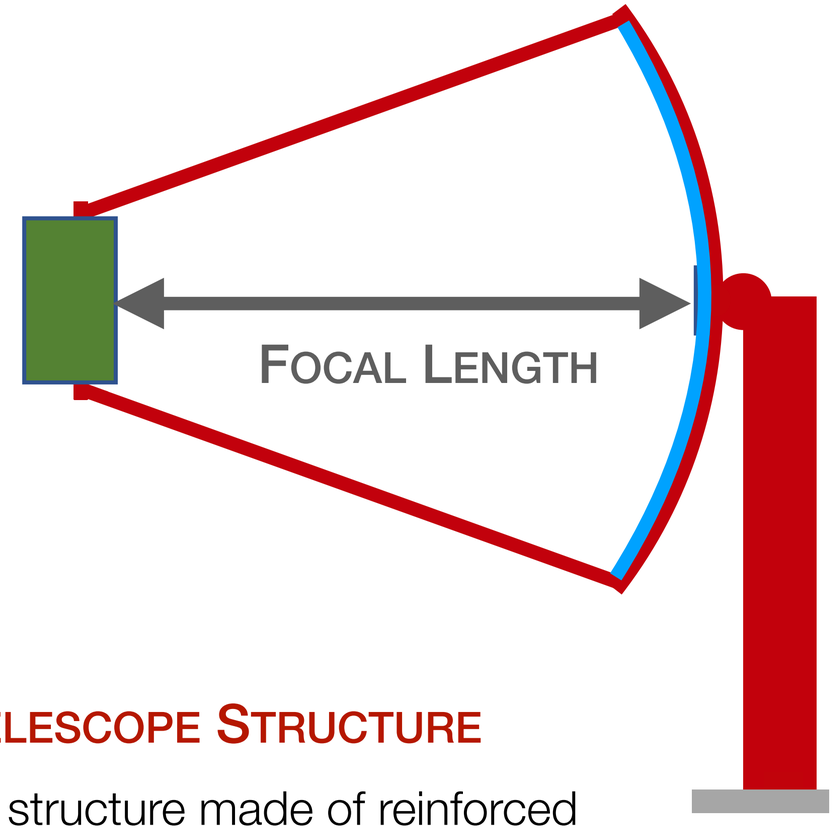
→ Solid angle of the sky covered

Telescope optics, camera pixel sizes

An IACT's prototype

Camera

Pixellated with optical detectors (PMTs, SiMPs). Fast (~ns)



Telescope Structure

Tubular structure made of reinforced carbon fibre and steel tubes or made of steel to ensure sufficient stiffness. Total weight of ~90 – 100t

Dish + Reflector (Mirrors)

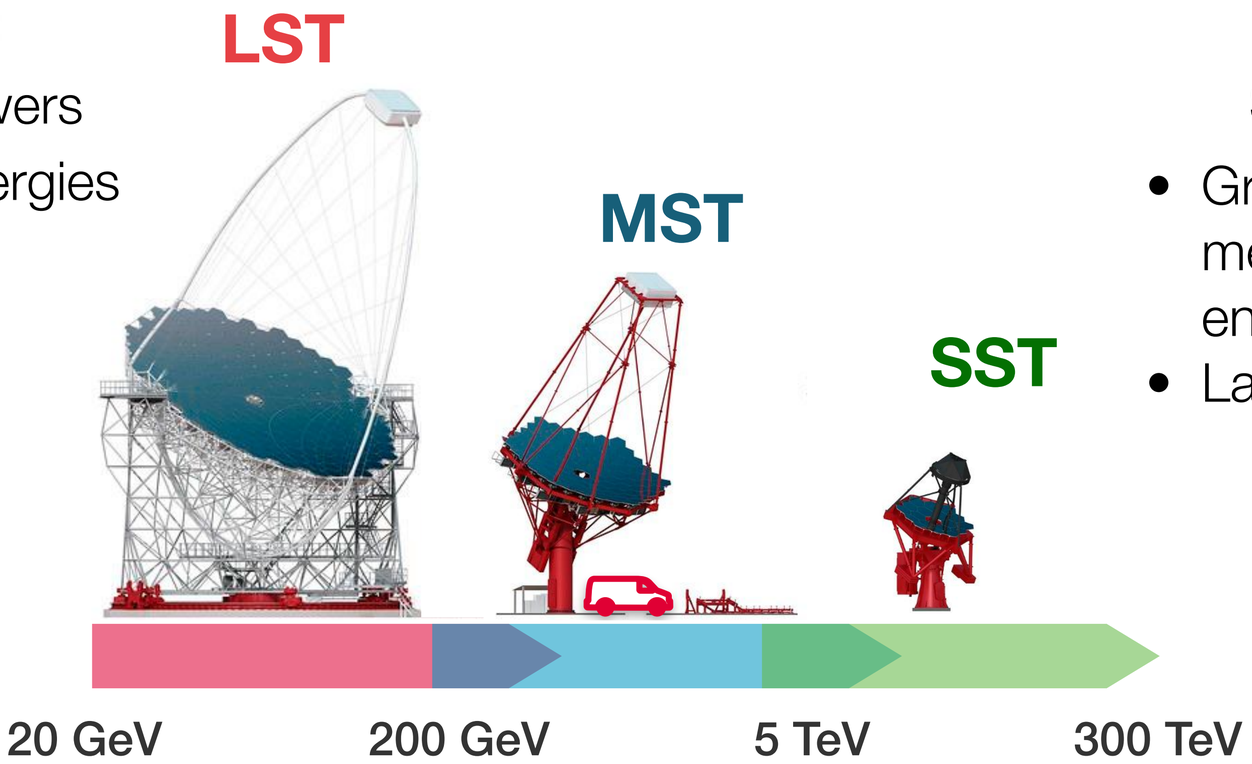
- **Parabolic design**
  - ➡ Larger off-axis PSF but perfectly isochronous (no time spread)
- **Davis-Cotton (DC) design**
  - ➡ Smaller off-axis PSF than parabolic but non-isochronous (few-ns time dispersion)
- **Modified Davis-Cotton (DC) design**
  - ➡ Reduced dish-induced signal dispersion
  - ➡ Almost DC-like PSF but improved isochronicity of reflector thus reduced time spread
- **Modified Schwarzschild-Couder dual-mirror**
  - ➡ Shorter focal length allowing large FOV and compact cameras

IACTs: how big? How many?

A compromise between size and number…

Bigger dishes:

- Image fainter showers
  → go to lower energies
- Smaller FOV (~4º)



Smaller dishes:

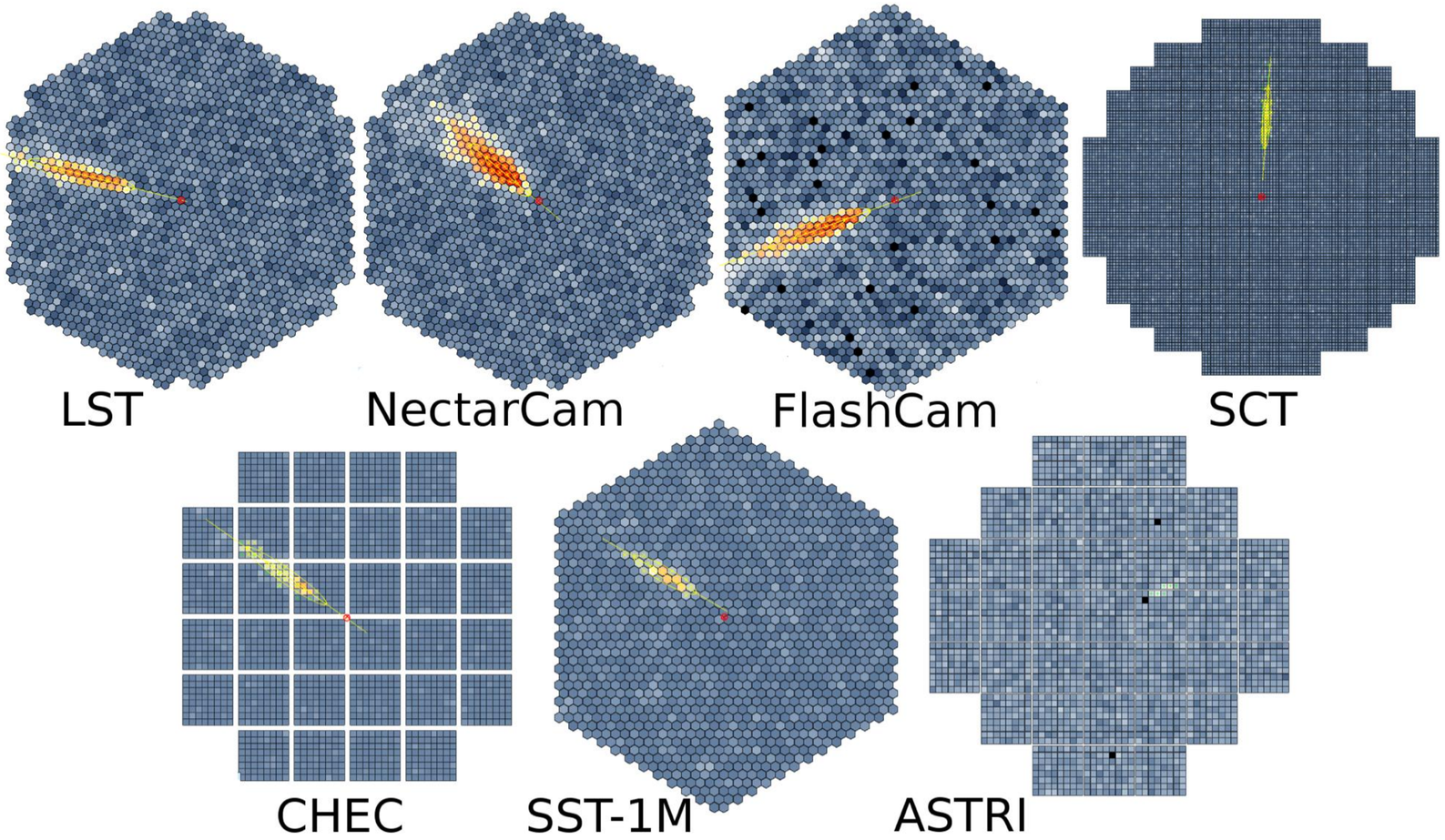
- Grater sensitivity at medium to high energies
- Larger FOV (~9º)

Ideally: Lots of big-dish telescopes, but money is always a limitation!

Cost increases with size of dish or number of telescopes

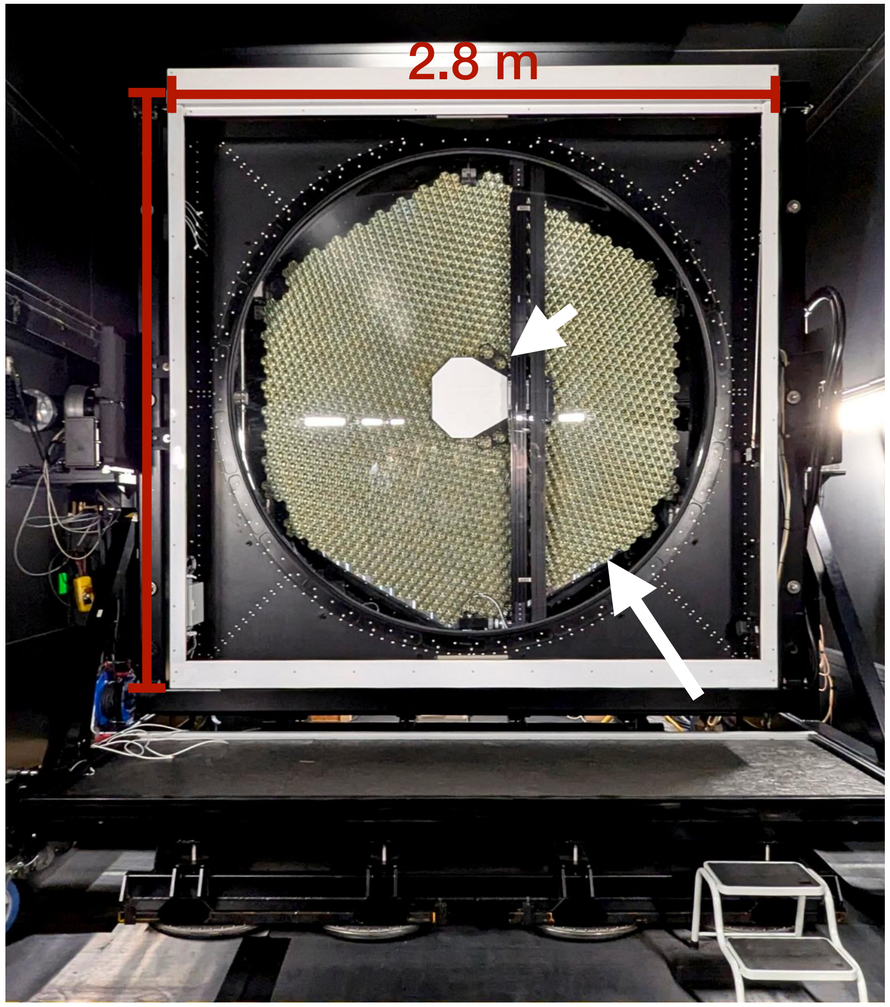
IACT cameras

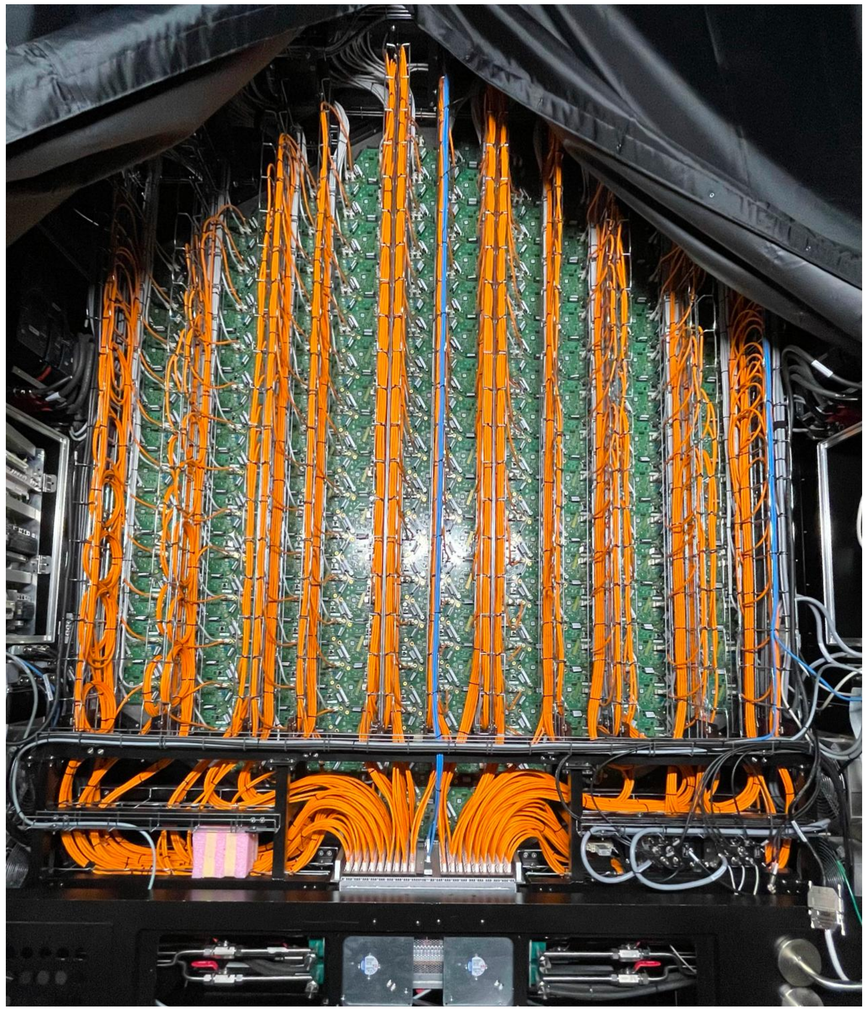
The CTAO designs



IACT cameras

NectarCAM camera at IRFU/CEA Paris-Saclay

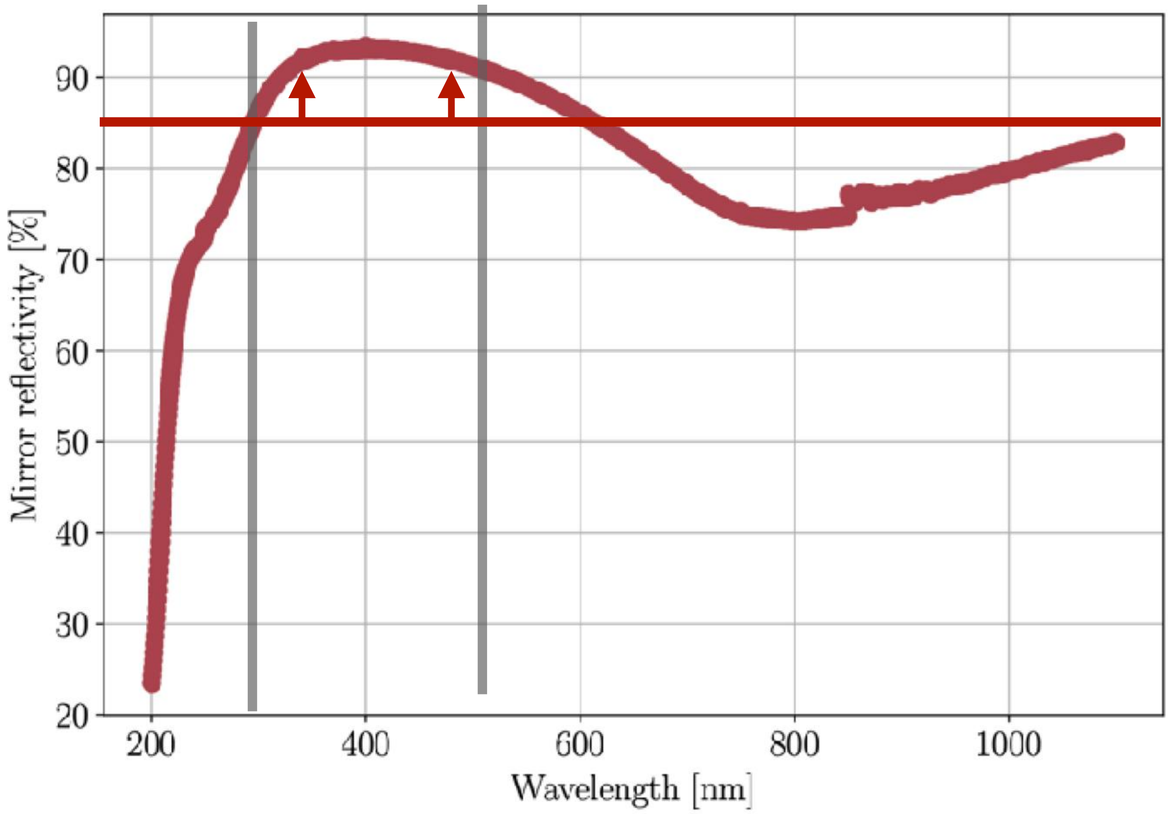




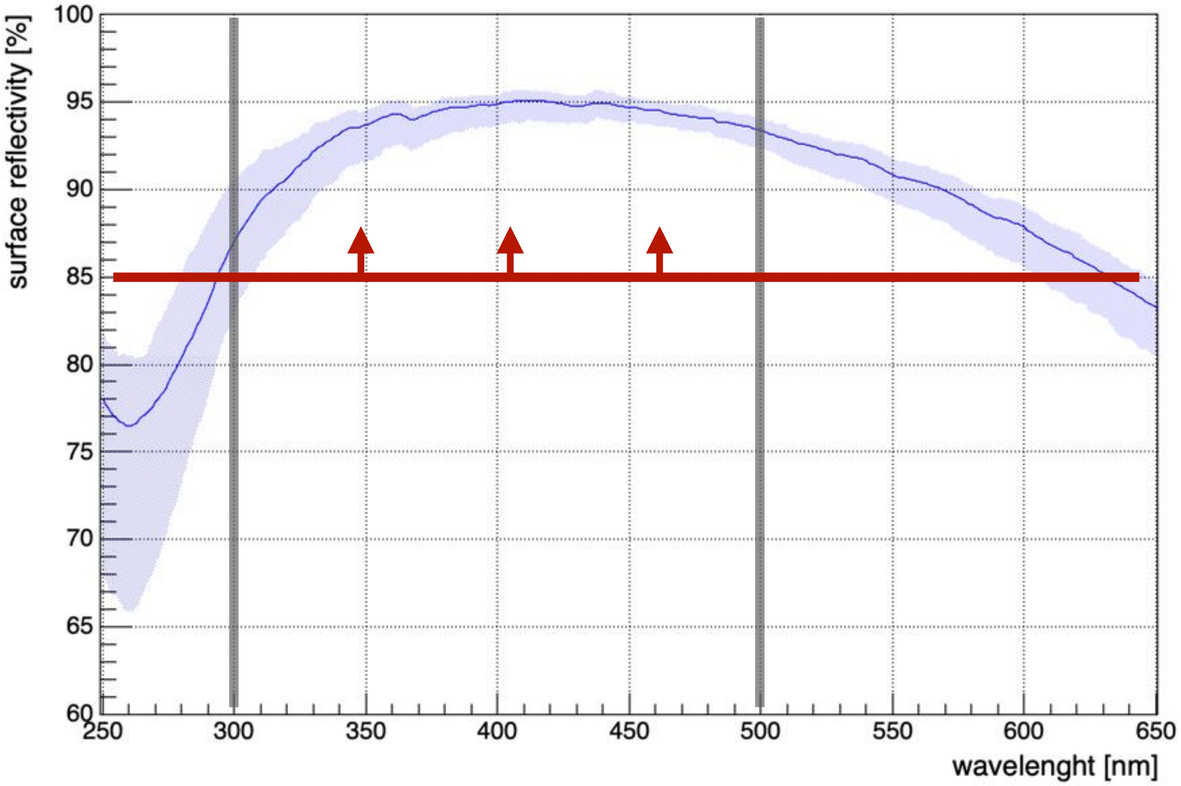
IACT mirrors

Reflectivity > 85% in [300, 500] nm

LST



MST

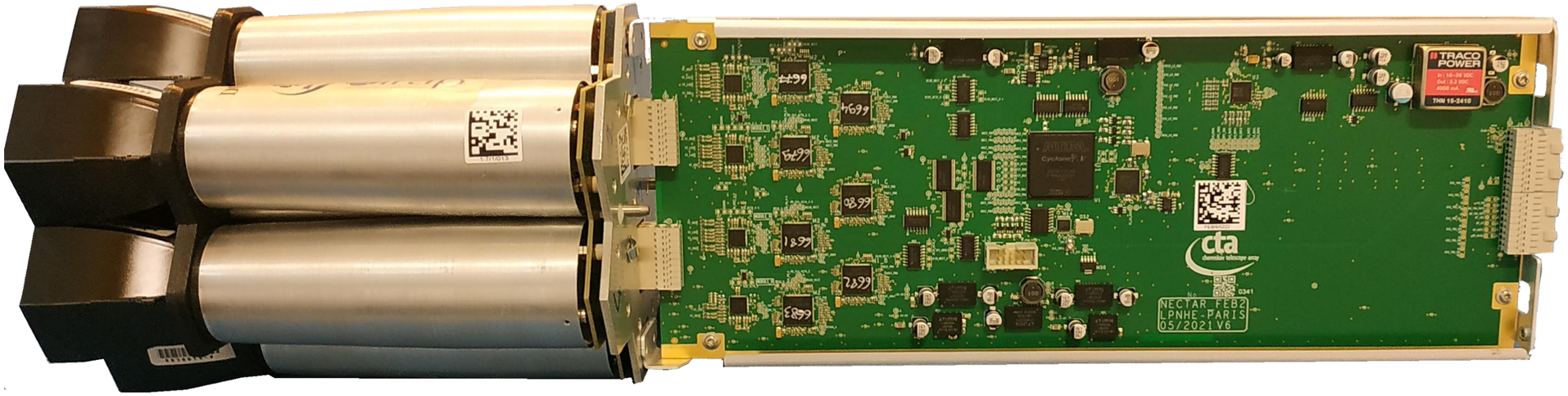


From photons to photoelectrons

Formation of the electric signal

Focal Plane Module

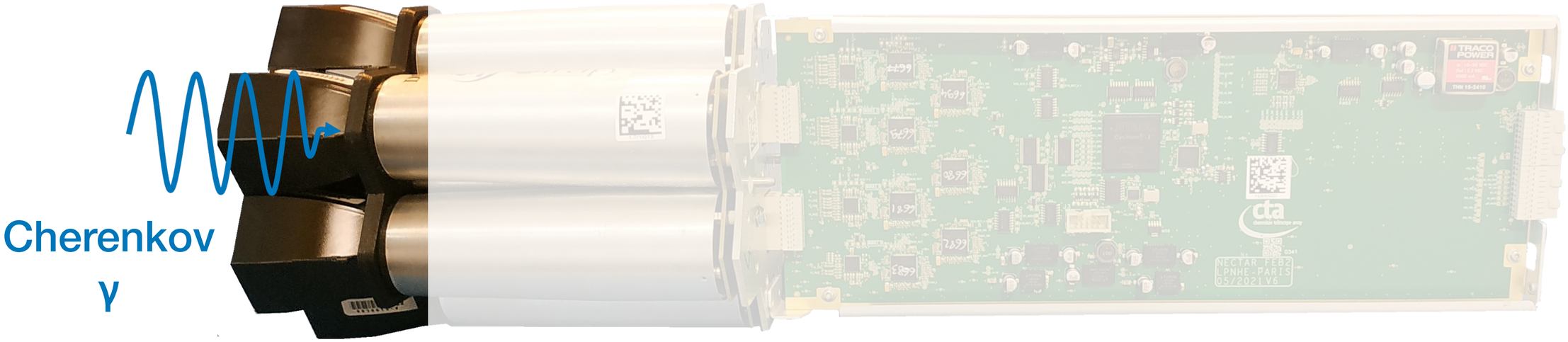
Front End Board



From photons to photoelectrons

Formation of the electric signal

Focal Plane Module **Front End Board**

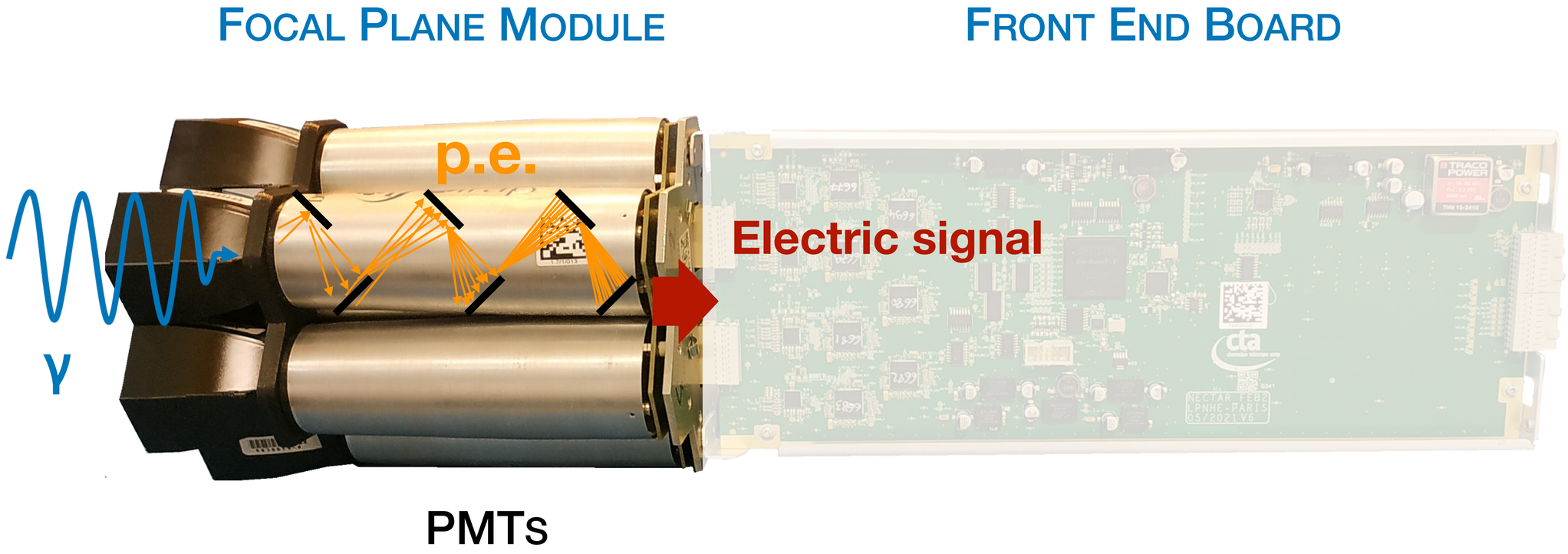


Winston Cones

1. Light deposited in the camera is first collected in the **light guides** and detected in the focal plane

From photons to photoelectrons

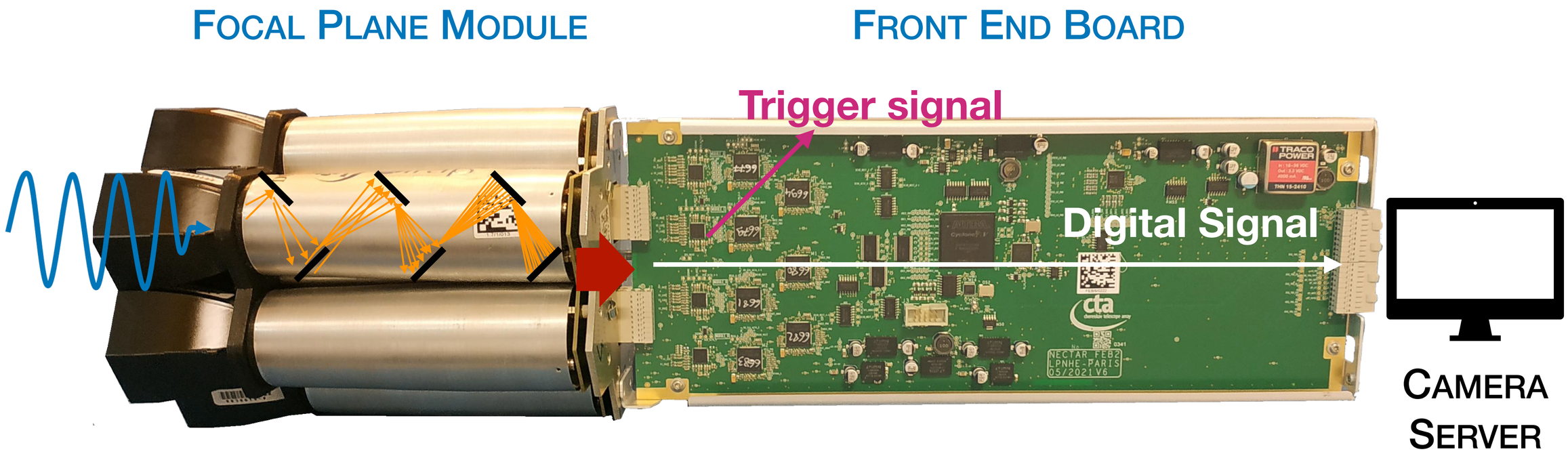
Formation of the electric signal



2. The signal is converted into electric signal by the **PMTs** and pre-amplified

From photons to photoelectrons

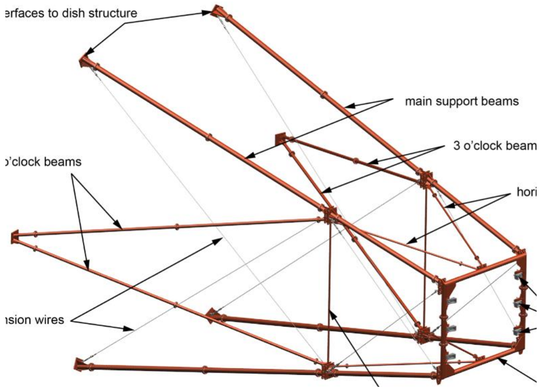
Formation of the electric signal



3. Signal is amplified, sampled and digitized + formation of trigger signal

From photons to photoelectrons

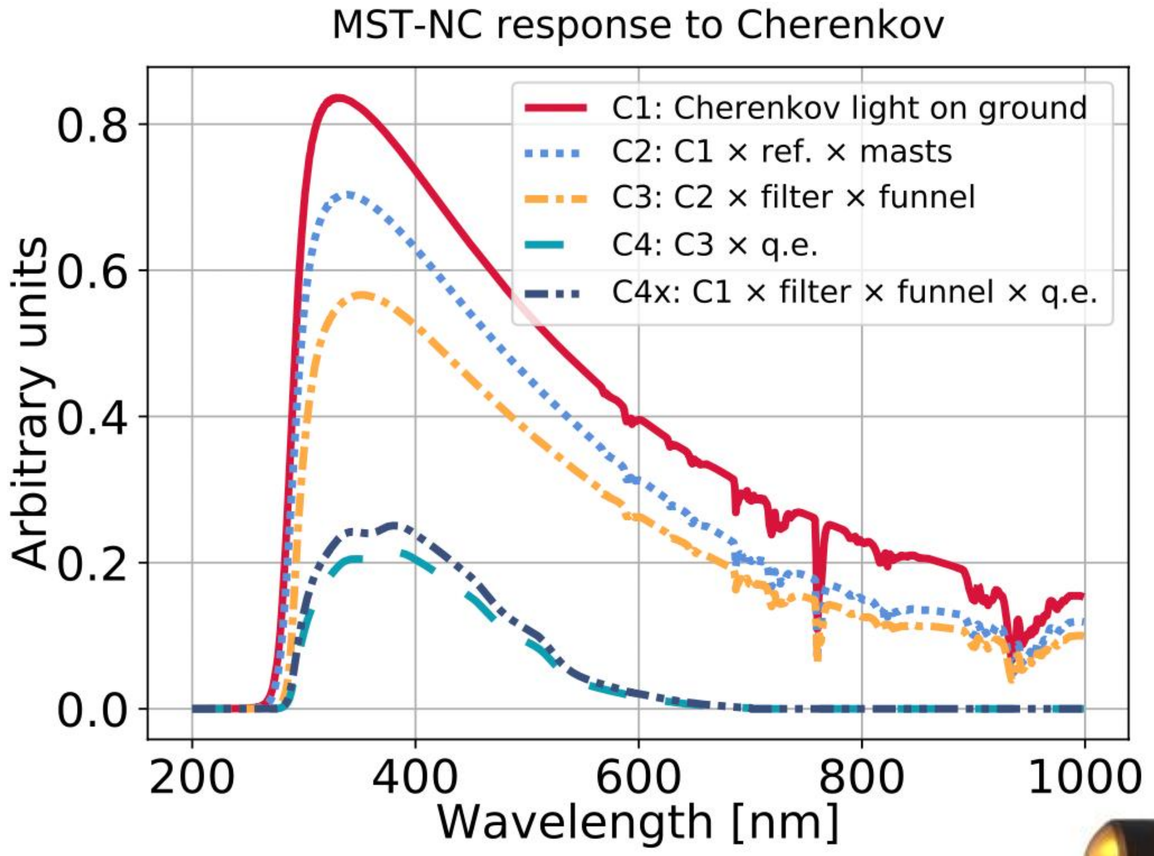
Impact of components on Cherenkov light detection efficiency



TELESCOPE STRUCTURE



MIRRORS





ENTRANCE WINDOW



LIGHT GUIDES

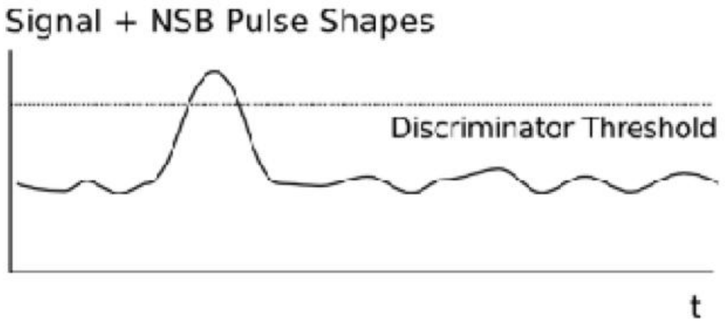


HAMAMATSU PMT

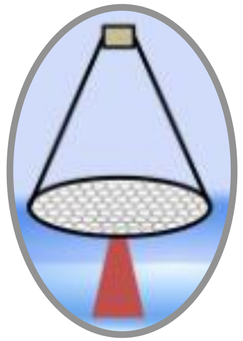
Triggering an IACT array

From the single pixels to an array trigger

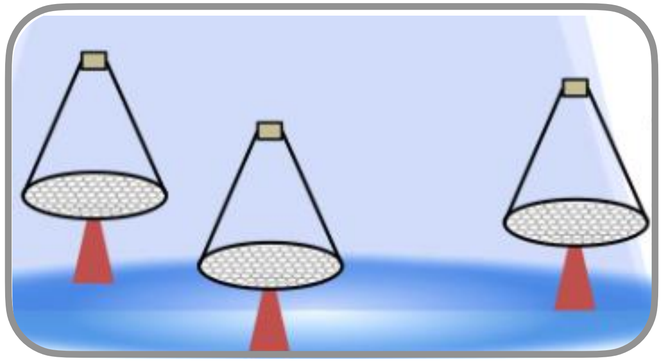
L1: pixel trigger ➡ **L2: telescope trigger** ➡ **L3: array trigger**



Is the signal in a single (or trigger sector) above a threshold?



Have enough pixels/ sectors triggered?

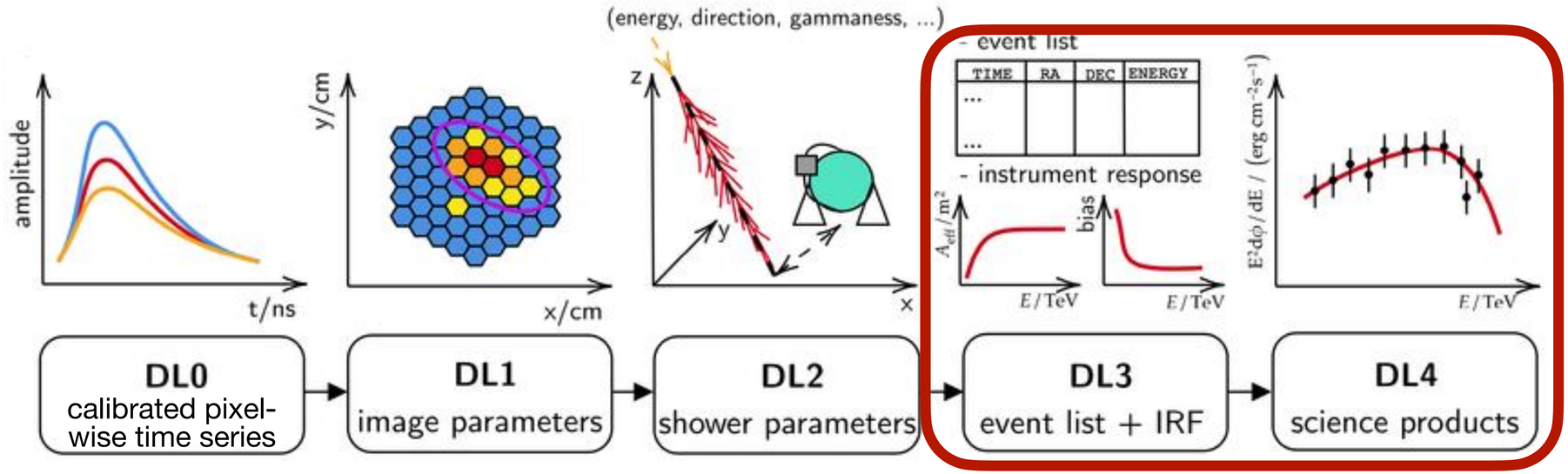


Has more than one telescope triggered within a small time window?

✓ Rejects most muons

✓ Drastically reduces triggers due to NSB fluctuations

From low-level to high-level IACTs data

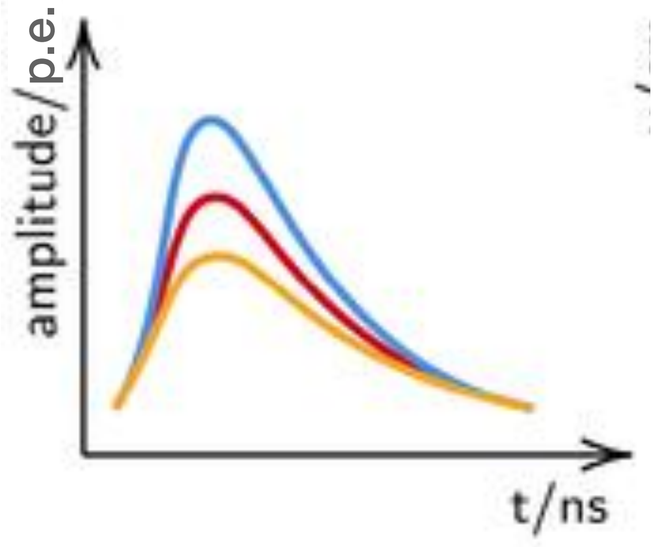


→ Jonathan's talk this afternoon!

Data levels

Definition of data levels (DLs) foreseen for CTAO

DL0



First level of calibrated data written to disk and long-term preserved.
Pixel-wise time series for each telescope.

DL1

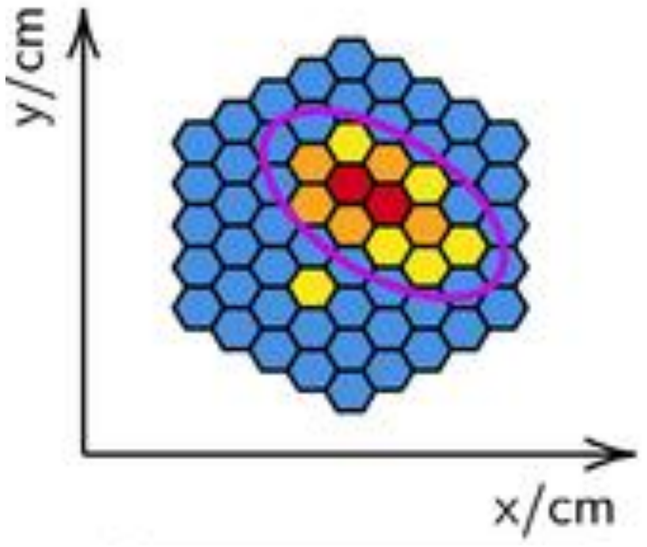
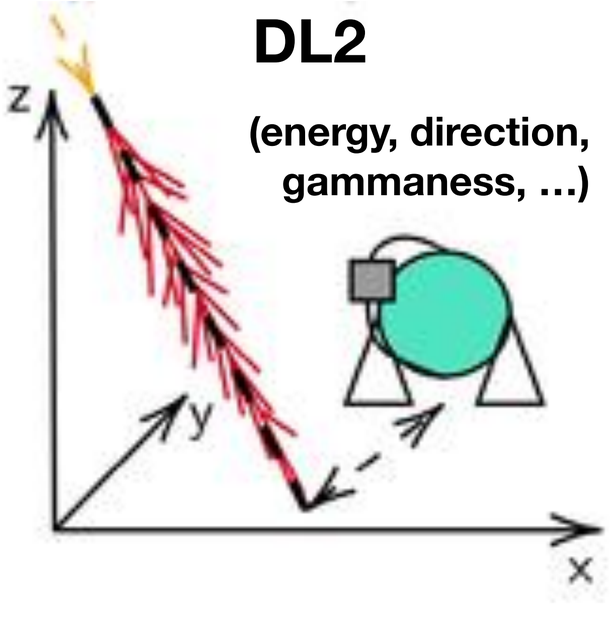


Image extraction and cleaning, and derived per-telescopes parameters.
Camera image parameters (Hillas parameters)

DL2



Reconstructed shower parameters.
Parameters from different telescopes combined to infer air shower properties

Data levels

From R0, through R1 to DL0

We still have uncalibrated waveforms in units of ADC counts (R0)

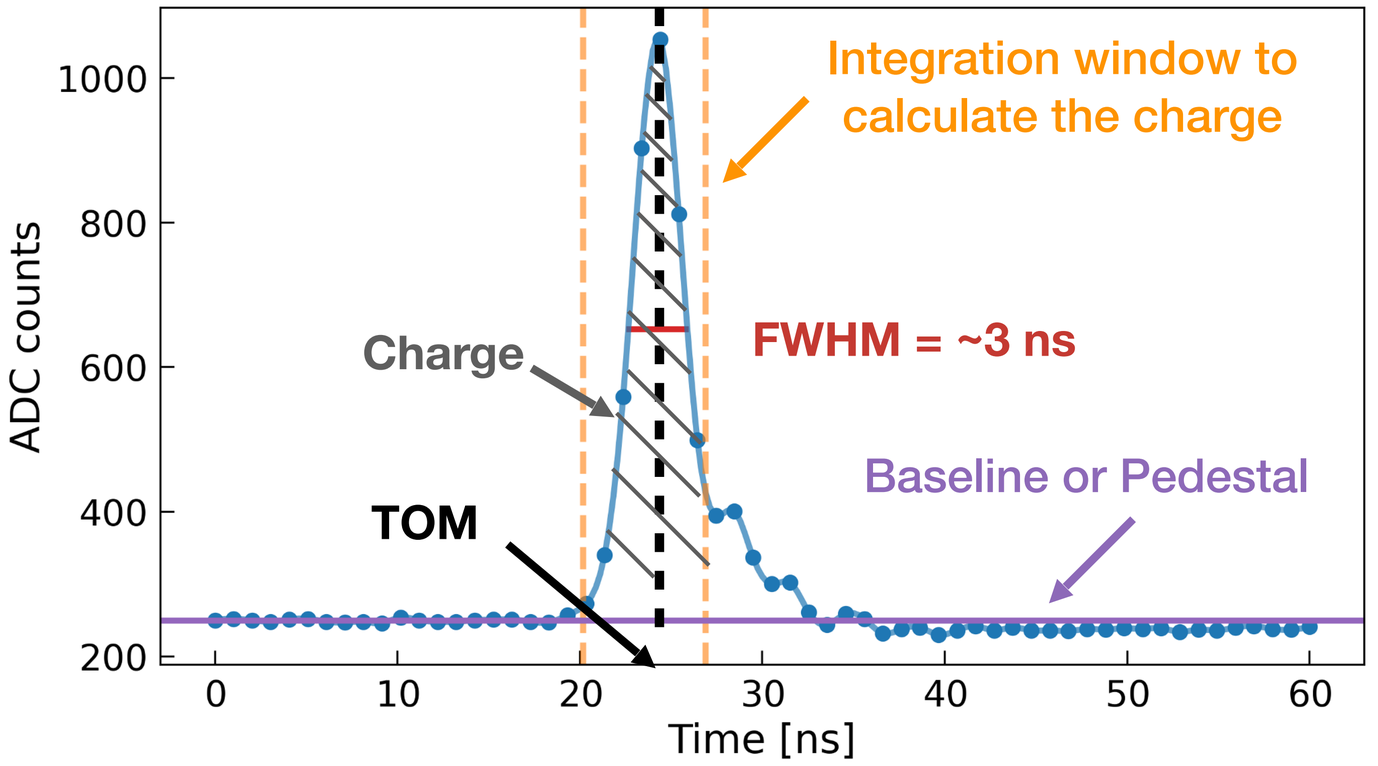
→ Calibration (R0) and data volume reduction (R1): transition to DL0

Low-level data processing

Row data calibration: charge extraction [ADC counts] and arrival time

1. Subtraction of the baseline (NSB + electronic noise)
2. Integration of charge around mean peak in ADC counts
3. Calculation of the signal arrival time: time of maximum (TOM)

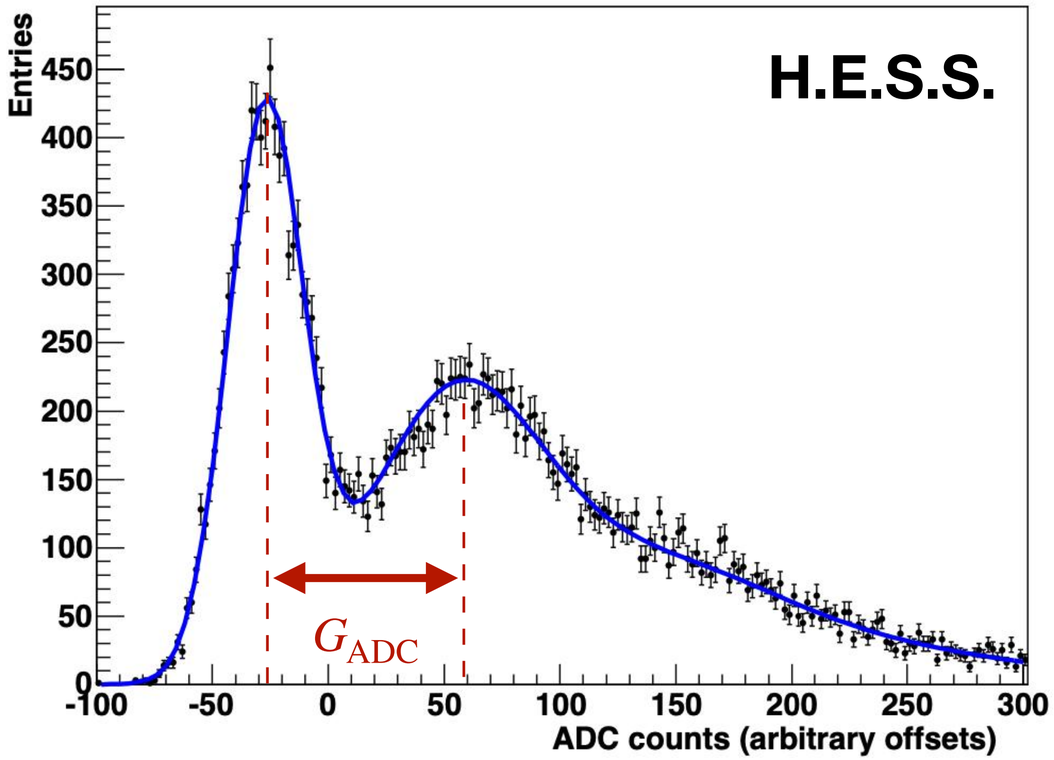
Typical waveform of a light pulse event (NectarCAM)



Pedestal calibration P to correct for baseline noise and charge extraction A_{ADC}

Low-level data processing

Row data calibration: from ADC counts to p.e.

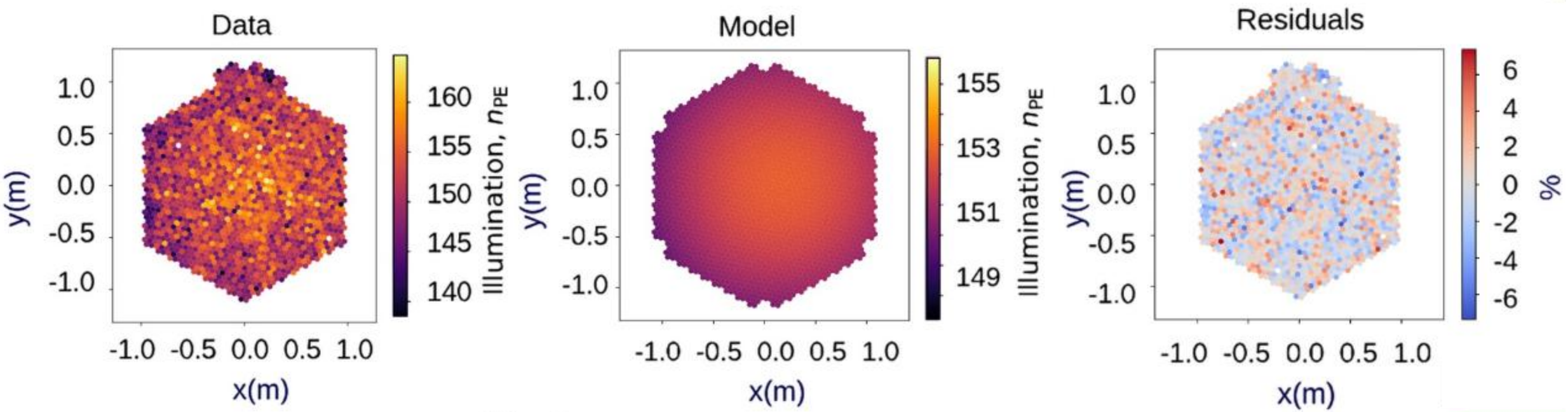


Gain calibration G_{ADC}: determination of the amplification gain of the photo-sensors for a single photo-electron

Low-level data processing

Row data calibration: correct for pixel-to-pixel variations

NectarCAM



[Mikhno+, PoS ICRC2025 759]

Flatfield calibration FF: homogenize the response of the sensors across the camera

Low-level data processing

Calibrated signal in p.e. + data volume reduction

High Gain

$$A^{\mathrm{HG}}\ [\mathrm{p.e.}] = \frac{\mathrm{A}^{\mathrm{HG}}_{\mathrm{ADC}} - \mathrm{P}^{\mathrm{HG}}_{\mathrm{ADC}}}{\mathrm{G}^{\mathrm{HG}}_{\mathrm{ADC}}} \times \mathrm{FF}$$

Low Gain

$$A^{\mathrm{LG}}\ [\mathrm{p.e.}] = \frac{\mathrm{A}^{\mathrm{LG}}_{\mathrm{ADC}} - \mathrm{P}^{\mathrm{LG}}_{\mathrm{ADC}}}{\mathrm{G}^{\mathrm{LG}}_{\mathrm{ADC}}} \times (\mathrm{HG/LG}) \times \mathrm{FF}$$

High gain/low gain ratio

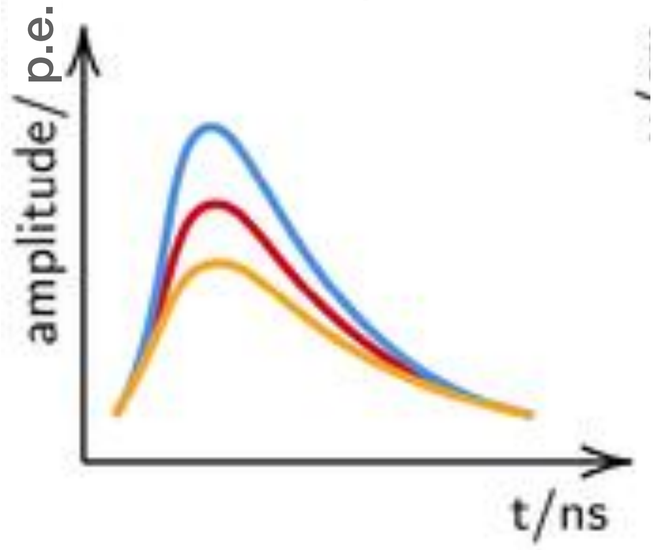
Gain selection + exclusion of “broken” pixels

(due to HV off/unstable, PMT without signal, stars, highly illuminated pixels)

Data levels

We are now at DL0!

DL0



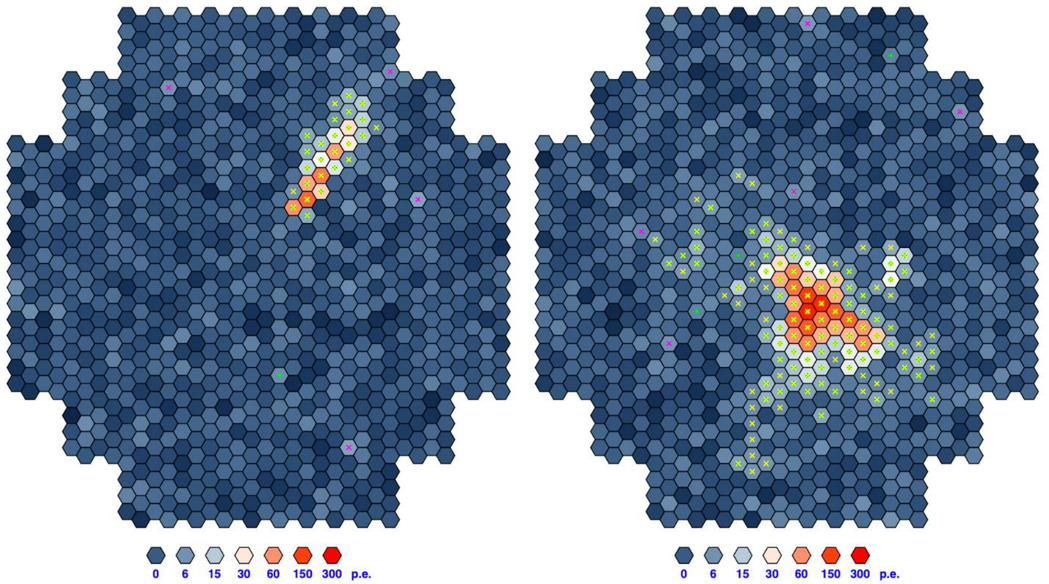
We have calibrated and filtered pixel-wise time-series in p.e. for each telescope

→ Now image extraction: transition to DL1

DL0 data processing

Image cleaning

After the calibration of the signals recorded in the individual camera pixels, a calibrated camera image is obtained



1.0 TeV gamma shower 2.6 TeV proton shower

- Most of the shower images contained in a dozen - few tens of pixels
- Rest of the pixels have a mixture of electronic noise and NSB/afterpulses

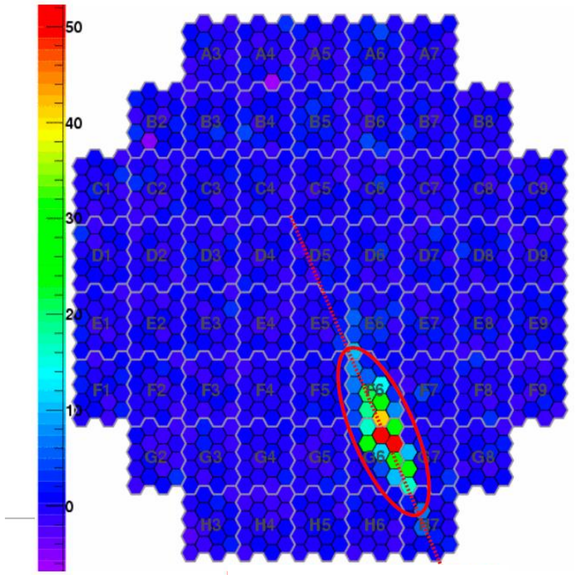
Cleaning algorithm profits from:

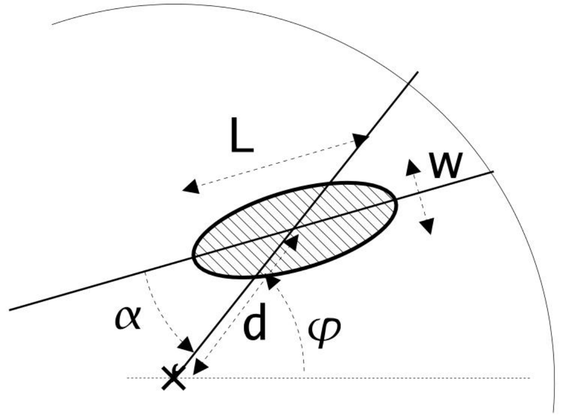
- Shower signal stick out from NSB (more of high energies)
- Shower signals are bundled in neighboring pixels
- Shower signals have time structure

DL0 data processing

Hillas parameters

- Images of gamma rays are elongated ellipses
- Hillas parameters = analytical representation of the image as a 2D Gaussian:
  - Width (***W***) and length (***L***)
  - Total image amplitude = total number of photoelectrons (***size*** or ***intensity***)
  - Nominal distance ***d*** between the centre of the camera and the (amplitude-weighted) centroid of the image
  - Azimuthal angle φ of the image centroid with respect to the camera centre
  - Orientation angle α of the ellipse in the camera

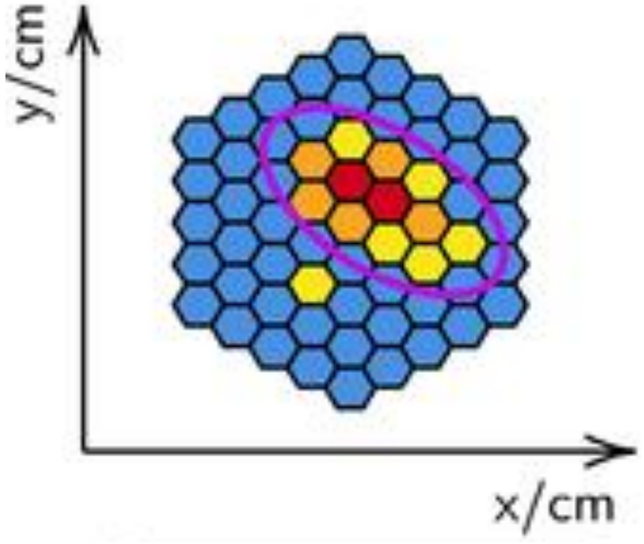




Data levels

We are now at DL1!

DL1

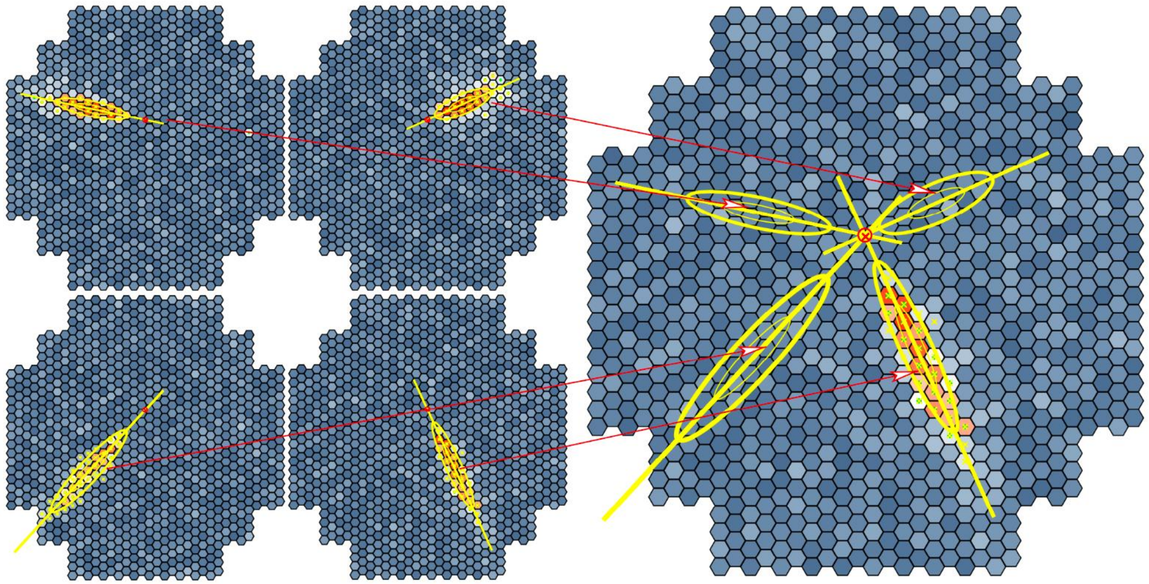


We have images and they are cleaned and parameterized

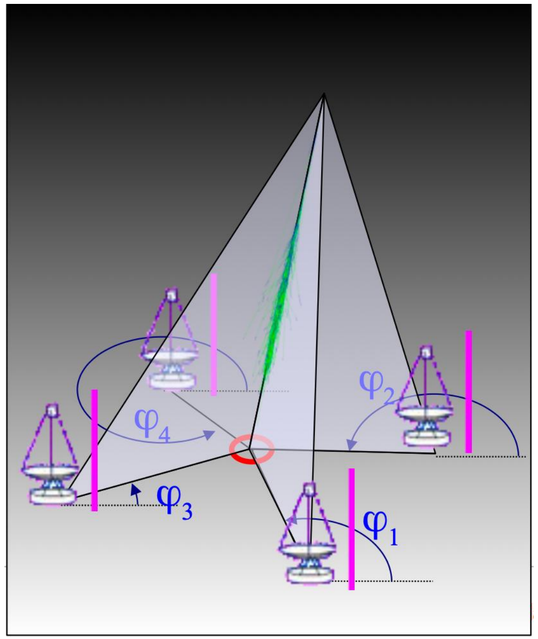
→ transition to DL2

Stereoscopic reconstruction

Event reconstruction: (improved) arrival direction



Shower direction given by intersection of major axes of the ellipses obtained in the different camera images



Shower axis given by intersection of planes corresponding to major axes of ellipses.
Impact parameter given by intersection of major axes on ground

Stereoscopic reconstruction

Event reconstruction: energy

- Hillas' parameters **size** and **distance** (which is proportional to impact parameter) can be used to determine the energy
- At fixed impact parameter, **E ≈ size**
- Energy derived from image intensity compared to MC
  - Hillas parameters derived from MC simulations of gamma-ray showers
  - Quantities derived for multiple image amplitudes, zenith angle and impact position and stored in look-up tables

Stereoscopic reconstruction

Event reconstruction: particle type (γ/background)

- As a gamma-ray observatory, for almost all science cases we can simplify to binary classification:

Signal, gamma rays ⇔ background, everything else

- Hillas parameters can be used for gamma/hadron separation:
  - **Width:** primary hadron rejection parameter. Hadronic showers are wider due to larger transverse momentum and sub-showers
  - **Length/size:** good veto for muon ring images
- MC simulation of large library of showers with different energies, distribution of impact points, etc and use it as a label dataset for supervised learning for signal and background:
  - Signal is almost always taken from simulated gamma-ray events
  - Background can either be simulations as well or taken from observations of “dark spots”

Stereoscopic reconstruction

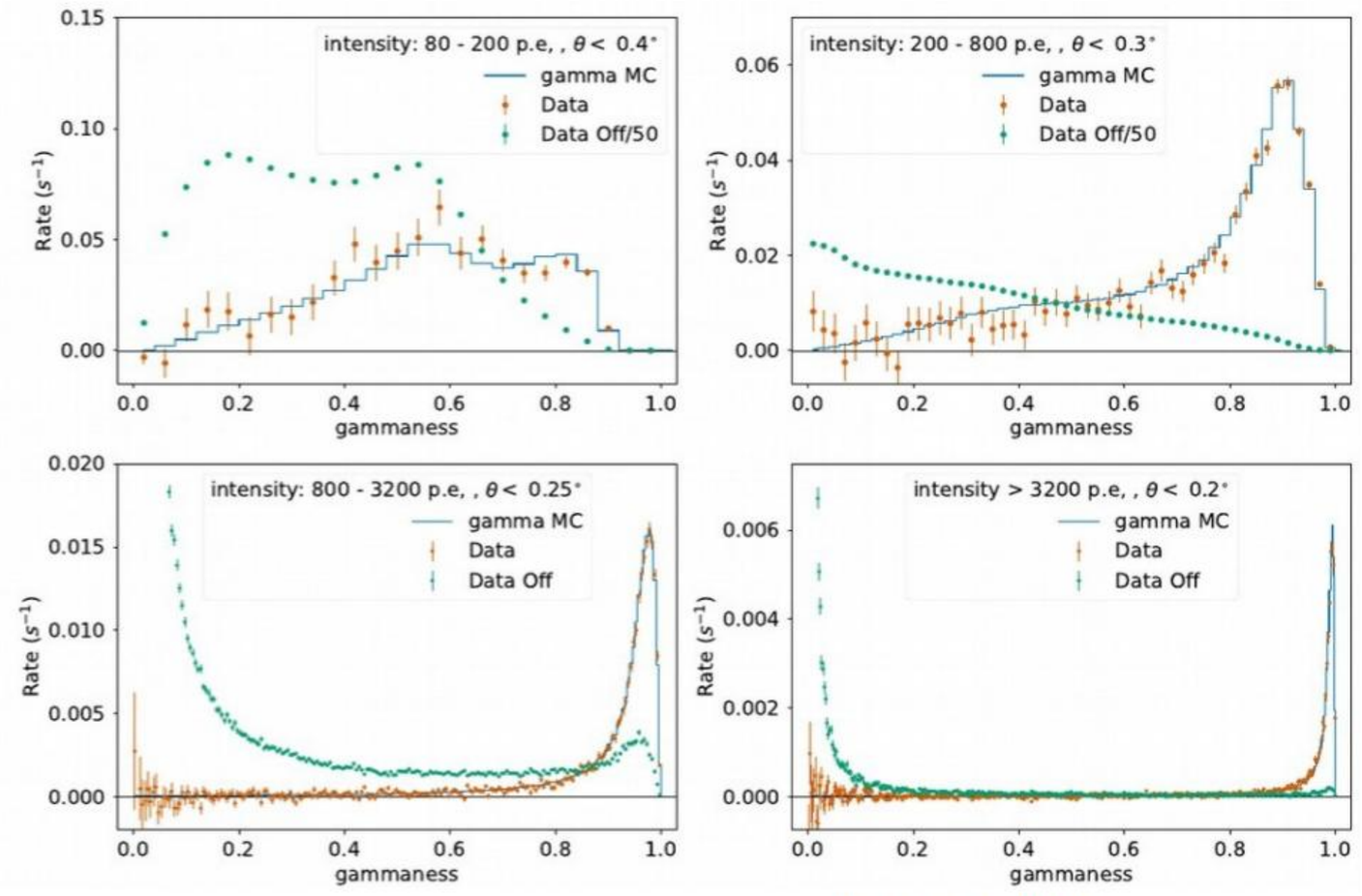
Gammaness

- By using training data (with known gamma-ray and cosmic-ray events), machine learning models or statistical methods (such as Random Forests or Boosted Decision Trees) are employed to develop a classifier that calculates the **probability that a given event is a gamma-ray**
- The classifier output is the **gammaness** parameter, typically ranging between 0 and 1, where:
  - Gammaness close to 1: Indicates a high probability of being a gamma-ray event
  - Gammaness close to 0: Suggests a higher likelihood of being a cosmic-ray background event
- No “physical” meaning — the values of gammaness will strongly depend on the training
- Efficiency of the cut = a cut that keeps a certain percentage of the gamma-rays

Stereoscopic reconstruction

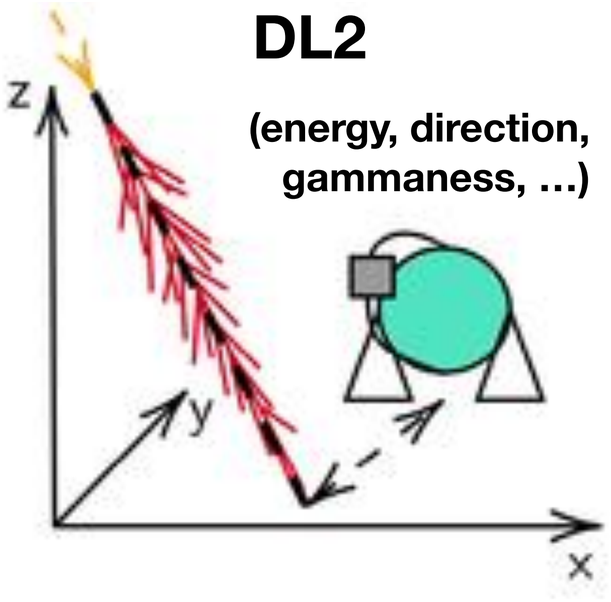
Gammaness

- High energies (larger images) separate better: a cut excludes most of the protons while keeping most of gammas
- At low energies the distributions heavily overlap



Data levels

We are now at DL2!

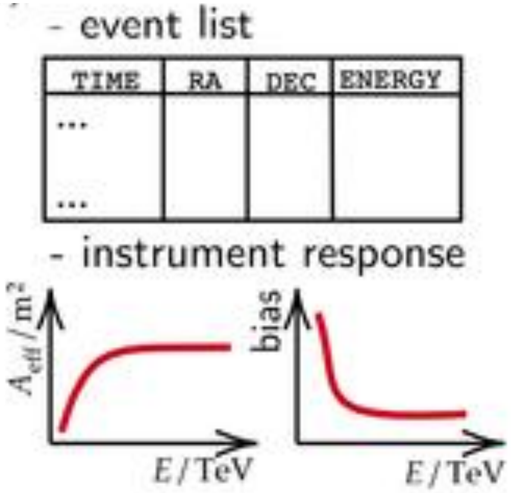


Data reconstruction of the intrinsic gamma-ray event parameters: energy, direction and gammaness

What is next?

A small spoiler to Jonathan's talk…

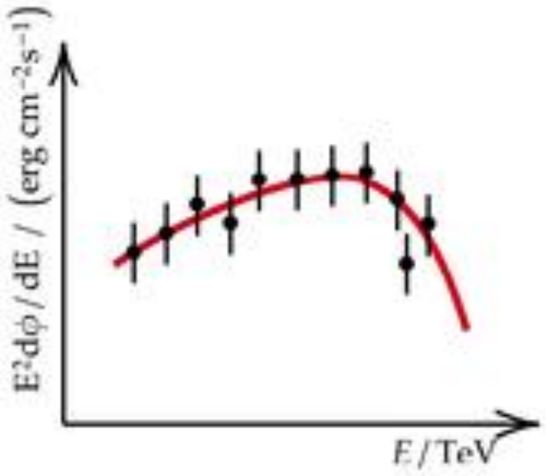
DL3



List of gamma-ray candidate events and associated IRFs

Reconstructed properties of events that pass certain event selection criteria, as well as the IRFs (effective area, PSF, etc. – typically obtained from MC) that detail the performance of the system

DL4



High-level science data products

From the list of selected events and the IRFs, "astronomical" science products such as sky maps, energy spectra, or light curves may be derived

What do we have?

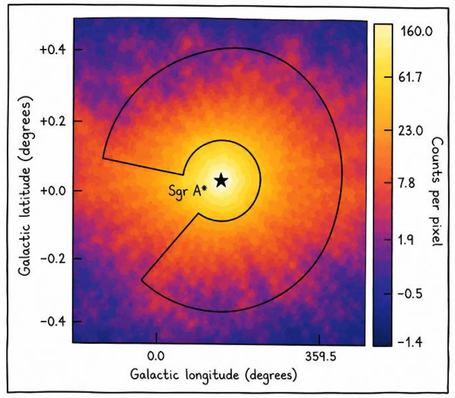
Gamma-like events

Points in space, time, and estimated energy that may be gamma rays

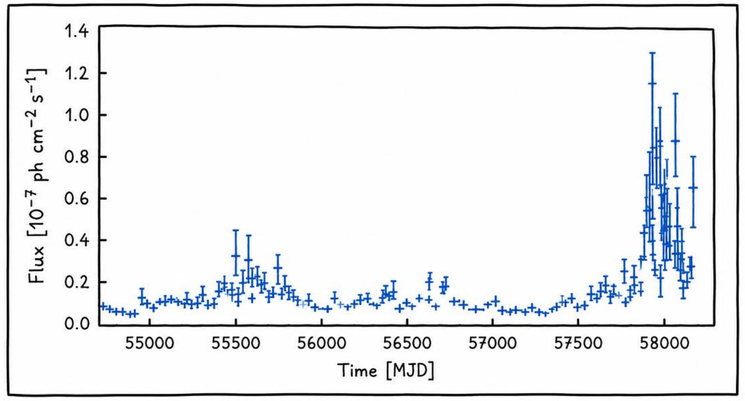
or may be mis-reconstructed cosmic rays

event id	time	ra	dec	E_{reco}
1	12	87.6	-23.7	5.6
2	150	87.2	-22.1	0.32
3	190	86.5	-23.4	0.45
4	2000	82.0	-23.2	0.57
5	7029	88.6	-24.1	2.4

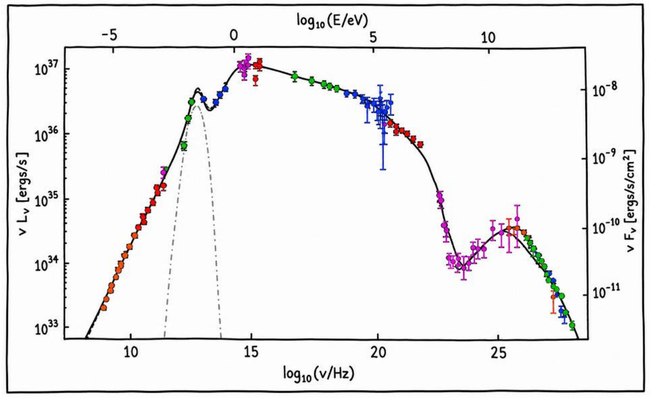
What do we want?



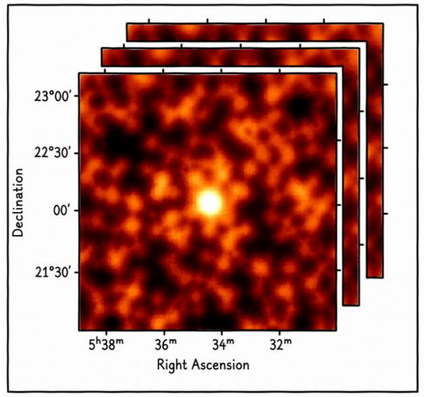
Images: γ-ray flux as a function of spatial coordinates



Light Curves: γ-ray flux from a region as a function of time

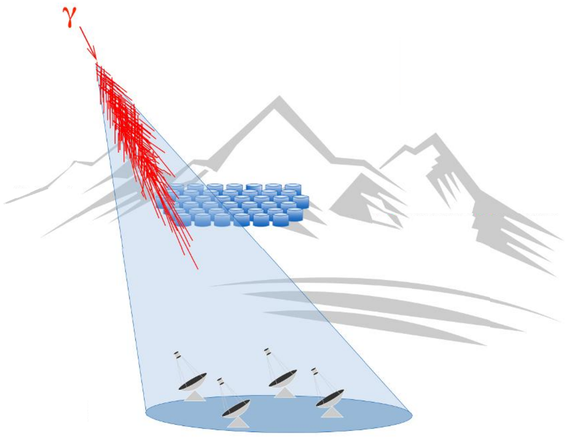


Spectra: γ-ray flux from a region as a function of energy

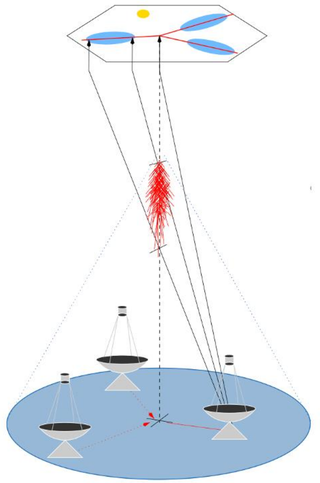


Or their combination **(data cubes)**

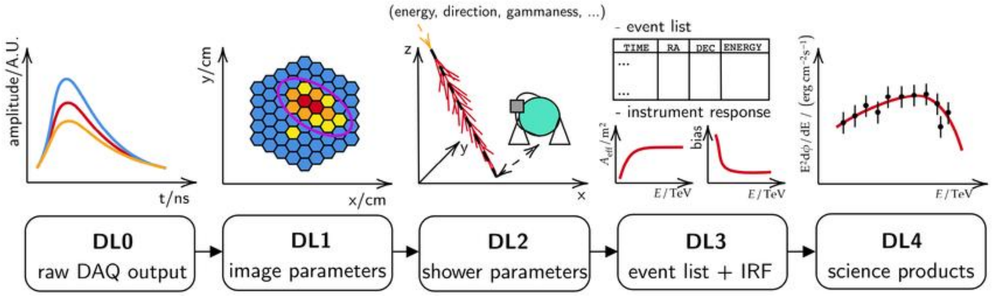
Take home messages



1. VHE gamma-ray produced in astrophysical source via non-thermal processes creates atmospheric EM showers detectable on the ground by WCD and IACTs



2. Cherenkov light produced by charged particles in the EM showers detected by IACTs via stereoscopic method



3. From camera raw data to ready-to-analyze data: waveform calibration → single telescope reconstruction → stereoscopic reconstruction